

Project Data Page:





Table of Contents

Cover Page	1
Vessel Design Summary:	2
Nozzle Summary:	5
Nozzle Schedule:	6
Bill of Material:	7
MDMT Summary:	8
Warnings and Errors:	9
Input Echo:	10
XY Coordinate Calculations:	19
Internal Pressure Calculations:	20
Element: Bottom D end	21
Element: Shell 1	24
Element: Shell 2	27
Element: Top D End	30
Shop Pressure Test Results:	33
External Pressure Calculations:	35
Element and Detail Weights:	36
Nozzle Flange MAWP:	39
Natural Frequency Calculation:	40
Wind Load Calculation:	41
Earthquake Load Calculation:	44
Wind/Earthquake Shear, Bending:	46
Wind/Earthquake Deflection:	47
Longitudinal Stress Constants:	48
Longitudinal Allowable Stresses:	49
Longitudinal Stresses due to:	50
Stress due to Combined Loads:	52
Skirt Stress Calculations:	60
Center of Gravity Calculation:	65
Basering Calculations:	66
Nozzle Calcs.: D1_3"	70
Nozzle Calcs.: L2_2"	78
Nozzle Calcs.: L4_2"	84
Nozzle Calcs.: B2_6"	90
Nozzle Calcs.: L6_2"	98
Nozzle Calcs.: M1_24"	104
Nozzle Calcs.: A1_20"	110
Nozzle Calcs.: L1_2"	121
Nozzle Calcs.: L3_2"	127
Nozzle Calcs.: L5_2"	133
Nozzle Calcs.: B1_24"	139



DESIGN CALCULATION

In Accordance with ASME Section VIII Division 2

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : ...\\A150NRL12HN001Ammonia Separator - 7 Percent.pvdb

Date of Analysis : Sep 18, 2026 1:41pm

PV Elite 28 U1, March 2026

Vessel Design Summary:

ASME Code, Section VIII Division 2, 2025

Diameter Spec : 2450.000 mm. ID	
Vessel Design Length, Tangent to Tangent	4650.00 mm.
Distance of Bottom Tangent above Grade	3300.00 mm.
Distance of Base above Grade	300.00 mm.
Specified Datum Line Distance	3000.03 mm.
Internal Design Temperature	85 °C
Internal Design Pressure	191.000 bars
External Design Temperature	85 °C
External Design Pressure	1.034 bars
Maximum Allowable Working Pressure	192 bars
External Max. Allowable Working Pressure	126 bars
Shop Test Pressure	259.603 bars
Required Minimum Design Metal Temperature	-40.0 °C
Warmest Computed Minimum Design Metal Temperature	-29.0 °C

Warning: Computed overall MDMT was higher than the required value !

Wind Design Code	ASCE/SEI 7-22
Earthquake Design Code	ASCE/SEI 7-22

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-516 70	...	...	K02700	+N+IT
Head	SA-516 70	...	...	K02700	+N+IT
Skirt	SA-36	...	...	K02600	
Skirt	SA-516 70	...	...	K02700	+N
Nozzle	SA-350 LF1	1	...	K03009	+IT
Nozzle Flg	SA-350 LF1	1	...	K03009	
Basering	SA-36	...	...	K02600	
Base Bolting	SA-36	...	...	K02600	

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized
 +Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (bars & mm.):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
Skirt 1	...	...	...	1.5000	N/A	N/S
Top Y Ring	...	...	...	1.5000	N/A	No
Bottom D end	191.214	1.03	192.000	1.5000	N/A	No
Shell 1	191.134	1.03	192.000	1.5000	N/A	No
Shell 2	191.029	1.03	192.000	1.5000	N/A	No
Top D End	191.000	1.03	193.000	1.5000	N/A	No

N/S in the Creep Range column specifies that the Creep Range Temperature was not specified in the ASME Code Section II Part D for this material.

Liquid Level: 3363.00 mm. Dens.: 0.001 kg./cm³ Sp. Gr.: 0.651

Element Types and Properties:

Element	"To" Elev	Element	Nominal	Finished	Reqd Thk	Reqd Thk	Long	Circ
---------	-----------	---------	---------	----------	----------	----------	------	------



Type	mm.	Length mm.	Thickness mm.	Thickness mm.	Internal mm.	External mm.	Eff	Eff
Skirt	-1000.0	2000.0	20.0	20.0	...	...	0.70	0.70
Skirt	-0.0	1000.0	20.0	20.0	...	...	0.70	0.55
Sphere	-0.0	0.0	86.0	80.0	79.4	5.1	1.00	1.00
Cylinder	1650.0	1650.0	157.0	157.0	156.0	12.6	1.00	1.00
Cylinder	4650.0	3000.0	157.0	157.0	155.9	12.6	1.00	1.00
Sphere	4650.0	0.0	86.0	80.0	79.3	5.1	1.00	1.00

Local Stress Analysis Results:

Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
D1_3"	WRC-107/537	0.999	n/a	Passed
B2_6"	WRC-107/537	0.999	n/a	Passed
A1_20"	WRC-297	0.999	Shell	Passed
B1_24"	WRC-107/537	0.997	n/a	Passed

Wind/Earthquake Shear, Bending:

From	To	Distance to Support mm.	Cumulative Wind Shear Kgf	Earthquake Shear Kgf	Wind Bending Kg-m.	Earthquake Bending Kg-m.
10	20	1000	7330.4	22972.8	46324.9	135628
20	30	2500	6746.32	22827.4	32247.9	89827.4
30	40	2999.98	6454.28	22684.9	25647.7	67071.5
40	60	3825	6454.27	20998.3	25647.5	67070.8
60	70	6150	5078.43	16749.6	16132.9	35928.1
70	80	7649.98	4072.2	5166.46	2406.76	3053.49

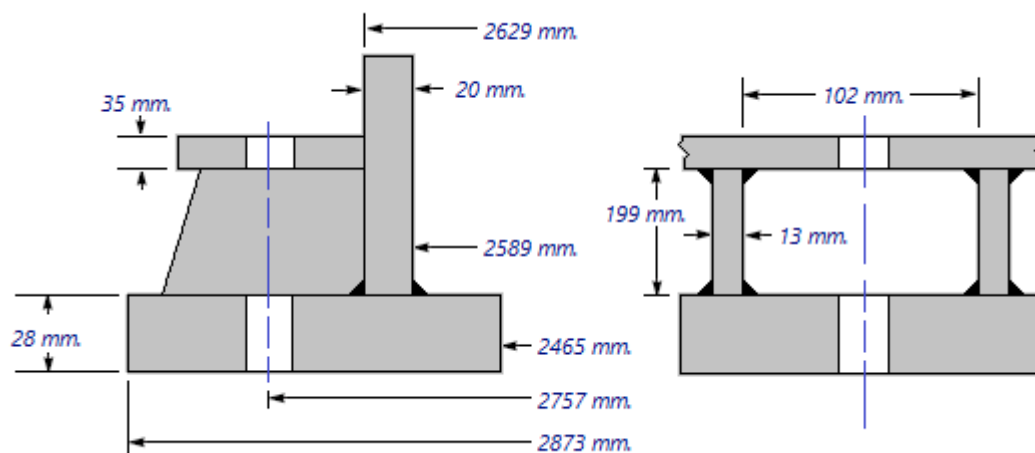
Abs Max of the all of the Stress Ratio's : 0.9347

Basing Data : Continuous Top Ring W/Gussets

Thickness of Basing 28.0000 mm.
 Inside Diameter of Basing 2465.0000 mm.
 Outside Diameter of Basing 2873.0000 mm.
 Nominal Diameter of Bolts 36.0000 mm.
 Diameter of Bolt Circle 2757.0000 mm.
 Number of Bolts 8

Thickness of Gusset Plates 13.0000 mm.
 Average Width of Gusset Plates 122.0000 mm.
 Height of Gussets 199.0000 mm.
 Distance between Gussets 102.0000 mm.
 Thickness of Top Plate or Ring 35.0000 mm.
 Radial Width of the Top Plate 122.0000 mm.
 Number of Gussets 16

Basing Sketch for Cylindrical Skirts:



Basing Cross Section View

Loads for Foundation/Support Design:

Total Wind Shear on Support	7330.	Kgf
Total Earthquake Shear on Support	22973.	Kgf
Wind Moment on Support	46415.	Kg-m.
Earthquake Moment on Support	135718.	Kg-m.
Maximum Initial Bolt Load	11400.	Kgf

Wind and Earthquake moments include the effects of user defined forces and moments if any exist in the job and were specified to act (compute loads and stresses) during these cases. Also included are moment effects due to eccentric weights if any are present in the input.

Weights:

Fabricated - Bare w/o removable internals	87633.3	kg.
Shop Test - Fabricated + water (full)	118402.9	kg.
Shipping - Fab. + rem. intls.+ shipping app.	96229.2	kg.
Erected - Fab. + rem. intls.+ insul. (etc)	96459.2	kg.
Empty - Fab. + intls. + details + wghts.	96459.2	kg.
Operating - Empty + operating liquid (no ca)	105702.9	kg.

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Nozzle Calculation Summary (bars & mm.):

Description	MAWP	Ext	MAPNC	Tbl 4.5.2	Weld	Pass/Fail
D1_3"	193	OK	...	...	OK	Passed
L2_2"	193	OK	...	OK	OK	Passed
L4_2"	193	OK	...	OK	OK	Passed
B2_6"	193	OK	...	OK	OK	Passed
L6_2"	193	OK	...	OK	OK	Passed
M1_24"	192	OK	...	...	OK	Passed
A1_20"	192	OK	...	OK	OK	Passed
L1_2"	192	OK	...	OK	OK	Passed
L3_2"	192	OK	...	OK	OK	Passed
L5_2"	192	OK	...	OK	OK	Passed
B1_24"	193	OK	...	OK	OK	Passed

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 192 Nozzle : L52"
 Minimum MAWP Shells/Flanges : 192 Element : Shell 1
 Minimum MAPnc Shells/Flanges : 211 Element : Shell 1

Computed Vessel M.A.W.P. : 192 bars

VIII-2 Nozzle to Gross Structural Discontinuity Check:

• *Error: Nozzle M1_24", was too close to a gross structural discontinuity.*

Please see ASME VIII-2 Section 4.5.13.1 for more information.

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	Y Coordinate mm.	Layout Angle deg	Inside Radius mm.
30	D1_3"	0.000	0.000	39.600
30	L2_2"	0.000	90.000	26.900
30	L4_2"	0.000	180.000	26.900
30	B2_6"	0.000	270.000	77.700
30	L6_2"	0.000	270.000	26.900
60	M1_24"	3750.000	0.000	306.300
60	A1_20"	2875.000	180.000	255.500
60	L1_2"	2200.000	0.000	26.900
60	L3_2"	2200.000	90.000	26.900
60	L5_2"	2200.000	270.000	26.900
70	B1_24"	0.000	0.000	306.300

The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.

lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

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Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia in	Wall Thk mm	Reinforcing Pad Diameter	Pad Thk mm	Cut Length mm	Flg Class
L2_2"	2.000 in	Actual	WNF	2.630	8.000	...	...	282.24	1500
L4_2"	2.000 in	Actual	WNF	2.630	8.000	...	...	282.24	1500
L6_2"	2.000 in	Actual	WNF	2.630	8.000	...	...	282.24	1500
L1_2"	2.000 in	Actual	WNF	2.630	8.000	...	...	361.54	1500
L3_2"	2.000 in	Actual	WNF	2.630	8.000	...	...	361.54	1500
L5_2"	2.000 in	Actual	WNF	2.630	8.000	...	...	361.54	1500
D1_3"	3.000 in	Actual	None	3.787	10.000	...	...	283.79	...
B2_6"	6.000 in	Actual	None	7.260	16.000	...	...	389.62	...
A1_20"	20.000 in	Actual	None	23.622	46.000	...	...	494.24	...
M1_24"	24.000 in	Actual	RTJWn	28.252	54.000	...	...	575.11	1500
B1_24"	24.000 in	Actual	None	28.252	54.000	...	...	573.47	...

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm.):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
L2_2"	SA-350 LF1	80.000	10.000	...	...	...
L4_2"	SA-350 LF1	80.000	10.000	...	...	...
L6_2"	SA-350 LF1	80.000	10.000	...	...	...
L1_2"	SA-350 LF1	157.000	10.000	...	...	...
L3_2"	SA-350 LF1	157.000	10.000	...	...	...
L5_2"	SA-350 LF1	157.000	10.000	...	...	...
D1_3"	SA-350 LF1	80.000	10.000	...	...	...
B2_6"	SA-350 LF1	80.000	10.000	...	...	...
A1_20"	SA-350 LF1	157.000	10.000	...	...	...
M1_24"	SA-350 LF1	157.000	10.000	...	...	...
B1_24"	SA-350 LF1	80.000	10.000	...	...	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum mm.	Layout Angle deg	Proj Outside mm.	Proj Inside mm.	Installed in Component
L2_2"	...	90.0	200.00	0.00	Bottom D end
L4_2"	...	180.0	200.00	0.00	Bottom D end
L6_2"	...	270.0	200.00	0.00	Bottom D end
L1_2"	2199.975	0.0	200.00	0.00	Shell 2
L3_2"	2199.975	90.0	200.00	0.00	Shell 2
L5_2"	2199.975	270.0	200.00	0.00	Shell 2
D1_3"	...	0.0	200.00	0.00	Bottom D end
B2_6"	...	270.0	300.00	0.00	Bottom D end
A1_20"	2874.975	180.0	250.00	0.00	Shell 2
M1_24"	3749.974	0.0	300.00	0.00	Shell 2
B1_24"	...	0.0	397.50	0.00	Top D End



Bill of Materials:

QTY	DESCRIPTION	MATERIAL
2	SPHERICAL HEAD: 86.0mm. THK X 2545.0mm. ID X 0.0mm.	SA-516 70
1	INSULATION: 1263mm X 40mm THK	CC
1	PLATFORM: 1000mm X 1500mm WIDE	
1	INSULATION: 1650mm X 40mm THK	CC
1	INSULATION: 3000mm X 40mm THK	CC
1	PLATFORM: 1000mm X 4814mm WIDE	
1	LADDER: 8800.0mm	...
1	INSULATION: 1264mm X 40mm THK	CC
1	CLASS 1500 GR 1.4, 24.0" BLIND FLANGE(S)	SA-350 LF1
1	BASERING: 2873.0mm. OD X 2465.0 ID X 28.0mm.	SA-36
8	BASERING BOLTS: 36.0mm. DIA	SA-36
8	BASERING BOLT NUTS: 36.0mm. DIA	...
1	BASERING TOP RING: 2873.0mm. OD X 2629.0 ID X 35.0mm.	...
16	BASERING GUSSET PLATES: 122.0mm. X 122.0mm. X 199.0mm. X 13.0mm.	...
2	CYLINDRICAL SKIRT SEGMENT 20.0mm. THK X 2828.4mm. X 2000.0mm.	SA-36
1	CYLINDRICAL SKIRT SEGMENT 20.0mm. THK X 2476.7mm. X 2000.0mm.	SA-36
1	CYLINDRICAL SKIRT SEGMENT 20.0mm. THK X 2824.1mm. X 1000.0mm.	SA-516 70
1	CYLINDRICAL SKIRT SEGMENT 20.0mm. THK X 2846.7mm. X 1000.0mm.	SA-516 70
1	CYLINDRICAL SKIRT SEGMENT 20.0mm. THK X 2462.6mm. X 1000.0mm.	SA-516 70
2	CYLINDRICAL SEGMENT 157.0mm. THK X 3057.3mm. X 1650.0mm.	SA-516 70
1	CYLINDRICAL SEGMENT 157.0mm. THK X 1582.1mm. X 1650.0mm.	SA-516 70
2	CYLINDRICAL SEGMENT 157.0mm. THK X 3848.4mm. X 3000.0mm.	SA-516 70
1	NAMEPLATE	...



Minimum Design Metal Temperature Results Summary (°C):

Description	Curve	Basic MDMT	Reduced MDMT	PWHT	Thickness ratio	Gov Thk mm.	E*
Notes							
Bottom D end	[9] !	-40		Yes			
Shell 1	[8] !	-40		Yes			
Shell 2	[8] !	-40		Yes			
Top D End	[9] !	-40		Yes			
D13"	[1] !	-29		Yes			
L22"	[1] !	-29		Yes			
Nozzle Flg	[5] !	-29	-34	Yes			
L42"	[1] !	-29		Yes			
Nozzle Flg	[5] !	-29	-34	Yes			
B26"	[1] !	-29		Yes			
L62"	[1] !	-29		Yes			
Nozzle Flg	[5] !	-29	-34	Yes			
M124"	[1] !	-29		Yes			
Nozzle Flg	[5] !	-29	-34	Yes			
A120"	[1] !	-29		Yes			
L12"	[1] !	-29		Yes			
Nozzle Flg	[5] !	-29	-43	Yes			
L32"	[1] !	-29		Yes			
Nozzle Flg	[5] !	-29	-34	Yes			
L52"	[1] !	-29		Yes			
Nozzle Flg	[5] !	-29	-34	Yes			
B124"	[1] !	-29		Yes			
Nozl. Bolting[22]		-48		No	1.000	64.000	1.00
Warmest MDMT:		-29	-34				

Required Minimum Design Metal Temperature -40.0 °C
 Warmest Computed Minimum Design Metal Temperature -29.0 °C

Warning: Computed overall MDMT was higher than the required value !

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per equation 3.1.
- [5] - ASME Flange MDMT Calcs; Thickness ratio per equation 3.3.
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per 3.11.2.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld



Class From To : Basic Element Checks.

Class From To: Check of Additional Element Data

There were no geometry errors or warnings.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



One or more nozzles in this model was specified using the actual thickness basis and a standard flange is also specified. In many cases a matching nominal diameter cannot be determined and the weight of the flange and optional blind flange cannot be determined. When actual thickness basis is specified, you must calculate the weight of the assembly (by hand) and enter it into the 'Overriding Weight' field in the nozzle dialog when there is a standard flange specified. However, for FVC nozzles that have their nominal diameter specified in the FVC dialog, entering the overriding weight for that nozzle is not required.



The level of precision is set to a value of less than 3. At this level you may not obtain proper resolution on some results. This value can be changed in Tools->Set Configuration Parameters->Default Values->Level of Precision.

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Units used in this Analysis (bars):

Name	System Unit	Constant	User Unit
Length	Feet	304.8000	mm.
Force	Pounds	0.4536	Kgf
Mass	Pounds	0.4536	kg.
Area	sq. inches	6.4516	cm ²
Moment	ft. lbs.	0.1383	Kg-m.
Stress	lbs./sq.in.	0.0069	N./mm ²
Temperature	Degrees F	0.5556	°C
Pressure	psig	0.0689	bars
Elast. Modulus	lbs./sq.in.	6.8946	KPa.
Pipe Density	lbs./cu.in.	0.0277	kg./cm ³
Ins. Density	lbs./cu.ft.	0.1600E-04	kg./cm ³
Fluid Density	lbs./cu.ft.	0.1600E-04	kg./cm ³
Wind Speed	miles/hr	1.6093	Km/hr
Tray Weight	lbs./sq.ft.	0.0005	Kgs/cm ²
Inertia	in.4	41.6231	cm4
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	4.8825	Kgs/m ²
Elevation	Feet	304.8000	mm.
Volume	in.3	16.3900	cm ³
Diameter	inches	25.4000	mm.
Thickness	inches	25.4000	mm.

PV Elite Vessel Analysis Program: Input Data

Design Internal Pressure (for Hydrotest)	191 bars
Design Internal Temperature	85.0 °C
Projection of Nozzle from Vessel Top	500 mm.
Projection of Nozzle from Vessel Bottom	500 mm.
Minimum Design Metal Temperature	-40.0 °C
Type of Construction	Welded
Special Service	None
Use Higher Longitudinal Stresses (Flag)	N
Select t for Internal Pressure (Flag)	N
Select t for External Pressure (Flag)	N
Select t for Axial Stress (Flag)	N
Select Location for Stiff. Rings (Flag)	N
Consider Vortex Shedding	N

Shop Pressure Test:

Type of Pressure Test	Hydrostatic
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	No

Load Case 1	NP+EW+1.1WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+1.1WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS
Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+1.1WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+1.1WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE
Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW
Load Case 16	IP+VE+EW
Load Case 17	NP+VO+OW
Load Case 18	FS+BS+IP+OW
Load Case 19	FS+BS+EP+OW
Load Case 20	BL+IP+OW



Wind Design Code	ASCE/SEI 7-22	
Wind Load Reduction Scale Factor	1.100	
Basic Wind Speed	[V] 180	Km/hr
Surface Roughness Category	D: Flat, unobstructed	
Type of Surface	Moderately Smooth	
Base Elevation	300	mm.
Percent Wind for Hydrotest	33.0	
Using User defined Wind Press. Vs Elev.	N	
Height of Hill or Escarpment	H or Hh 0	mm.
Distance Upwind of Crest	Lh 0	mm.
Distance from Crest to the Vessel	x 0	mm.
Type of Terrain (Hill, Escarpment)	Flat	
Damping Factor (Beta) for Wind (Ope)	0.0100	
Damping Factor (Beta) for Wind (Empty)	0.0000	
Damping Factor (Beta) for Wind (Filled)	0.0000	
Site Elevation above Sea Level	0	mm.

Seismic Design Code	ASCE/SEI 7-22	
Importance Factor	1.500	
Force Modification Factor R	1.500	
Site Class	D	
Component Elevation Ratio	z/h 0.000	
Amplification Factor	Ap 0.000	
Force Factor	0.000	
Consider Vertical Acceleration	No	
Minimum Acceleration Multiplier	0.000	
User Value of Sds (used if > 0)	0.000	
User Value of Sd1 (used if > 0)	0.000	
Component resonance duct. fact.	[CAR] 0.000	
Component strength factor	[Rpo] 0.000	
Overstrength factor	[Omega0] 0.000	
Response Acceleration Factor	[SMS] 0.326	
Response Acceleration Factor	[SM1] 0.139	
Building Importance Factor	[Ie] 1.000	
Site Latitude	0.000	deg
Site Longitude	0.000	deg

M.A.W.P + Static Head to Nozzle	Y
Consider MAP New and Cold in Noz. Design	N
Consider External Loads for Nozzle Des.	Y
Use ASME VIII-1 Appendix 1-9	N

Perform Blast Load Analysis	No
Material Database Year	Current w/Addenda or Code Year

Configuration Directives:

g0 is taken as min(g0, attached shell thk)	Yes
Using Metric Material Databases, ASME II D	Yes
Using ASME VIII-1 Appendix 46	No
Calculate B31.3 type stress for Nozzles with Loads	Yes
Reduce the MDMT due to lower membrane stress	Yes
Consider Longitudinal Stress in MDMT Calculations	Yes

Complete Listing of Vessel Elements and Details:

Element#	1/6
Element From Node	10
Element To Node	20
Element Type	Skirt Support
Description	Skirt 1
Distance "FROM" to "TO"	2000 mm.
Skirt Outside Diameter	2629 mm.
Diameter of Skirt at Base	2629 mm.
Skirt Thickness	20 mm.
Internal Corrosion Allowance	0.75 mm.
Nominal Thickness	20 mm.
External Corrosion Allowance	0.75 mm.
Design Temperature	85 °C
Internal Pressure	



Design Temperature External Pressure	85 °C
Effective Diameter Multiplier	1.36
Material Name	SA-36
Allowable Stress, Ambient	165 N./mm ²
Allowable Stress, Operating	153.14 N./mm ²
Allowable Stress, Hydrotest	237.5 N./mm ²
Material Density	0.00775 kg./cm ³
P Number Thickness	0 mm.
Yield Stress, Operating	229.57 N./mm ²
MDMT Curve	A
External Pressure Chart Name	CS-2
UNS Number	K02600
Product Form	Plate; bar; shapes
Efficiency, Longitudinal Seam	0.7
Efficiency, Head-to-Skirt or Circ. Seam	0.7
Weld is pre-Heated	No

Element#	2/6
Element From Node	20
Element To Node	30
Element Type	Skirt Support
Description	Top Y Ring
Distance "FROM" to "TO"	1000 mm.
Skirt Outside Diameter	2629 mm.
Diameter of Skirt at Base	2629 mm.
Skirt Thickness	20 mm.
Internal Corrosion Allowance	0.75 mm.
Nominal Thickness	20 mm.
External Corrosion Allowance	0.75 mm.
Design Temperature Internal Pressure	85 °C
Design Temperature External Pressure	85 °C
Effective Diameter Multiplier	1.36
Material Name	SA-516 70 +N
Allowable Stress, Ambient	175 N./mm ²
Allowable Stress, Operating	161.14 N./mm ²
Allowable Stress, Hydrotest	247 N./mm ²
Material Density	0.00775 kg./cm ³
P Number Thickness	0 mm.
Yield Stress, Operating	242 N./mm ²
MDMT Curve	B
External Pressure Chart Name	CS-2
UNS Number	K02700
Product Form	Plate
Efficiency, Longitudinal Seam	0.7
Efficiency, Head-to-Skirt or Circ. Seam	0.55
Weld is pre-Heated	No

Element#	3/6
Element From Node	30
Element To Node	40
Element Type	Spherical
Description	Bottom D end
Distance "FROM" to "TO"	0 mm.
Inside Diameter	2545 mm.
Element Thickness	80 mm.
Internal Corrosion Allowance	1.5 mm.
Nominal Thickness	86 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	191 bars
Design Temperature Internal Pressure	85 °C
Design External Pressure	1.0342 bars
Design Temperature External Pressure	85 °C
Effective Diameter Multiplier	1.36
Material Name	SA-516 70 +N+IT
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0



Weld is pre-Heated	No
Element From Node	30
Detail Type	Liquid
Detail ID	Liquid 30
Dist. from "FROM" Node / Offset dist	-1263 mm.
Height/Length of Liquid	1263 mm.
Liquid Density	0.0006495 kg./cm ³
Element From Node	30
Detail Type	Insulation
Detail ID	Ins: 50
Dist. from "FROM" Node / Offset dist	-1263 mm.
Height/Length of Insulation	1263 mm.
Thickness of Insulation	40 mm.
Density	0.40000E-04 kg./cm ³
Element From Node	30
Detail Type	Nozzle
Detail ID	D13"
Dist. from "FROM" Node / Offset dist	0 mm.
Nozzle Diameter	3 in.
Nozzle Schedule	None
Nozzle Class	None
Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	86.286 Kg
Grade of Attached Flange	None
Nozzle Matl	SA-350 LF1 +IT
Element From Node	30
Detail Type	Nozzle
Detail ID	L22"
Dist. from "FROM" Node / Offset dist	1200 mm.
Nozzle Diameter	2 in.
Nozzle Schedule	None
Nozzle Class	1500
Layout Angle	90.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	57.761 Kg
Grade of Attached Flange	GR 1.4
Nozzle Matl	SA-350 LF1 +IT
Element From Node	30
Detail Type	Nozzle
Detail ID	L42"
Dist. from "FROM" Node / Offset dist	1200 mm.
Nozzle Diameter	2 in.
Nozzle Schedule	None
Nozzle Class	1500
Layout Angle	180.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	41.35 Kg
Grade of Attached Flange	GR 1.4
Nozzle Matl	SA-350 LF1 +IT
Element From Node	30
Detail Type	Nozzle
Detail ID	B26"
Dist. from "FROM" Node / Offset dist	700 mm.
Nozzle Diameter	6 in.
Nozzle Schedule	None
Nozzle Class	None
Layout Angle	270.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	253.38 Kg
Grade of Attached Flange	None
Nozzle Matl	SA-350 LF1 +IT
Element From Node	30



Detail Type	Nozzle
Detail ID	L62"
Dist. from "FROM" Node / Offset dist	1200 mm.
Nozzle Diameter	2 in.
Nozzle Schedule	None
Nozzle Class	1500
Layout Angle	270.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	41.35 Kg
Grade of Attached Flange	GR 1.4
Nozzle Matl	SA-350 LF1 +IT

Element From Node	30
Detail Type	Weight
Detail ID	Btm Dish SF
Dist. from "FROM" Node / Offset dist	0 mm.
Miscellaneous Weight	2300 Kg
Offset from Element Centerline	0 mm.

Element From Node	30
Detail Type	Weight
Detail ID	vortex Breaker
Dist. from "FROM" Node / Offset dist	0 mm.
Miscellaneous Weight	5 Kg
Offset from Element Centerline	700 mm.

Element#	4/6
Element From Node	40
Element To Node	60
Element Type	Cylinder
Description	Shell 1
Distance "FROM" to "TO"	1650 mm.
Inside Diameter	2450 mm.
Element Thickness	157 mm.
Internal Corrosion Allowance	1.5 mm.
Nominal Thickness	157 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	191 bars
Design Temperature Internal Pressure	85 °C
Design External Pressure	1.0342 bars
Design Temperature External Pressure	85 °C
Effective Diameter Multiplier	1.36
Material Name	SA-516 70 +N+IT
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	40
Detail Type	Platform
Detail ID	PLAT:[1 OF 1]
Dist. from "FROM" Node / Offset dist	0 mm.
Platform Start Angle (degrees)	0.0
Platform End Angle (degrees)	360.0
Platform Wind Area	60000 cm ²
Platform Weight	3553.9 Kg
Platform Railing Weight	0 Kg/mm.
Platform Grating Weight	0.017 Kgs/cm ²
Platform Width	1500 mm.
Platform Height	1000 mm.
Platform Clearance or End Offset	85 mm.
Platform Force Coefficient	2.0
Ladder Layout Angle	0.0
Ladder Start Elevation	0 mm.
Ladder End Elevation	0 mm.
Unit Weight of Ladder	0 Kg/mm.
Platform Length (top head platform)	0 mm.

Element From Node	40
-------------------	----



Detail Type	Liquid
Detail ID	Liquid 40
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Liquid	1650 mm.
Liquid Density	0.0006495 kg./cm ³

Element From Node	40
Detail Type	Insulation
Detail ID	Ins: 50
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Insulation	1650 mm.
Thickness of Insulation	40 mm.
Density	0.40000E-04 kg./cm ³

Element#	5/6
Element From Node	60
Element To Node	70
Element Type	Cylinder
Description	Shell 2
Distance "FROM" to "TO"	3000 mm.
Inside Diameter	2450 mm.
Element Thickness	157 mm.
Internal Corrosion Allowance	1.5 mm.
Nominal Thickness	157 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	191 bars
Design Temperature Internal Pressure	85 °C
Design External Pressure	1.0342 bars
Design Temperature External Pressure	85 °C
Effective Diameter Multiplier	1.36
Material Name	SA-516 70 +N+IT
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	60
Detail Type	Liquid
Detail ID	Liquid: 60
Dist. from "FROM" Node / Offset dist	60 mm.
Height/Length of Liquid	450 mm.
Liquid Density	0.0006495 kg./cm ³

Element From Node	60
Detail Type	Insulation
Detail ID	Ins: 50
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Insulation	3000 mm.
Thickness of Insulation	40 mm.
Density	0.40000E-04 kg./cm ³

Element From Node	60
Detail Type	Nozzle
Detail ID	M124"
Dist. from "FROM" Node / Offset dist	2100 mm.
Nozzle Diameter	24 in.
Nozzle Schedule	None
Nozzle Class	1500
Layout Angle	0.0
Blind Flange (Y/N)	Y
Weight of Nozzle (Used if > 0)	5005.5 Kg
Grade of Attached Flange	GR 1.4
Nozzle Matl	SA-350 LF1 +IT

Element From Node	60
Detail Type	Nozzle
Detail ID	A120"
Dist. from "FROM" Node / Offset dist	1225 mm.
Nozzle Diameter	20 in.



Nozzle Schedule	None
Nozzle Class	None
Layout Angle	180.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	1506.5 Kgf
Grade of Attached Flange	None
Nozzle Matl	SA-350 LF1 +IT

Element From Node	60
Detail Type	Nozzle
Detail ID	L12"
Dist. from "FROM" Node / Offset dist	550 mm.
Nozzle Diameter	2 in.
Nozzle Schedule	None
Nozzle Class	1500
Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	77.859 Kgf
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-350 LF1 +IT

Element From Node	60
Detail Type	Nozzle
Detail ID	L32"
Dist. from "FROM" Node / Offset dist	550 mm.
Nozzle Diameter	2 in.
Nozzle Schedule	None
Nozzle Class	1500
Layout Angle	90.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	77.859 Kgf
Grade of Attached Flange	GR 1.4
Nozzle Matl	SA-350 LF1 +IT

Element From Node	60
Detail Type	Nozzle
Detail ID	L52"
Dist. from "FROM" Node / Offset dist	550 mm.
Nozzle Diameter	2 in.
Nozzle Schedule	None
Nozzle Class	1500
Layout Angle	270.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	77.859 Kgf
Grade of Attached Flange	GR 1.4
Nozzle Matl	SA-350 LF1 +IT

Element From Node	60
Detail Type	Weight
Detail ID	Demister WT
Dist. from "FROM" Node / Offset dist	2775 mm.
Miscellaneous Weight	230 Kgf
Offset from Element Centerline	0 mm.

Element From Node	60
Detail Type	Weight
Detail ID	FEED PIPE WT
Dist. from "FROM" Node / Offset dist	1225 mm.
Miscellaneous Weight	260 Kgf
Offset from Element Centerline	0 mm.

Element From Node	60
Detail Type	Weight
Detail ID	Lifting WT
Dist. from "FROM" Node / Offset dist	700 mm.
Miscellaneous Weight	2800 Kgf
Offset from Element Centerline	0 mm.

Element From Node	60
Detail Type	Weight



Detail ID	CLIP WT	
Dist. from "FROM" Node / Offset dist	700	mm.
Miscellaneous Weight	900	Kgf
Offset from Element Centerline	0	mm.

Element From Node	60	
Detail Type	Weight	
Detail ID	Manway Davit	
Dist. from "FROM" Node / Offset dist	2100	mm.
Miscellaneous Weight	50	Kgf
Offset from Element Centerline	1800	mm.

Element From Node	60	
Detail Type	Weight	
Detail ID	demister suppor	
Dist. from "FROM" Node / Offset dist	2775	mm.
Miscellaneous Weight	30	Kgf
Offset from Element Centerline	0	mm.

Element#	6/6	
Element From Node	70	
Element To Node	80	
Element Type	Spherical	
Description	Top D End	
Distance "FROM" to "TO"	0	mm.
Inside Diameter	2545	mm.
Element Thickness	80	mm.
Internal Corrosion Allowance	1.5	mm.
Nominal Thickness	86	mm.
External Corrosion Allowance	0	mm.
Design Internal Pressure	191	bars
Design Temperature Internal Pressure	85	°C
Design External Pressure	1.0342	bars
Design Temperature External Pressure	85	°C
Effective Diameter Multiplier	1.36	
Material Name	SA-516 70 +N+IT	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	70	
Detail Type	Platform	
Detail ID	PLAT 2	
Dist. from "FROM" Node / Offset dist	1400	mm.
Platform Start Angle (degrees)	0.0	
Platform End Angle (degrees)	0.0	
Platform Wind Area	167806	cm ²
Platform Weight	4381.7	Kgf
Platform Railing Weight	0	Kgf/mm.
Platform Grating Weight	0.017	Kgs/cm ²
Platform Width	4814	mm.
Platform Height	1000	mm.
Platform Clearance or End Offset	2407	mm.
Platform Force Coefficient	2.0	
Ladder Layout Angle	45.0	
Ladder Start Elevation	0	mm.
Ladder End Elevation	8800	mm.
Unit Weight of Ladder	0.05	Kgf/mm.
Platform Length (top head platform)	4814	mm.

Element From Node	70	
Detail Type	Insulation	
Detail ID	Ins: 50	
Dist. from "FROM" Node / Offset dist	0	mm.
Height/Length of Insulation	1264	mm.
Thickness of Insulation	40	mm.
Density	0.40000E-04	kg./cm ³



Element From Node	70
Detail Type	Nozzle
Detail ID	B124"
Dist. from "FROM" Node / Offset dist	0 mm.
Nozzle Diameter	24 in.
Nozzle Schedule	None
Nozzle Class	None
Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	1246.7 Kg
Grade of Attached Flange	None
Nozzle Matl	SA-350 LF1 +IT

Element From Node	70
Detail Type	Weight
Detail ID	Top dish SF
Dist. from "FROM" Node / Offset dist	0 mm.
Miscellaneous Weight	2300 Kg
Offset from Element Centerline	0 mm.

XY Coordinate Calculations:

From	To	X (Horiz.) mm.	Y (Vert.) mm.	DX (Horiz.) mm.	DY (Vert.) mm.
Skirt 1		...	2000	...	2000
Top Y Ring		...	3000	...	1000
Bottom D end		...	3000	...	...
Shell 1		...	4650	...	1650
Shell 2		...	7650	...	3000
Top D End		...	7650	...	...

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

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Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd bars	Nominal Thickness mm.	Total Corr Allowance mm.	Element Diameter mm.	Allowable Stress(SE) N./mm ²	Element#
Skirt 1		...	20	1.5	2589	...	1
Top Y Ring		...	20	1.5	2589	...	2
Bottom D end		191.21	86	1.5	2545	161.14	3
Shell 1		191.13	157	1.5	2450	161.14	4
Shell 2		191.03	157	1.5	2450	161.1	5
Top D End		191	86	1.5	2545	161.14	6

Element Required Thickness and MAWP :

From	To	Design Pressure bars	M.A.W.P. Corroded bars	M.A.P. New & Cold bars	Minimum Thickness mm.	Required Thickness mm.	Element#
Skirt 1		...	No Calc	No Calc	20	No Calc	1
Top Y Ring		...	No Calc	No Calc	20	No Calc	2
Bottom D end		191	192.477	213.383	80	79.3803	3
Shell 1		191	192.204	211.018	157	155.967	4
Shell 2		191	192.261	211.022	157	155.919	5
Top D End		191	192.691	213.383	80	79.2904	6

Minimum 192 211

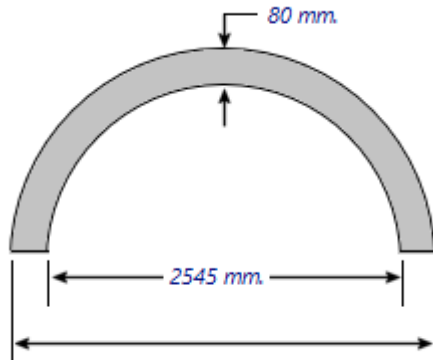
MAWP: 192 bars, limited by: Shell 1.

Elements Suitable for Design Internal Pressure.

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: Bottom D end

Spherical Head Design Information:					
Design Pressure	P	191.214 bars	Int. Design Temperature		85 °C
Material		SA-516 70	Ext. Design Temperature		85 °C
Length		0.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.5000 mm.	Allowable Stress (ope)	S	161.1 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	175.0 N./mm ²
Inside Diameter	Di	2545.000 mm.	Actual Stress		150.5 N./mm ²
Spec. Min. Thk t _e , or s _a		80.0000 mm.	Spec. Nominal Thk.		86.000 mm.
Surface Area		115957.6 cm ²	SG of Contents		0.651
Element Volume		4316282.0 cm ³	Weight of Contents		2806.0 Kg
Empty Weight		7250.1 Kg	Operating Weight		10056.1 Kg
			Comp. All. Ope. Stress		122.7 N./mm ²
Radiography Information:					
...			...		
Circ. Joint Efficiency		100 %	Long. Joint Eff.	E or v	100 %

Tolerance for Formed Heads per 4.3.2.2:

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 31.813 mm.
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 15.906 mm.

Material treatment options: +N+IT

Spherical Shell Calculation - Section 4.3.5.1

Computed Minimum Required Thickness [t]:
 $= D/2 \cdot (\exp(0.5P/(S \cdot E)) - 1) + c_i + c_o$
 $= 2548.0/2 \cdot (\exp(0.5 \cdot 191.214/(161.141 \cdot 1.0)) - 1) + 1.5 + 0.0$
 $= 77.8803 + 1.5 + 0.0 = 79.3803 \text{ mm.}$

Computed Maximum Allowable Working Pressure [MAWP]:
 $= 2 \cdot S \cdot E \cdot \ln((2 \cdot t + D)/D) - P_{liq}$
 $= 2 \cdot 161.1 \cdot 1.0 \cdot \ln((2 \cdot 78.5 + 2548.0)/2548.0) - 0.214$
 $= 192.691 - 0.214 = 192.4768 \text{ bars}$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:
 $= 2 \cdot S \cdot E \cdot \ln((2 \cdot t + D)/D)$
 $= 2 \cdot 174.997 \cdot 1.0 \cdot \ln((2 \cdot 80.0 + 2545.0)/2545.0)$
 $= 213.3833 \text{ bars}$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D^2 / (E (D_o^2 - D^2))$$

$$= 191.214 \cdot 2548.0^2 / (1.0 (2705.0^2 - 2548.0^2))$$

$$= 150.535 \text{ N./mm}^2$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= 0.5 \cdot P / (E \cdot \ln((2 \cdot t + D) / D))$$

$$= 0.5 \cdot 191.214 / (1.0 \cdot \ln((2 \cdot 78.5 + 2548.0) / 2548.0))$$

$$= 159.906 \text{ N./mm}^2$$

Percent Fiber Elongation per Table 6.1

$$\text{For Heads formed from Segments } (75 \cdot t_{\text{nom}} / R_f) \cdot (1 - R_f / R_o) \quad 4.903 \%$$

$$\text{For one Piece doubly Curved Heads } 100 \cdot \ln(D_b / (D_f - 2t)) \quad 41.197 \%$$

Where D_b is the blank diameter for one piece doubly curved surfaces [D_b]:

$$= \text{square root}(4 \cdot \text{Head Surface Area} / 3.14159)$$

$$= \text{square root}(4 \cdot 115957.6 / 3.14159)$$

$$= 3842.418 \text{ mm.}$$

MDMT Calculations in the Spherical Portion:

Note: This Element/Detail was specified as being Impact Tested.

Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

Tolerance for Formed Heads per 4.3.2.2:

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 31.813 mm.
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 15.906 mm.

Material treatment options: +N+IT

EXTERNAL PRESSURE RESULTS for: Bottom D end

Spherical Head From 30 to 40 Ext. Chart: CS-2 at 85 °C

Maximum External Pressure per ASME Division 2 Section 4.4:

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$$= 0.075 \cdot E_y (t / R_o)$$

$$= 0.075 \cdot 0.19904\text{E}+09 (78.5 / 1352.5)$$

$$= 866.491 \text{ N./mm}^2$$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$$= F_{he} / S_y$$

$$= 866.5 / 242.0$$

$$= 3.581$$

Engineering Yield to Tensile Stress ratio [R]:

$$= \sigma_{ys} / \sigma_{uts}$$

$$= 242.0 / 482.6$$

$$= 0.501$$

Ratio [AE]:

$$= F_{he} / E$$

$$= 866.5 / 0.19904\text{E}+09$$

$$= 0.004353$$

Factor [m1]:

$$= (\ln(r) + (\epsilon_{psp} - \epsilon_{sys})) / \ln(\ln(1 + \epsilon_{psp}) / \ln(1 + \epsilon_{sys}))$$

$$= (\ln(0.501) + (0.00002 - 0.002)) / \ln(\ln(1 + 0.00002) / \ln(1 + 0.002))$$

$$= 0.150$$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$$= \sigma_{ys} (1 + \epsilon_{sys}) / \ln(1 + \epsilon_{sys}) m_1$$

$$= 242.0 (1 + 0.002) / \ln(1 + 0.002) 0.150373$$

= 617.442 N./mm²

Curve Fitting Exponent for, Ferritic Steel [m2]:

= $m2f(m2f2 - R)$
 = $0.6(1.0 - 0.501)$
 = 0.299

Curve Fitting Constant [A2]:

= $(\sigma_{uts} \cdot \exp(m2)) / (m2 m2)$
 = $(482.6 \cdot \exp(0.299)) / (0.299 0.299)$
 = 934.000 N./mm²

Material Parameter for Stress-Strain Curve Model [K]:

= $1.5 \cdot R1.5 - 0.5 \cdot R2.5 - R3.5$
 = $1.5 \cdot 0.5011.5 - 0.5 \cdot 0.5012.5 - 0.5013.5$
 = 0.354

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	241	-2.016	0.267E-05	0.242E-05	0.751E-05	-0.638E-05	0.149E+08	0.119E-01
2	120	-4.846	0.531E-07	0.531E-07	0.148E-05	-0.148E-05	0.164E+09	0.362E-02
3	180	-3.431	0.525E-06	0.522E-06	0.383E-05	-0.380E-05	0.634E+08	0.150E-02
4	211	-2.723	0.125E-05	0.123E-05	0.550E-05	-0.531E-05	0.315E+08	0.234E-02
5	196	-3.077	0.825E-06	0.817E-06	0.462E-05	-0.455E-05	0.451E+08	0.500E-05
6	203	-2.900	0.102E-05	0.101E-05	0.505E-05	-0.493E-05	0.378E+08	0.104E-02
7	199	-2.989	0.919E-06	0.908E-06	0.483E-05	-0.474E-05	0.413E+08	0.486E-03
8	197	-3.033	0.871E-06	0.861E-06	0.472E-05	-0.464E-05	0.432E+08	0.234E-03
9	197	-3.055	0.848E-06	0.839E-06	0.467E-05	-0.459E-05	0.441E+08	0.113E-03
10	196	-3.066	0.836E-06	0.828E-06	0.465E-05	-0.457E-05	0.446E+08	0.534E-04
11	196	-3.072	0.831E-06	0.822E-06	0.463E-05	-0.456E-05	0.448E+08	0.241E-04
12	196	-3.074	0.828E-06	0.820E-06	0.463E-05	-0.455E-05	0.450E+08	0.953E-05
13	196	-3.076	0.827E-06	0.818E-06	0.462E-05	-0.455E-05	0.450E+08	0.226E-05
14	196	-3.076	0.826E-06	0.817E-06	0.462E-05	-0.455E-05	0.451E+08	0.137E-05
15	196	-3.076	0.826E-06	0.818E-06	0.462E-05	-0.455E-05	0.450E+08	0.441E-06
16	196	-3.076	0.826E-06	0.818E-06	0.462E-05	-0.455E-05	0.451E+08	0.467E-06
17	196	-3.076	0.826E-06	0.818E-06	0.462E-05	-0.455E-05	0.451E+08	0.132E-07

Factor of Safety [FS]:

Since $Fic > 0.55 Fy$ and $Fic < Fy$
 = $2.407 - 0.741 \cdot Fic / Fy$
 = $2.407 - 0.741 \cdot 196.12/242.0$
 = 1.806

Allowable Circumferential Compressive Stress [Fha]:

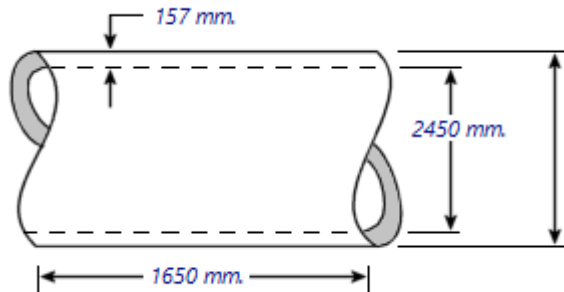
= Fic/FS
 = $196.117/1.806$
 = 108.563 N./mm²

Allowable External Pressure at the given Thickness [Pa]:

= $2 \cdot Fha[t/Ro]$
 = $2 \cdot 108.563[78.5/1352.5]$
 = 126.014 bars

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: Shell 1

Cylindrical Shell Design Information:					
Design Pressure	P	191.134 bars	Int. Design Temperature		85 °C
Material		SA-516 70	Ext. Design Temperature		85 °C
Length		1650.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.5000 mm.	Allowable Stress (ope)	S	161.1 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	175.0 N./mm ²
Inside Diameter	Di	2450.000 mm.	Actual Stress		150.8 N./mm ²
Spec. Min. Thk t _e , or s _a		157.0000 mm.	Spec. Nominal Thk.		157.000 mm.
Surface Area		143275.5 cm ²	SG of Contents		0.651
Element Volume		7780076.0 cm ³	Weight of Contents		5058.2 Kg
Empty Weight		16443.9 Kg	Operating Weight		21502.1 Kg
			Comp. All. Ope. Stress		122.7 N./mm ²

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +N+IT

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 2453.0 \cdot (\exp(191.134/(161.141 \cdot 1.0)) - 1) + 1.5 + 0.0$$

$$= 154.4674 + 1.5 + 0.0 = 155.9674 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 161.141 \cdot 1.0 \cdot \ln((2 \cdot 155.5 + 2453.0) / 2453.0) - 0.134$$

$$= 192.338 - 0.134 = 192.2043 \text{ bars}$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 174.997 \cdot 1.0 \cdot \ln((2 \cdot 157.0 + 2450.0) / 2450.0)$$

$$= 211.0181 \text{ bars}$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E (D_o - D))$$

$$= 191.134 \cdot 2453.0 / (1.0 (2764.0 - 2453.0))$$

$$= 150.7649 \text{ N./mm}^2$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$

$$= 191.134 / (1.0 \cdot \ln((2 \cdot 155.5 + 2453.0) / 2453.0))$$

$$= 160.1320 \text{ N./mm}^2$$

Percent Fiber Elongation per Table 6.1

$$\text{Cylinders formed from Plate} \quad (50 \cdot t_{\text{nom}} / R_f) \cdot (1 - R_f / R_o) \quad 6.022 \%$$

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

Note:
 Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

Material treatment options: +N+IT

EXTERNAL PRESSURE RESULTS for: Shell 1

Cylindrical Shell From 40 to 60 Ext. Chart: CS-2 at 85 °C

Maximum External Pressure per ASME Division 2 Section 4.4:

Determine Parameter [Mx]:

$$= L / \sqrt{(R_o \cdot t)}$$

$$= 5498.333 / \sqrt{(1382.0 \cdot 155.5)}$$

$$= 11.861$$

Compute the Factor [Ch]:

Since $M_x > 1.5$ and $M_x < 13$, $Ch = 0.92 / (M_x - 0.579)$

$$= 0.92 / (11.861 - 0.579)$$

$$= 0.0815$$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$$= 1.6 \cdot Ch \cdot E_y(t / D_o)$$

$$= 1.6 \cdot 0.082 \cdot 0.19766E+09(155.5 / 2764.0)$$

$$= 1451.026 \text{ N./mm}^2$$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$$= F_{he} / F_y$$

$$= 1451.0 / 242.0$$

$$= 5.996$$

Predicted Buckling Stress [Fic]:

Engineering Yield to Tensile Stress ratio [R]:

$$= \sigma_{ys} / \sigma_{uts}$$

$$= 242.0 / 482.6$$

$$= 0.501$$

Ratio [AE]:

$$= F_{he} / E$$

$$= 1451.0 / 0.19766E+09$$

$$= 0.007340$$

Factor [m1]:

$$= (\ln(r) + (\epsilon_{psp} - \epsilon_{sys})) / \ln(\ln(1 + \epsilon_{psp}) / \ln(1 + \epsilon_{sys}))$$

$$= (\ln(0.501) + (0.00002 - 0.002)) / \ln(\ln(1 + 0.00002) / \ln(1 + 0.002))$$

$$= 0.150$$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$$= \sigma_{ys}(1 + \epsilon_{sys}) / \ln(1 + \epsilon_{sys}) m_1$$

$$= 242.0(1 + 0.002) / \ln(1 + 0.002) \cdot 0.150373$$

$$= 617.442 \text{ N./mm}^2$$

Curve Fitting Exponent for, Ferritic Steel [m2]:

$$= m_2 f(m_2 f_2 - R)$$

$$= 0.6(1.0 - 0.501)$$

$$= 0.299$$

Curve Fitting Constant [A2]:
 $= (\sigma_{uts} \cdot \exp(m_2)) / (m_2 m_2)$
 $= (482.6 \cdot \exp(0.299)) / (0.299 \cdot 0.299)$
 $= 934.000 \text{ N./mm}^2$

Material Parameter for Stress-Strain Curve Model [K]:
 $= 1.5 \cdot R1.5 - 0.5 \cdot R2.5 - R3.5$
 $= 1.5 \cdot 0.5011.5 - 0.5 \cdot 0.5012.5 - 0.5013.5$
 $= 0.354$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	241	-2.016	0.267E-05	0.242E-05	0.751E-05	-0.638E-05	0.149E+08	0.889E-02
2	120	-4.846	0.531E-07	0.531E-07	0.148E-05	-0.148E-05	0.163E+09	0.660E-02
3	180	-3.431	0.525E-06	0.522E-06	0.383E-05	-0.380E-05	0.632E+08	0.448E-02
4	211	-2.723	0.125E-05	0.123E-05	0.550E-05	-0.531E-05	0.315E+08	0.639E-03
5	226	-2.370	0.185E-05	0.177E-05	0.646E-05	-0.599E-05	0.218E+08	0.305E-02
6	218	-2.546	0.153E-05	0.148E-05	0.597E-05	-0.567E-05	0.262E+08	0.996E-03
7	214	-2.635	0.139E-05	0.135E-05	0.573E-05	-0.549E-05	0.288E+08	0.132E-03
8	213	-2.679	0.132E-05	0.129E-05	0.561E-05	-0.540E-05	0.301E+08	0.264E-03
9	213	-2.657	0.135E-05	0.132E-05	0.567E-05	-0.545E-05	0.294E+08	0.687E-04
10	214	-2.646	0.137E-05	0.133E-05	0.570E-05	-0.547E-05	0.291E+08	0.311E-04
11	214	-2.652	0.136E-05	0.133E-05	0.568E-05	-0.546E-05	0.293E+08	0.189E-04
12	214	-2.649	0.136E-05	0.133E-05	0.569E-05	-0.547E-05	0.292E+08	0.606E-05
13	214	-2.650	0.136E-05	0.133E-05	0.569E-05	-0.546E-05	0.292E+08	0.645E-05
14	214	-2.649	0.136E-05	0.133E-05	0.569E-05	-0.546E-05	0.292E+08	0.200E-06
15	214	-2.649	0.136E-05	0.133E-05	0.569E-05	-0.546E-05	0.292E+08	0.293E-05
16	214	-2.649	0.136E-05	0.133E-05	0.569E-05	-0.546E-05	0.292E+08	0.136E-05
17	214	-2.649	0.136E-05	0.133E-05	0.569E-05	-0.546E-05	0.292E+08	0.582E-06

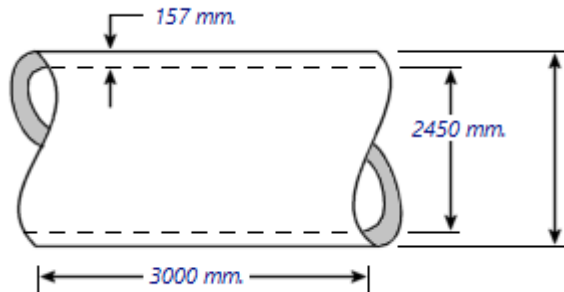
Factor of Safety [FS]:
 Since $Fic > 0.55 F_y$ and $Fic < F_y$
 $= 2.407 - 0.741 \cdot Fic / F_y$
 $= 2.407 - 0.741 \cdot 214.32/242.0$
 $= 1.751$

Allowable Circumferential Compressive Stress [Fha]:
 $= Fic/FS$
 $= 214.315/1.751$
 $= 122.412 \text{ N./mm}^2$

Allowable External Pressure at the given Thickness [Pa]:
 $= 2 \cdot Fha(t / D_o)$
 $= 2 \cdot 1224.053(155.5/2764.0)$
 $= 137.728 \text{ bars}$

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: Shell 2

Cylindrical Shell Design Information:					
Design Pressure	P	191.029 bars	Int. Design Temperature		85 °C
Material		SA-516 70	Ext. Design Temperature		85 °C
Length		3000.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.5000 mm.	Allowable Stress (ope)	S	161.1 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	175.0 N./mm ²
Inside Diameter	Di	2450.000 mm.	Actual Stress		150.7 N./mm ²
Spec. Min. Thk t _e , or sa		157.0000 mm.	Spec. Nominal Thk.		157.000 mm.
Surface Area		260500.9 cm ²	SG of Contents		0.651
Element Volume		14145593.0 cm ³	Weight of Contents		1379.5 Kg
Empty Weight		29896.4 Kg	Operating Weight		31275.9 Kg
			Comp. All. Ope. Stress		122.7 N./mm ²

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +N+IT

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$
 $= 0.5 \cdot 2453.0 \cdot (\exp(191.029/(161.1 \cdot 1.0)) - 1) + 1.5 + 0.0$
 $= 154.419 + 1.5 + 0.0 = 155.919 \text{ mm. Equation 4.3.1}$

Computed Maximum Allowable Working Pressure [MAWP]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$
 $= 161.1 \cdot 1.0 \cdot \ln((2 \cdot 155.5 + 2453.0) / 2453.0) - 0.029$
 $= 192.289 - 0.029 = 192.2606 \text{ bars}$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D)$
 $= 175.0 \cdot 1.0 \cdot \ln((2 \cdot 157.0 + 2450.0) / 2450.0)$
 $= 211.0217 \text{ bars}$

Computed Stress Intensity at Design Pressure [Si]:
 $= P \cdot D / (E (D_o - D))$
 $= 191.029 \cdot 2453.0 / (1.0 (2764.0 - 2453.0))$
 $= 150.6819 \text{ N./mm}^2$

Computed Membrane Stress at Design Pressure [Sm]:
 $= P / (E \cdot \ln((2 \cdot t + D) / D))$

$$= 191.029 / (1.0 \cdot \ln((2 \cdot 155.5 + 2453.0) / 2453.0))$$

$$= 160.0439 \text{ N./mm}^2$$

Percent Fiber Elongation per Table 6.1

$$\text{Cylinders formed from Plate} \quad (50 \cdot t_{\text{nom}} / R_f) \cdot (1 - R_f / R_o) \quad 6.022 \%$$

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

Note:
 Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

Material treatment options: +N+IT

EXTERNAL PRESSURE RESULTS for: Shell 2

Cylindrical Shell From 60 to 70 Ext. Chart: CS-2 at 85 °C

Maximum External Pressure per ASME Division 2 Section 4.4:

Determine Parameter [Mx]:

$$= L / \sqrt{(R_o \cdot t)}$$

$$= 5498.333 / \sqrt{(1382.0 \cdot 155.5)}$$

$$= 11.861$$

Compute the Factor [Ch]:

Since $M_x > 1.5$ and $M_x < 13$, $Ch = 0.92 / (M_x - 0.579)$

$$= 0.92 / (11.861 - 0.579)$$

$$= 0.0815$$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$$= 1.6 \cdot Ch \cdot E_y(t / D_o)$$

$$= 1.6 \cdot 0.082 \cdot 0.19766E+09(155.5 / 2764.0)$$

$$= 1451.026 \text{ N./mm}^2$$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$$= F_{he} / F_y$$

$$= 1451.0 / 242.0$$

$$= 5.996$$

Predicted Buckling Stress [Fic]:

Engineering Yield to Tensile Stress ratio [R]:

$$= \sigma_{ys} / \sigma_{uts}$$

$$= 242.0 / 482.6$$

$$= 0.501$$

Ratio [AE]:

$$= F_{he} / E$$

$$= 1451.0 / 0.19766E+09$$

$$= 0.007340$$

Factor [m1]:

$$= (\ln(r) + (\epsilon_{psp} - \epsilon_{sys})) / \ln(\ln(1 + \epsilon_{psp}) / \ln(1 + \epsilon_{sys}))$$

$$= (\ln(0.501) + (0.00002 - 0.002)) / \ln(\ln(1 + 0.00002) / \ln(1 + 0.002))$$

$$= 0.150$$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$$= \sigma_{ys}(1 + \epsilon_{sys}) / \ln(1 + \epsilon_{sys}) m_1$$

$$= 242.0(1 + 0.002) / \ln(1 + 0.002) \cdot 0.150371$$

$$= 617.440 \text{ N./mm}^2$$

Curve Fitting Exponent for, Ferritic Steel [m2]:

$$= m_2 f(m_2 f_2 - R)$$

$$= 0.6(1.0 - 0.501)$$

$$= 0.299$$

Curve Fitting Constant [A2]:
 $= (\sigma_{uts} \cdot \exp(m_2)) / (m_2 m_2)$
 $= (482.6 \cdot \exp(0.299)) / (0.299 \cdot 0.299)$
 $= 933.996 \text{ N./mm}^2$

Material Parameter for Stress-Strain Curve Model [K]:
 $= 1.5 \cdot R1.5 - 0.5 \cdot R2.5 - R3.5$
 $= 1.5 \cdot 0.5011.5 - 0.5 \cdot 0.5012.5 - 0.5013.5$
 $= 0.354$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	241	-2.016	0.267E-05	0.242E-05	0.751E-05	-0.638E-05	0.149E+08	0.889E-02
2	120	-4.846	0.531E-07	0.531E-07	0.148E-05	-0.148E-05	0.163E+09	0.660E-02
3	180	-3.431	0.525E-06	0.522E-06	0.383E-05	-0.380E-05	0.632E+08	0.448E-02
4	211	-2.723	0.125E-05	0.123E-05	0.550E-05	-0.531E-05	0.315E+08	0.639E-03
5	226	-2.370	0.185E-05	0.177E-05	0.646E-05	-0.599E-05	0.218E+08	0.305E-02
6	218	-2.547	0.153E-05	0.148E-05	0.597E-05	-0.567E-05	0.262E+08	0.995E-03
7	214	-2.635	0.139E-05	0.135E-05	0.573E-05	-0.549E-05	0.288E+08	0.132E-03
8	213	-2.679	0.132E-05	0.129E-05	0.561E-05	-0.540E-05	0.301E+08	0.265E-03
9	213	-2.657	0.135E-05	0.132E-05	0.567E-05	-0.545E-05	0.294E+08	0.692E-04
10	214	-2.646	0.137E-05	0.133E-05	0.570E-05	-0.547E-05	0.291E+08	0.306E-04
11	214	-2.652	0.136E-05	0.133E-05	0.568E-05	-0.546E-05	0.293E+08	0.195E-04
12	214	-2.649	0.136E-05	0.133E-05	0.569E-05	-0.546E-05	0.292E+08	0.552E-05
13	214	-2.650	0.136E-05	0.133E-05	0.569E-05	-0.546E-05	0.292E+08	0.699E-05
14	214	-2.650	0.136E-05	0.133E-05	0.569E-05	-0.546E-05	0.292E+08	0.734E-06
15	214	-2.649	0.136E-05	0.133E-05	0.569E-05	-0.546E-05	0.292E+08	0.239E-05
16	214	-2.649	0.136E-05	0.133E-05	0.569E-05	-0.546E-05	0.292E+08	0.830E-06
17	214	-2.649	0.136E-05	0.133E-05	0.569E-05	-0.546E-05	0.292E+08	0.480E-07

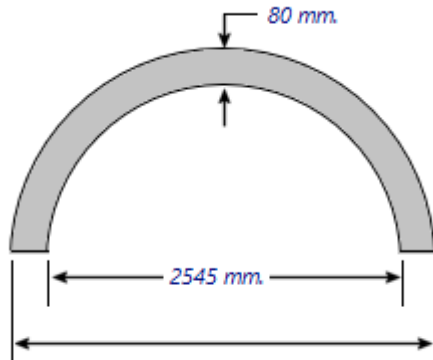
Factor of Safety [FS]:
 Since $Fic > 0.55 F_y$ and $Fic < F_y$
 $= 2.407 - 0.741 \cdot Fic / F_y$
 $= 2.407 - 0.741 \cdot 214.32/242.0$
 $= 1.751$

Allowable Circumferential Compressive Stress [Fha]:
 $= Fic/FS$
 $= 214.315/1.751$
 $= 122.412 \text{ N./mm}^2$

Allowable External Pressure at the given Thickness [Pa]:
 $= 2 \cdot Fha(t / D_o)$
 $= 2 \cdot 1224.048(155.5/2764.0)$
 $= 137.728 \text{ bars}$

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: Top D End

Spherical Head Design Information:					
Design Pressure	P	191.000 bars	Int. Design Temperature		85 °C
Material		SA-516 70	Ext. Design Temperature		85 °C
Length		0.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.5000 mm.	Allowable Stress (ope)	S	161.1 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	175.0 N./mm ²
Inside Diameter	Di	2545.000 mm.	Actual Stress		150.4 N./mm ²
Spec. Min. Thk t _e , or s _a		80.0000 mm.	Spec. Nominal Thk.		86.000 mm.
Surface Area		115957.6 cm ²	SG of Contents		0.000
Element Volume		4316282.0 cm ³	Weight of Contents		0.0 Kgf
Empty Weight		7250.1 Kgf	Operating Weight		7250.1 Kgf
			Comp. All. Ope. Stress		122.7 N./mm ²
Radiography Information:					
...			...		
Circ. Joint Efficiency		100 %	Long. Joint Eff.	E or v	100 %

Tolerance for Formed Heads per 4.3.2.2:

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 31.813 mm.
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 15.906 mm.

Material treatment options: +N+IT

Spherical Shell Calculation - Section 4.3.5.1

Computed Minimum Required Thickness [t]:
 $= D/2 \cdot \exp(0.5P/(S \cdot E)) - 1 + c_i + c_o$
 $= 2548.0/2 \cdot (\exp(0.5 \cdot 191.0/(161.141 \cdot 1.0)) - 1) + 1.5 + 0.0$
 $= 77.7904 + 1.5 + 0.0 = 79.2904 \text{ mm.}$

Computed Maximum Allowable Working Pressure [MAWP]:
 $= 2 \cdot S \cdot E \cdot \ln((2 \cdot t + D)/D) - P_{liq}$
 $= 2 \cdot 161.1 \cdot 1.0 \cdot \ln((2 \cdot 78.5 + 2548.0)/2548.0) - 0.0$
 $= 192.691 - 0.0 = 192.6913 \text{ bars}$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:
 $= 2 \cdot S \cdot E \cdot \ln((2 \cdot t + D)/D)$
 $= 2 \cdot 174.997 \cdot 1.0 \cdot \ln((2 \cdot 80.0 + 2545.0)/2545.0)$
 $= 213.3833 \text{ bars}$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D^2 / (E (D_o^2 - D^2))$$

$$= 191.0 \cdot 2548.0^2 / (1.0 (2705.0^2 - 2548.0^2))$$

$$= 150.366 \text{ N./mm}^2$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= 0.5 \cdot P / (E \cdot \ln((2 \cdot t + D) / D))$$

$$= 0.5 \cdot 191.0 / (1.0 \cdot \ln((2 \cdot 78.5 + 2548.0) / 2548.0))$$

$$= 159.727 \text{ N./mm}^2$$

Percent Fiber Elongation per Table 6.1

$$\text{For Heads formed from Segments } (75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o) \quad 4.903 \%$$

$$\text{For one Piece doubly Curved Heads } 100 \cdot \ln(D_b / (D_f - 2t)) \quad 41.197 \%$$

Where D_b is the blank diameter for one piece doubly curved surfaces [D_b]:

$$= \text{square root}(4 \cdot \text{Head Surface Area} / 3.14159)$$

$$= \text{square root}(4 \cdot 115957.6 / 3.14159)$$

$$= 3842.418 \text{ mm.}$$

MDMT Calculations in the Spherical Portion:

Note: This Element/Detail was specified as being Impact Tested.

Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

Tolerance for Formed Heads per 4.3.2.2:

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 31.813 mm.
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 15.906 mm.

Material treatment options: +N+IT

EXTERNAL PRESSURE RESULTS for: Top D End

Spherical Head From 70 to 80 Ext. Chart: CS-2 at 85 °C

Maximum External Pressure per ASME Division 2 Section 4.4:

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$$= 0.075 \cdot E_y (t / R_o)$$

$$= 0.075 \cdot 0.19904E+09 (78.5 / 1352.5)$$

$$= 866.491 \text{ N./mm}^2$$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$$= F_{he} / S_y$$

$$= 866.5 / 242.0$$

$$= 3.581$$

Engineering Yield to Tensile Stress ratio [R]:

$$= \sigma_{ys} / \sigma_{uts}$$

$$= 242.0 / 482.6$$

$$= 0.501$$

Ratio [AE]:

$$= F_{he} / E$$

$$= 866.5 / 0.19904E+09$$

$$= 0.004353$$

Factor [m1]:

$$= (\ln(r) + (\epsilon_{psp} - \epsilon_{sys})) / \ln(\ln(1 + \epsilon_{psp}) / \ln(1 + \epsilon_{sys}))$$

$$= (\ln(0.501) + (0.00002 - 0.002)) / \ln(\ln(1 + 0.00002) / \ln(1 + 0.002))$$

$$= 0.150$$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$$= \sigma_{ys} (1 + \epsilon_{sys}) / \ln(1 + \epsilon_{sys}) m_1$$

$$= 242.0 (1 + 0.002) / \ln(1 + 0.002) 0.150373$$

= 617.442 N./mm²

Curve Fitting Exponent for, Ferritic Steel [m2]:

= $m2f(m2f2 - R)$
 = $0.6(1.0 - 0.501)$
 = 0.299

Curve Fitting Constant [A2]:

= $(\sigma_{uts} \cdot \exp(m2)) / (m2 m2)$
 = $(482.6 \cdot \exp(0.299)) / (0.299 0.299)$
 = 934.000 N./mm²

Material Parameter for Stress-Strain Curve Model [K]:

= $1.5 \cdot R1.5 - 0.5 \cdot R2.5 - R3.5$
 = $1.5 \cdot 0.5011.5 - 0.5 \cdot 0.5012.5 - 0.5013.5$
 = 0.354

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	241	-2.016	0.267E-05	0.242E-05	0.751E-05	-0.638E-05	0.149E+08	0.119E-01
2	120	-4.846	0.531E-07	0.531E-07	0.148E-05	-0.148E-05	0.164E+09	0.362E-02
3	180	-3.431	0.525E-06	0.522E-06	0.383E-05	-0.380E-05	0.634E+08	0.150E-02
4	211	-2.723	0.125E-05	0.123E-05	0.550E-05	-0.531E-05	0.315E+08	0.234E-02
5	196	-3.077	0.825E-06	0.817E-06	0.462E-05	-0.455E-05	0.451E+08	0.500E-05
6	203	-2.900	0.102E-05	0.101E-05	0.505E-05	-0.493E-05	0.378E+08	0.104E-02
7	199	-2.989	0.919E-06	0.908E-06	0.483E-05	-0.474E-05	0.413E+08	0.486E-03
8	197	-3.033	0.871E-06	0.861E-06	0.472E-05	-0.464E-05	0.432E+08	0.234E-03
9	197	-3.055	0.848E-06	0.839E-06	0.467E-05	-0.459E-05	0.441E+08	0.113E-03
10	196	-3.066	0.836E-06	0.828E-06	0.465E-05	-0.457E-05	0.446E+08	0.534E-04
11	196	-3.072	0.831E-06	0.822E-06	0.463E-05	-0.456E-05	0.448E+08	0.241E-04
12	196	-3.074	0.828E-06	0.820E-06	0.463E-05	-0.455E-05	0.450E+08	0.953E-05
13	196	-3.076	0.827E-06	0.818E-06	0.462E-05	-0.455E-05	0.450E+08	0.226E-05
14	196	-3.076	0.826E-06	0.817E-06	0.462E-05	-0.455E-05	0.451E+08	0.137E-05
15	196	-3.076	0.826E-06	0.818E-06	0.462E-05	-0.455E-05	0.450E+08	0.441E-06
16	196	-3.076	0.826E-06	0.818E-06	0.462E-05	-0.455E-05	0.451E+08	0.467E-06
17	196	-3.076	0.826E-06	0.818E-06	0.462E-05	-0.455E-05	0.451E+08	0.132E-07

Factor of Safety [FS]:

Since $Fic > 0.55 Fy$ and $Fic < Fy$
 = $2.407 - 0.741 \cdot Fic / Fy$
 = $2.407 - 0.741 \cdot 196.12/242.0$
 = 1.806

Allowable Circumferential Compressive Stress [Fha]:

= Fic/FS
 = $196.117/1.806$
 = 108.563 N./mm²

Allowable External Pressure at the given Thickness [Pa]:

= $2 \cdot Fha[t/Ro]$
 = $2 \cdot 108.563[78.5/1352.5]$
 = 126.014 bars



Hydrostatic Test Pressure Results:

Test Pressure at high point in vessel per Equation 8.1
 $= \gamma_{St/s} \cdot MAWP(St/S)$
 $= 1.25 \cdot 192.2043(1.081)$
 $= 259.603 \text{ bars}$

Horizontal Test performed per: $\gamma_{St/s} \cdot MAWP(St/S)$

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & bars):

From To	Stress	Allowable	Ratio	Pressure
Bottom D end	213.1	247.0	0.863	259.85
Shell 1	215.5	247.0	0.872	259.84
Shell 2	215.5	247.0	0.872	259.84
Top D End	213.1	247.0	0.863	259.85

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
D1_3"	Nozzle	138.00	127.70	1.081
L2_2"	Nozzle	138.00	127.70	1.081
L4_2"	Nozzle	138.00	127.71	1.081
B2_6"	Nozzle	138.00	127.70	1.081
L6_2"	Nozzle	138.00	127.71	1.081
M1_24"	Nozzle	138.00	127.70	1.081
A1_20"	Nozzle	138.00	127.70	1.081
L1_2"	Nozzle	138.00	127.70	1.081
L3_2"	Nozzle	138.00	127.70	1.081
L5_2"	Nozzle	138.00	127.70	1.081
B1_24"	Nozzle	138.00	127.70	1.081

Minimum 1.081

Stress ratios for Pressurized Vessel Elements (N./mm²):

Description	Ambient	Operating	Ratio
Bottom D end	175.00	161.14	1.086
Shell 1	175.00	161.14	1.086
Shell 2	175.00	161.10	1.086
Top D End	175.00	161.14	1.086

Minimum 1.086

Hoop Stress in Nozzle Wall during Pressure Test (N./mm²):

Description	Ambient	Operating	Ratio
D1_3"	111.50	194.70	0.573
L2_2"	94.91	194.70	0.487
L4_2"	94.91	194.75	0.487
B2_6"	136.34	194.70	0.700
L6_2"	94.91	194.75	0.487
M1_24"	159.32	194.70	0.818
A1_20"	156.12	194.70	0.802
L1_2"	94.91	194.70	0.487
L3_2"	94.91	194.70	0.487
L5_2"	94.91	194.70	0.487
B1_24"	159.32	194.70	0.818



Elements Suitable for Test Pressure.

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External Pressure Calculation Results Summary:

ASME Code, Section VIII Division 2, 2025

External Pressure Calculations:

From	To	Section Length mm.	Outside Diameter mm.	Corroded Thickness mm.	Factor A	Factor B N./mm ²
10	20	3000	...	...	No Calc	No Calc
20	30	3000	...	...	No Calc	No Calc
30	40	No Calc	2705	78.5	No Calc	No Calc
40	60	5498.33	2764	155.5	No Calc	No Calc
60	70	5498.33	2764	155.5	No Calc	No Calc
70	80	No Calc	2705	78.5	No Calc	No Calc

External Pressure Calculations:

From	To	External Actual T. mm.	External Required T. mm.	External Design Pressure bars	External M.A.W.P. bars
10	20	...	No Calc	...	No Calc
20	30	...	No Calc	...	No Calc
30	40	80	5.0606	1.03419	126.014
40	60	157	12.6065	1.03419	137.728
60	70	157	12.6065	1.03419	137.728
70	80	80	5.0606	1.03419	126.014

Minimum

126

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm.	Allowable Length Bet. Stiffeners mm.	Ring Inertia Required cm ⁴	Ring Inertia Available cm ⁴
10	20	3000	No Calc	No Calc	No Calc
20	30	3000	No Calc	No Calc	No Calc
30	40	No Calc	No Calc	No Calc	No Calc
40	60	5498.33	25399974	No Calc	No Calc
60	70	5498.33	25399974	No Calc	No Calc
70	80	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

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Element and Detail Weights:

From	To	Element Metal Wgt. kg.	Element ID Volume cm ³	Corroded Metal Wgt. kg.	Corroded ID Volume cm ³	Extra due Misc % kg.
10	20	3237.42	...	3046.86	...	226.619
20	30	1270.52	...	1175.25	...	88.9367
30	40	7250.11	4316282	7131.69	4331564	507.508
40	60	16443.9	7780076	16296.2	7799140	1151.07
60	70	29896.4	14145593	29627.8	14180256	2092.75
70	80	7250.11	4316282	7131.69	4331564	507.508
Total		65348	30558232.00	64409	30642524.00	4574

Weight of Details:

From	Type	Weight of Detail kg.	X Offset, Dtl. Cent. mm.	Y Offset, Dtl. Cent. mm.	Z Offset Dtl. Cent. mm.	Description
30	Liqd	2805.97	...	-631.5	...	Liquid 30
30	Insl	19.5869	...	-631.5	...	Ins: 50
30	Nozl	92.3257	...	-1332.5	...	D1_3"
30	Nozl	61.8043	...	-1678.1	-1200	L2_2"
30	Nozl	44.2445	-1200	-483.387	...	L4_2"
30	Nozl	271.12	...	-1172.66	700	B2_6"
30	Nozl	44.2445	...	-1578.1	1200	L6_2"
30	Wght	2300	...	...	...	Btm Dish SF
30	Wght	5	...	...	700	vortex Breaker
40	Plat	3802.7	0.19047E-04	...	...	PLAT:[1 OF 1]
40	Liqd	5058.17	...	825	...	Liquid 40
40	Insl	23.2831	...	825	...	Ins: 50
60	Liqd	1379.5	...	285	...	Liquid: 60
60	Insl	42.3329	...	1500	...	Ins: 50
60	Nozl	5355.84	1375	2100	...	M1_24"
60	Nozl	1611.96	-1350.01	1225	...	A1_20"
60	Nozl	83.3095	1325	550	...	L1_2"
60	Nozl	83.3095	...	550	-1325	L3_2"
60	Nozl	83.3095	...	550	1325	L5_2"
60	Wght	230	...	2775	...	Demister WT
60	Wght	260	...	1225	...	FEED PIPE WT
60	Wght	2800	...	700	...	Lifting WT
60	Wght	900	...	700	...	CLIP WT
60	Wght	50	1800	2100	...	Manway Davit
60	Wght	30	...	2775	...	demister suppor
70	Plat	4688.43	...	1400	...	PLAT 2
70	Insl	19.5869	...	632	...	Ins: 50
70	Nozl	1333.95	...	1431.26	...	B1_24"
70	Wght	2300	...	...	...	Top dish SF

Total Weight of Each Detail Type:

Platforms	8491.1
Liquid	9243.6
Insulation	104.8
Nozzles	9065.4
Weights	8875.0
Sum of the Detail Weights	35780.0 kg.

Weight Summation Results: (kg.)

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	69922.9	69922.9	69922.9	69922.9	69922.9	69922.9
Nozzles	9065.4	9065.4	9065.4	9065.4	9065.4	9065.4
Wld Weights	8645.0	8645.0	8645.0	8645.0	8645.0	8645.0



Platforms	...	...	8491.1	8491.1	8491.1	8491.1
Insulation	...	...	104.8	104.8	104.8	104.8
Empty Weights	...	...	...	230.0	230.0	...
Test Weights	...	230.0	...	...	...	...
Ope Weights	...	...	...	...	...	230.0
Ope. Liquid	...	...	...	...	...	9243.6
Test Liquid	...	30539.6	...	...	...	...
Totals	87633.3	118402.9	96229.2	96459.2	96459.2	105702.9

Field Installation Options:

Miscellaneous Weight Percent: 7.0 %

Note that the above value for the miscellaneous weight percent has been applied to the shells/heads/flange/tubesheets/tubes etc. in the weight calculations for metallic components.

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	87633.3 kg.
Shop Test Wt.	- Fabricated weight + water (full)	118402.9 kg.
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	96229.2 kg.
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	96459.2 kg.
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	96459.2 kg.
Operating Wt.	- Empty weight + operating liq. uncorroded	105702.9 kg.

Weights used in Vortex Shedding Calculations:

Mass of the Upper 1/3 of the Vertical Vessel (operating)	55048.4 kg.
Mass of the Upper 1/3 of the Vertical Vessel (filled w/water)	44403.7 kg.
Mass of the Upper 1/3 of the Vertical Vessel (empty)	55235.5 kg.

Surface Areas of Elements:

From	To	Outside Surface Area cm ²	Inside Surface Area cm ²
10	20	165185	162672
20	30	82592.5	81335.8
30	40	115958	101741
40	60	143275	126999
60	70	260501	230907
70	80	115958	101741

Total	883469.0	805395.2 cm ²
Total	951.0	866.9 ft ²

Element and Detail Weights:

From	To	Total Ele. Empty Wgt. kg.	Total. Ele. Oper. Wgt. kg.	Total. Ele. Hydro. Wgt. kg.	Total Dtl. Offset Mom. Kg-m.	Oper. Wgt. No Liquid kg.
10	20	3464.04	3464.04	3464.04	...	3464.04
20	30	1359.46	1359.46	1359.46	...	1359.46
30	40	10595.9	13401.9	14909.6	53.0945	10595.9
40	60	21420.9	26479.1	29196.3	0.72431E-04	21420.9
60	70	43519.2	44898.7	57656.2	5388.62	43519.2
70	80	16099.6	16099.6	20413.2	...	16099.6

Cumulative Vessel Weight

From	To	Cumulative Ope Wgt. No Liquid kg.	Cumulative Oper. Wgt. kg.	Cumulative Hydro. Wgt. kg.
10	20	96459.2	105703	126999



20	30	92995.2	102239	123535
30	40	91635.7	100879	122175
40	60	81039.8	87477.4	107266
60	70	59618.8	60998.3	78069.4
70	80	16099.6	16099.6	20413.2

The cumulative operating weights no liquid in the column above are the cumulative operating weights minus the operating liquid weight minus any weights absent in the empty condition.

Cumulative Vessel Moment

From	To	Cumulative Empty Mom. Kg-m.	Cumulative Oper. Mom. Kg-m.	Cumulative Hydro. Mom. Kg-m.
10	20	5441.72	5441.72	5441.72
20	30	5441.72	5441.72	5441.72
30	40	5441.72	5441.72	5441.72
40	60	5388.62	5388.62	5388.62
60	70	5388.62	5388.62	5388.62
70	80	...	...	...

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Nozzle Flange MAWP Results: (bars & °C)

Nozzle Description	Flange Rating		Des. Temp	Class	Grade/Group	Equiv. Press	Max Pressure		
	Ope.	Ambient					UG-44	50%	DNV
L2_2"	198.48	212.90	85	1500	GR 1.4	...	...	...	...
L4_2"	198.48	212.90	85	1500	GR 1.4	...	...	...	...
L6_2"	198.48	212.90	85	1500	GR 1.4	...	...	...	...
M1_24"	198.48	212.90	85	1500	GR 1.4	...	...	...	...
L1_2"	238.12	255.10	85	1500	GR 1.1	...	...	...	...
L3_2"	198.48	212.90	85	1500	GR 1.4	...	...	...	...
L5_2"	198.48	212.90	85	1500	GR 1.4	...	...	...	...
Min Rating	198.483	212.900, [for Core Elements]					...	...	...

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

Large Diameter Flange Pressure Ratings are per ASME B16.47 2025 Edition

Warning:

There are nozzles in this model, but no flange MAWPs were de-rated. Be sure this is what you intended. There is a check box in the nozzle dialog that instructs PV Elite to perform the de-rating for each nozzle flange. See ASME VIII-2, 4.16.12 for more information.

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The Natural Frequencies for the vessel have been computed iteratively by solving a system of matrices. These matrices describe the mass and the stiffness of the vessel. This is the generalized eigenvalue/eigenvector problem and is referenced in some mathematical texts.

The Natural Frequency for the Vessel (Empty.) is 9.73135 Hz.

The Natural Frequency for the Vessel (Ope...) is 9.51231 Hz.

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Wind Load Calculation Results:

Input Values:

Wind Design Code	ASCE/SEI 7-22
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed	[V] 180 Km/hr
Surface Roughness Category	D: Flat, unobstructed
Type of Surface	Moderately Smooth
Base Elevation	300 mm.
Percent Wind for Hydrotest	33.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 mm.
Distance Upwind of Crest Lh	0 mm.
Distance from Crest to the Vessel x	0 mm.
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm.

Wind Analysis Results

The importance factor is not used in the calculation of wind pressure vs. elevation for the selected wind code.

Static Gust-Effect Factor, Operating Case [G]:
 $= \min(0.85, 0.925((1 + 1.7 \cdot g_Q \cdot I_{zbar} \cdot Q) / (1 + 1.7 \cdot g_V \cdot I_{zbar})))$
 $= \min(0.85, 0.925((1 + 1.7 \cdot 3.4 \cdot 0.171 \cdot 0.952) / (1 + 1.7 \cdot 3.4 \cdot 0.171)))$
 $= \min(0.85, 0.903)$
 $= 0.850$

Natural Frequency of Vessel (Operating)	9.512 Hz
Natural Frequency of Vessel (Empty)	9.731 Hz
Natural Frequency of Vessel (Test)	8.880 Hz

User Entered Importance Factor is	1.000
Force Coefficient	[Cf] 0.540
Structure Height to Diameter ratio	3.385
Height to top of Structure	9042.500 mm.

This is classified as a rigid structure. Static analysis performed.

Sample Calculation for the First Element:

The ASCE code performs calculations in Imperial Units. The wind pressure is therefore computed in these units.

Value of $[\alpha]$ and $[Z_g]$:
 Exposure Category: D from Table 26.11-1
 $\alpha = 11.5$; $Z_g = 589788$ mm.

Effective Height $[z]$:
 $= \text{Centroid Height} + \text{Vessel Base Elevation}$
 $= 1000.0 + 300.0 = 1300.0$ mm.
 $= 4.265$ ft. Imperial Units

Velocity Pressure coefficient evaluated at height z $[K_z]$:
 Because z (4.265 ft.) < 15 ft.
 $= 2.41(15 / Z_g)^{2/\alpha}$
 $= 2.41(15 / 1935.0)^{2/11.5}$
 $= 1.035$

Type of Hill: No Hill

Wind Directionality Factor $[K_d]$:
 $= 1$, per Table 26.6-1

As there is No Hill Present: $[K_{zt}]$:

$K1 = 0, K2 = 0, K3 = 0$

Topographical Factor [Kzt]:

$$= (1 + K1 \cdot K2 \cdot K3)^2$$

$$= (1 + 0.0 \cdot 0.0 \cdot 0.0)^2$$

$$= 1.0$$

Velocity Pressure evaluated at height z, Imperial Units [qz]:

$$= \max(16, 0.00256 \cdot Kz \cdot Kzt \cdot Kd \cdot Ke \cdot V(\text{mph})^2)$$

$$= \max(16, 0.00256 \cdot 1.035 \cdot 1.0 \cdot 1.0 \cdot 1.0 \cdot 111.85^2)$$

$$= 33.1 \text{ psf } [161.849] \text{ Kgs/m}^2$$

Force on the first element [F]:

$$= qz \cdot G \cdot Cf \cdot \text{WindArea}$$

$$= 33.149 \cdot 0.85 \cdot 0.54 \cdot 76.971$$

$$= 1170.6 \text{ lbs. } [531.0] \text{ Kg}$$

Element	Hgt (z) mm.	K1	K2	K3	Kz	Kzt	qz Kgs/m ²
Skirt 1	1300.0	0.000	0.000	0.000	1.035	1.000	161.849
Top Y Ring	2800.0	0.000	0.000	0.000	1.035	1.000	161.849
Bottom D end	3300.0	0.000	0.000	0.000	1.035	1.000	161.849
Shell 1	4125.0	0.000	0.000	0.000	1.035	1.000	161.849
Shell 2	6450.0	0.000	0.000	0.000	1.099	1.000	171.832
Top D End	8541.0	0.000	0.000	0.000	1.154	1.000	180.431

Wind Vibration Calculations

This evaluation is based on work by Kanti Mahajan and Ed Zorilla

Nomenclature

Cf - Correction factor for natural frequency
 D - Average outer insulated diameter of vessel, mm.
 Df - Damping Factor < 0.75 Unstable, > 0.95 Stable
 Dr - Average outer insulated of top half of vessel, mm.
 f - Natural frequency of vibration (Hertz)
 f1 - Natural frequency of bare vessel based on a unit value of $(D/L^2)(10^4)$
 L - Total height of structure mm.
 Lc - Total length of conical section(s) of vessel, mm.
 tb - Uncorroded plate thickness at bottom of vessel, mm.
 V30 - Design Wind Speed provided by user, Km/hr
 Vc - Critical wind velocity, Km/hr
 Vw - Maximum wind speed at top of structure, Km/hr
 W - Total corroded weight of structure, Kg
 Ws - Cor. vessel weight excl. weight of parts which do not effect stiff. Kg
 Z - Maximum amplitude of vibration at top of vessel, mm.
 D1 - Logarithmic decrement (taken as 0.03 for Welded Structures)
 Vp - Vib. Chance, $\leq 0.320\text{E-}06$ (High); $0.320\text{E-}06 < 0.400\text{E-}06$ (Probable)
 P30 - wind pressure 30 feet above the base

Check other Conditions and Basic Assumptions:

:: Total Cone Length / Total Length < 0.5
 :: $0.0/7650.0 = 0.0$

:: $[D / L^2] \cdot 10^4 < 8$ (English Units, ft.)
 :: $[9.05/25.1^2] \cdot 10^4 = 143.731$ [Geometry Violation]

Compute the vibration possibility. If $Vp > 0.400\text{E-}06$ no chance. [Vp]:

$$= W / (L \cdot Dr^2)$$

$$= 104698 / (7650.0 \cdot 2817.333^2)$$

$$= 0.17243\text{E-}05$$

Since Vp is $> 0.400\text{E-}06$ no further vibration analysis is required !

Platform Load Calculations



ID	Wind Area cm ²	Elevation mm.	Pressure Kgs/m ²	Force Kgf	Cf
PLAT:[1 OF 1]	60000.00	3300.00	161.85	971.10	2.00
PLAT 2	167806.39	9349.97	182.86	3068.45	2.00

Wind Loads on Masses/Equipment/Piping

ID	Wind Area cm ²	Elevation mm.	Pressure Kgs/m ²	Force Kgf
Btm Dish SF	0.00	3299.97	161.85	0.00
vortex Breake	0.00	3299.97	161.85	0.00
Demister WT	0.00	7725.00	177.08	0.00
FEED PIPE WT	0.00	6175.00	170.65	0.00
Lifting WT	0.00	5650.00	168.40	0.00
CLIP WT	0.00	5650.00	168.40	0.00
Manway Davit	0.00	7050.00	174.30	0.00
demister supp	0.00	7725.00	177.08	0.00
Top dish SF	0.00	7949.97	178.00	0.00

The Natural Frequency for the Vessel (Ope...) is 9.51231 Hz.

Wind Load Calculation:

From	To	Wind Height mm.	Wind Diameter mm.	Wind Area cm ²	Wind Pressure Kgs/m ²	Element Wind Load Kgf
10	20	1300	3575.44	71508.8	161.849	584.08
20	30	2800	3575.44	35754.4	161.849	292.04
30	40	3299.98	3787.6	1.15446	161.849	0.0094296
40	60	4125	3867.84	63819.4	161.849	1375.84
60	70	6450	3867.84	116035	171.832	1006.22
70	80	8540.98	3787.6	41424.9	180.431	4072.2

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

Be sure the wind speed is in accordance with the specified wind design code.

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Earthquake Load Calculation Results:

Earthquake Load Calculation:

Input Values:

Seismic Design Code		ASCE/SEI 7-22	
Importance Factor		1.500	
Force Modification Factor R		1.500	
Site Class		D	
Component Elevation Ratio	z/h	0.000	
Amplification Factor	Ap	0.000	
Force Factor		0.000	
Consider Vertical Acceleration		No	
Minimum Acceleration Multiplier		0.000	
User Value of Sds (used if > 0)		0.000	
User Value of Sd1 (used if > 0)		0.000	
Component resonance duct. fact.	[CAR]	0.000	
Component strength factor	[Rpo]	0.000	
Overstrength factor	[Omega0]	0.000	
Response Acceleration Factor	[SMS]	0.326	
Response Acceleration Factor	[SM1]	0.139	
Building Importance Factor	[Ie]	1.000	
Site Latitude		0.000	deg
Site Longitude		0.000	deg

Seismic Analysis Results:

$$SDS = 2/3 * Sms = 2/3 * 0.326 = 0.217$$

$$SD1 = 2/3 * Sm1 = 2/3 * 0.139 = 0.093$$

Check Approximate Fundamental Period from 12.8-7 [Ta]:
 = $C_t \cdot h_{nx}$ where $C_t = 0.020$, $x = 0.75$ and h_n = Structural Height (ft.)
 = $0.020(29.27330.75)$
 = 0.252 seconds

The Coefficient C_u from Table 12.8-1 is : 1.700

Fundamental Period (1/Frequency) [T]:
 = (1/Natural Frequency) = (1/9.512)
 = 0.105

Check the Value of T which is the smaller of $C_u \cdot T_a$ and T:
 = Minimum Value of ($1.7 \cdot 0.252$, 0.105) per 12.8.2
 = 0.105

Seismic Response Coefficient per equation 12.8-2 [Cs]:
 = $Sds / (R / I)$
 = $0.217 / (1.5 / 1.5)$
 = 0.217

Maximum value of C_s per equation 12.8-3 [Cs]:
 = $Sd1 / (T(R / I_e))$
 = $0.093 / (0.105(1.5 / 1.5))$
 = 0.881

Minimum value of C_s per equation 15.4-1 [Cs]:
 = $\max((0.044 \cdot SDS \cdot I_e), 0.030)$
 = $\max((0.044 \cdot 0.2173 \cdot 1.5), 0.030)$
 = 0.030

Total Base Shear [V]:
 = $C_s \cdot W$ (Equation 12.8-1):
 = $0.2173 \cdot 105702.9$
 = 22972.756 Kgf

Distribute the Base shear force to each element according to the equations
 $F_x = C_{vx} \cdot V$ (eqn. 12.8-11) and the vertical distribution factor
 $C_{vx} = W_x \cdot h_x^k / (\text{Sum of } W_i \cdot h_i^k)$ and k is an exponent which is related

to the period of Vibration.

In this case, the value of k was 1.0

The Natural Frequency for the Vessel (Ope...) is 9.51231 Hz.

Earthquake Load Calculation:

From	To	Earthquake Height mm.	Earthquake Weight Kgf	Element Ope Load Kgf	Element Emp Load Kgf	Element Vertical Load Kgf
10	20	1000	3464.04	145.312	142.004	...
20	30	2500	1359.46	142.569	139.323	...
30	40	2999.98	13401.9	1686.57	1303.1	...
40	60	3825	26479.1	4248.67	3358.83	...
60	70	6150	44898.7	11583.2	10971.7	...
70	80	7649.98	16099.6	5166.46	5048.85	...

The following table is for the Operating Case.

Wind/Earthquake Shear, Bending:

From	To	Distance to Support mm.	Cumulative Wind Shear Kgf	Earthquake Shear Kgf	Wind Bending Kg-m.	Earthquake Bending Kg-m.
10	20	1000	7330.4	22972.8	46324.9	135628
20	30	2500	6746.32	22827.4	32247.9	89827.4
30	40	2999.98	6454.28	22684.9	25647.7	67071.5
40	60	3825	6454.27	20998.3	25647.5	67070.8
60	70	6150	5078.43	16749.6	16132.9	35928.1
70	80	7649.98	4072.2	5166.46	2406.76	3053.49

Note:

The Wind Shears/Moments and the Earthquake Shears/Moments calculated and printed in the Wind/Earthquake Shear and Bending report have been factored by the input Scalar/Load reductions factors of;
 Wind: 1.100; Earthquake: 1.000.

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Wind Deflection Calculations:

The following table is for Wind Deflection; Operating Case.

Wind Load Deflection:

From	To	Cumulative Shear Kgf	Centroid Deflection mm.	Elem. End Deflection mm.	Elem. Ang. Rotation
10	20	7330.4	0.0080557	0.030483	0.28761E-04
20	30	6746.32	0.046292	0.064753	0.39375E-04
30	40	6454.28	0.064754	0.064754	0.39375E-04
40	60	6454.27	0.09761	0.13111	0.40933E-04
60	70	5078.43	0.19321	0.25623	0.42167E-04
70	80	4072.2	0.25623	0.25623	0.42167E-04

Critical Wind Velocity for Tower Vibration:

From	To	1st Crit. Wind Speed Km/hr	2nd Crit. Wind Speed Km/hr
10	20	610.543	3815.89
20	30	610.543	3815.89
30	40	646.772	4042.32
40	60	660.473	4127.96
60	70	660.473	4127.96
70	80	646.772	4042.32

Allowable deflection at the Tower Top (Ope)(6.000"/100ft. Criteria)
 Allowable deflection : 38.250 Actual deflection : 0.256 mm.

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Longitudinal Stress Constants:

From	To	Metal Area New cm ²	Metal Area Corroded cm ²	Section Modulus New mm. ³	Section Modulus Corroded mm. ³
10	20	1639.28	1516.35	106115208	98212424
20	30	1639.28	1516.35	106115208	98212424
30	40	6597.34	6477.34	420536288	413344352
40	60	12858.5	12743	793317120	787037440
60	70	12858.5	12743	793317120	787037440
70	80	6597.34	6477.34	420536288	413344352

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Longitudinal Allowable Stresses:

From	To	Tensile N./mm ²	Hydrotest Tensile N./mm ²	Compressive N./mm ²	Hydrotest Compressive N./mm ²
10	20	107.199	237.496	-100.785	-102.609
20	30	88.6276	246.996	-100.785	-102.609
30	40	161.141	246.996	-122.731	-122.731
40	60	161.141	246.996	-122.731	-122.731
60	70	161.1	247	-122.731	-122.731
70	80	161.141	246.996	-122.731	-122.731

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Longitudinal Stress Report



Longitudinal Operating and Empty Stresses are computed in the corroded condition. Stresses due to loads in the hydrostatic test cases have been computed in the new and cold condition.

Longitudinal Pressure Stresses due to:

From	To	Longitudinal Stress Internal Pressure N./mm ²	Longitudinal Stress External Pressure N./mm ²	Longitudinal Stress Hydrotest Pressure N./mm ²
10	20	...	...	...
20	30	...	...	...
30	40	150.366	-0.9176	200.185
40	60	70.839	-0.48699	95.1845
60	70	70.839	-0.48699	95.1845
70	80	150.366	-0.9176	200.185

Longitudinal Stresses due to Weight Loads for these Conditions:

From	To	Wght. Str. Empty N./mm ²	Wght. Str. Operating N./mm ²	Wght. Str. Hydrotest N./mm ²	Wght. Str. Emp. Mom. N./mm ²	Wght. Str. Opr. Mom. N./mm ²
10	20	-6.23839	-6.83621	...	0.54336	0.54336
20	30	-6.01436	-6.61218	...	0.54336	0.54336
30	40	-1.38738	-1.38738	...	0.12911	0.12911
40	60	-0.62367	-0.62367	...	0.067143	0.067143
60	70	-0.45882	-0.45882	...	0.067143	0.067143
70	80	-0.24375	-0.24375	...	...	...

Longitudinal Stresses due to Weight Loads and Bending for these Conditions:

From	To	Wght. Str. Hyd. Mom. N./mm ²	Bend. Str. Oper. Wind N./mm ²	Bend. Str. Oper. Equ. N./mm ²	Bend. Str. Hyd. Wind N./mm ²	Bend. Str. Hyd. Equ. N./mm ²
10	20	...	4.6256	13.5427	...	...
20	30	...	3.22	8.96938	...	...
30	40	...	0.60849	1.59128	...	...
40	60	...	0.31957	0.83571	...	...
60	70	...	0.20102	0.44767	...	...
70	80	...	0.057101	0.072444	...	...

Longitudinal Stresses due to these Conditions:

From	To	Vortex Shedding Operating Case N./mm ²	Vortex Shedding Empty Case N./mm ²	Vortex Shedding Test Case N./mm ²	Earthquake Empty Case N./mm ²
10	20	...	...	...	12.6145
20	30	...	...	...	8.44209
30	40	...	...	...	1.51354
40	60	...	...	...	0.79489
60	70	...	...	...	0.43098
70	80	...	...	...	0.070795

Longitudinal Stresses due to Applied Axial Forces:

From	To	Longitudinal Stress Y Forces Wind N./mm ²	Longitudinal Stress Y Forces Seismic N./mm ²
------	----	--	---

10	20	...	...
20	30	...	...
30	40	...	...
40	60	...	...
60	70	...	...
70	80	...	...

Longitudinal Stresses due to User Forces and Moments:

From	To	Wind For/Mom Corroded N./mm ²	Earthquake For/Mom Corroded N./mm ²	Wind For/Mom No Corrosion N./mm ²	Earthquake For/Mom No Corrosion N./mm ²
10	20	...	...	...	...
20	30	...	...	...	...
30	40	...	...	...	...
40	60	...	...	...	...
60	70	...	...	...	...
70	80	...	...	...	...

Bending Stresses due to Blast Load:

From	To	Bending Stress N./mm ²
10	20	...
20	30	...
30	40	...
40	60	...
60	70	...
70	80	...

Stress Combination Load Cases for Vertical Vessels:

Load Case Definition Key

IP = Longitudinal Stress due to Internal Pressure
 EP = Longitudinal Stress due to External Pressure
 HP = Longitudinal Stress due to Hydrotest Pressure
 NP = No Pressure
 EW = Longitudinal Stress due to Weight (No Liquid)
 OW = Longitudinal Stress due to Weight (Operating)
 HW = Longitudinal Stress due to Weight (Hydrotest)
 WI = Bending Stress due to Wind Moment (Operating)
 EQ = Bending Stress due to Earthquake Moment (Operating)
 EE = Bending Stress due to Earthquake Moment (Empty)
 HI = Bending Stress due to Wind Moment (Hydrotest)
 HE = Bending Stress due to Earthquake Moment (Hydrotest)
 WE = Bending Stress due to Wind Moment (Empty) (no CA)
 WF = Bending Stress due to Wind Moment (Filled) (no CA)
 CW = Longitudinal Stress due to Weight (Empty) (no CA)
 VO = Bending Stress due to Vortex Shedding Loads (Ope)
 VE = Bending Stress due to Vortex Shedding Loads (Emp)
 VF = Bending Stress due to Vortex Shedding Loads (Test No CA.)
 FW = Axial Stress due to Vertical Forces for the Wind Case
 FS = Axial Stress due to Vertical Forces for the Seismic Case
 BW = Bending Stress due to Lat. Forces for the Wind Case, Corroded
 BS = Bending Stress due to Lat. Forces for the Seismic Case, Corroded
 BN = Bending Stress due to Lat. Forces for the Wind Case, UnCorroded
 BU = Bending Stress due to Lat. Forces for the Seismic Case, UnCorroded
 BL = Bending Stress due to Blast Loads, Corroded

General Notes:

Case types HI and HE are in the Un-Corroded condition.

Case types WE, WF, and CW are in the Un-Corroded condition.

A blank stress and stress ratio indicates that the corresponding stress comprising those components that did not contribute to that type of stress.

For Division 2, the overall stress intensity is computed in accordance with the Section VIII Division II Example problem manual PTB-3, example 4.3.6, stress due to combined loads.
 The stresses are combined per equation 4.3.44 to determine the overall stress intensity.
 The unity check is from Section 4.4.12 equations 4.4.85 and 4.4.90.

An asterisk (*) in the final column denotes overstress.

Note: Performing Combined Stress Analysis per VIII-2 paragraph 4.3.10.2.

Analysis of Load Case 1 : NP+EW+WI+FW+BW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	11.41	107.20	-11.41	109.90	0.1064	0.1038	0.0571
20	9.78	88.63	-9.78	114.33	0.1103	0.0855	0.0529
30	2.12	161.14	-2.12	108.56	0.0132	0.0196	0.0862
40	1.01	161.14	-1.01	194.08	0.0063	0.0052	0.0033
60	0.73	161.10	-0.73	194.08	0.0045	0.0037	0.0024
70	0.30	161.14	-0.30	108.56	0.0019	0.0028	0.0061

Analysis of Load Case 2 : NP+EW+EE+FS+BS

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	19.40	107.20	-19.40	109.90	0.1809	0.1765	0.1163
20	15.00	88.63	-15.00	114.33	0.1692	0.1312	0.0758
30	3.03	161.14	-3.03	145.17	0.0188	0.0209	0.1005
40	1.49	161.14	-1.49	194.08	0.0092	0.0077	0.0044
60	0.96	161.10	-0.96	194.08	0.0059	0.0049	0.0026
70	0.31	161.14	-0.31	108.56	0.0020	0.0029	0.0082



Analysis of Load Case 3 : NP+OW+WI+FW+BW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	12.01	107.20	-12.01	109.90	0.1120	0.1092	0.0625
20	10.38	88.63	-10.38	114.33	0.1171	0.0908	0.0582
30	2.12	161.14	-2.12	108.56	0.0132	0.0196	0.0918
40	1.01	161.14	-1.01	194.08	0.0063	0.0052	0.0033
60	0.73	161.10	-0.73	194.08	0.0045	0.0037	0.0024
70	0.30	161.14	-0.30	108.56	0.0019	0.0028	0.0061

Analysis of Load Case 4 : NP+OW+EQ+FS+BS

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	20.92	107.20	-20.92	109.90	0.1952	0.1904	0.1245
20	16.12	88.63	-16.12	114.33	0.1819	0.1410	0.0803
30	3.11	161.14	-3.11	145.17	0.0193	0.0214	0.1082
40	1.53	161.14	-1.53	194.08	0.0095	0.0079	0.0047
60	0.97	161.10	-0.97	194.08	0.0060	0.0050	0.0027
70	0.32	161.14	-0.32	108.56	0.0020	0.0029	0.0083

Analysis of Load Case 5 : NP+HW+HI

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	0.00	237.50	0.00	102.61	0.0000	-0.0000	...
20	0.00	247.00	0.00	102.61	0.0000	-0.0000	0.0012
30	0.00	247.00	0.00	122.73	0.0000	-0.0000	...
40	0.00	247.00	0.00	122.73	0.0000	-0.0000	...
60	0.00	247.00	0.00	122.73	0.0000	-0.0000	...
70	0.00	247.00	0.00	122.73	0.0000	-0.0000	...

Analysis of Load Case 6 : NP+HW+HE

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	0.00	237.50	0.00	102.61	0.0000	-0.0000	...
20	0.00	247.00	0.00	102.61	0.0000	-0.0000	...
30	0.00	247.00	0.00	122.73	0.0000	-0.0000	...
40	0.00	247.00	0.00	122.73	0.0000	-0.0000	...
60	0.00	247.00	0.00	122.73	0.0000	-0.0000	...
70	0.00	247.00	0.00	122.73	0.0000	-0.0000	...

Analysis of Load Case 7 : IP+OW+WI+FW+BW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	12.01	107.20	-12.01	109.90	0.1120	0.1092	0.0625
20	10.38	88.63	-10.38	114.33	0.1171	0.0908	0.0582
30	150.13	161.14		108.56	0.9317		0.0918
40	130.68	161.14		194.08	0.8110		0.0033
60	130.60	161.10		194.08	0.8107		0.0024
70	150.27	161.14		122.73	0.9326		...

Analysis of Load Case 8 : IP+OW+EQ+FS+BS

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	20.92	107.20	-20.92	109.90	0.1952	0.1904	0.1245
20	16.12	88.63	-16.12	114.33	0.1819	0.1410	0.0803
30	150.62	161.14		145.17	0.9347		0.1082
40	130.71	161.14		194.08	0.8111		0.0047
60	130.61	161.10		194.08	0.8107		0.0027
70	150.28	161.14		122.73	0.9326		...

Analysis of Load Case 9 : EP+OW+WI+FW+BW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
-----------	------------------	-----------------------	--------------------	-----------------------	-------------	-------------	-------------



10	12.01	107.20	-12.01	109.90	0.1120	0.1092	0.0625
20	10.38	88.63	-10.38	114.33	0.1171	0.0908	0.0582
30	3.04	161.14	-3.04	108.56	0.0189	0.0280	0.0918
40	1.50	161.14	-1.50	194.08	0.0093	0.0077	0.0100
60	1.21	161.10	-1.21	194.08	0.0075	0.0063	0.0095
70	1.22	161.14	-1.22	145.17	0.0076	0.0084	0.0012

Analysis of Load Case 10 : EP+OW+EQ+FS+BS

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	20.92	107.20	-20.92	109.90	0.1952	0.1904	0.1245
20	16.12	88.63	-16.12	114.33	0.1819	0.1410	0.0803
30	4.03	161.14	-4.03	145.17	0.0250	0.0277	0.1082
40	2.01	161.14	-2.01	194.08	0.0125	0.0104	0.0113
60	1.46	161.10	-1.46	194.08	0.0091	0.0075	0.0100
70	1.23	161.14	-1.23	145.17	0.0077	0.0085	0.0029

Analysis of Load Case 11 : HP+HW+HI

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	0.00	237.50	0.00	102.61	0.0000	-0.0000	...
20	0.00	247.00	0.00	102.61	0.0000	-0.0000	0.0012
30	206.95	247.00		122.73	0.8379		...
40	187.04	247.00		122.73	0.7573		...
60	187.04	247.00		122.73	0.7572		...
70	206.95	247.00		122.73	0.8379		...

Analysis of Load Case 12 : HP+HW+HE

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	0.00	237.50	0.00	102.61	0.0000	-0.0000	...
20	0.00	247.00	0.00	102.61	0.0000	-0.0000	...
30	206.95	247.00		122.73	0.8379		...
40	187.04	247.00		122.73	0.7573		...
60	187.04	247.00		122.73	0.7572		...
70	206.95	247.00		122.73	0.8379		...

Analysis of Load Case 13 : IP+WE+EW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	6.78	107.20	-6.78	109.90	0.0633	0.0617	0.0571
20	6.56	88.63	-6.56	114.33	0.0740	0.0574	0.0529
30	149.83	161.14		108.56	0.9298		0.0566
40	130.67	161.14		194.08	0.8109		0.0033
60	130.59	161.10		194.08	0.8106		0.0024
70	150.24	161.14		122.73	0.9324		...

Analysis of Load Case 14 : IP+WF+CW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	6.32	107.20	-6.32	109.90	0.0590	0.0575	0.0578
20	6.12	88.63	-6.12	114.33	0.0690	0.0535	0.0538
30	149.78	161.14		108.56	0.9295		0.0563
40	130.67	161.14		194.08	0.8109		0.0032
60	130.59	161.10		194.08	0.8106		0.0024
70	150.25	161.14		122.73	0.9324		...

Analysis of Load Case 15 : IP+VO+OW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	7.38	107.20	-7.38	109.90	0.0688	0.0671	0.0625
20	7.16	88.63	-7.16	114.33	0.0807	0.0626	0.0582
30	149.83	161.14		108.56	0.9298		0.0621

40	130.67	161.14	194.08	0.8109	0.0033
60	130.59	161.10	194.08	0.8106	0.0024
70	150.24	161.14	122.73	0.9324	...

Analysis of Load Case 16 : IP+VE+EW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	6.78	107.20	-6.78	109.90	0.0633	0.0617	0.0571
20	6.56	88.63	-6.56	114.33	0.0740	0.0574	0.0529
30	149.83	161.14		108.56	0.9298		0.0566
40	130.67	161.14		194.08	0.8109		0.0033
60	130.59	161.10		194.08	0.8106		0.0024
70	150.24	161.14		122.73	0.9324		...

Analysis of Load Case 17 : NP+VO+OW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	7.38	107.20	-7.38	109.90	0.0688	0.0671	0.0625
20	7.16	88.63	-7.16	114.33	0.0807	0.0626	0.0582
30	1.52	161.14	-1.52	108.56	0.0094	0.0140	0.0621
40	0.69	161.14	-0.69	194.08	0.0043	0.0036	0.0033
60	0.53	161.10	-0.53	194.08	0.0033	0.0027	0.0024
70	0.24	161.14	-0.24	108.56	0.0015	0.0022	0.0042

Analysis of Load Case 18 : FS+BS+IP+OW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	7.38	107.20	-7.38	109.90	0.0688	0.0671	0.0625
20	7.16	88.63	-7.16	114.33	0.0807	0.0626	0.0582
30	149.83	161.14		108.56	0.9298		0.0621
40	130.67	161.14		194.08	0.8109		0.0033
60	130.59	161.10		194.08	0.8106		0.0024
70	150.24	161.14		122.73	0.9324		...

Analysis of Load Case 19 : FS+BS+EP+OW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	7.38	107.20	-7.38	109.90	0.0688	0.0671	0.0625
20	7.16	88.63	-7.16	114.33	0.0807	0.0626	0.0582
30	2.43	161.14	-2.43	108.56	0.0151	0.0224	0.0621
40	1.18	161.14	-1.18	194.08	0.0073	0.0061	0.0033
60	1.01	161.10	-1.01	194.08	0.0063	0.0052	0.0024
70	1.16	161.14	-1.16	108.56	0.0072	0.0107	0.0082

Analysis of Load Case 20 : BL+IP+OW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	7.38	107.20	-7.38	109.90	0.0688	0.0671	0.0625
20	7.16	88.63	-7.16	114.33	0.0807	0.0626	0.0582
30	149.83	161.14		108.56	0.9298		0.0621
40	130.67	161.14		194.08	0.8109		0.0033
60	130.59	161.10		194.08	0.8106		0.0024
70	150.24	161.14		122.73	0.9324		...

Absolute Maximum of the all of the Stress Ratio's 0.9347

Governing Element: Bottom D end
 Governing Load Case 8 : IP+OW+EQ+FS+BS

Element Description	Long Pressure Stress	Weight + Axial Load Stress	Eccentric Moment Stress	Wind or Earthquake Stress	Combined Stress	Which Side
Skirt 1	...	-6.8	0.5	13.5	7.2	+ Tens
Skirt 1	...	-6.8	-0.5	-13.5	-20.9	- Comp



Top Y Ring	...	-6.6	0.5	9.0	2.9	+ Tens
Top Y Ring	...	-6.6	-0.5	-9.0	-16.1	- Comp
Bottom D end	150.4	-1.4	0.1	1.6	150.7	+ Tens
Bottom D end	150.4	-1.4	-0.1	-1.6	...	- Comp
Shell 1	70.8	-0.6	0.1	0.8	71.1	+ Tens
Shell 1	70.8	-0.6	-0.1	-0.8	...	- Comp
Shell 2	70.8	-0.5	0.1	0.4	70.9	+ Tens
Shell 2	70.8	-0.5	-0.1	-0.4	...	- Comp
Top D End	150.4	-0.2	...	0.1	150.2	+ Tens
Top D End	150.4	-0.2	...	-0.1	...	- Comp

Additional Results per Section 4.4, see 4.4.15

Nomenclature :

L - Section Length mm.

Fxa - Allowable comp mem stress due to compression with lamc ≤ 0.15 , N./mm²

Fba - Allowable all comp mem stress due to bending moment, N./mm²

Fva - Allowable shear stress when only shear is present, N./mm²

Fha - Allowable hoop comp memb stress under ext pressure alone, N./mm²

Fxha - Allowable axial comp w/hoop compression and lamc ≤ 0.15 , N./mm²

Fhxa - Allowable hoop comp w/axial compression and lamc ≤ 0.15 , N./mm²

Faha - Allowable axial comp w/hoop compression and lamc > 0.15 , N./mm²

Fbha - Allowable axial comp from bending w/hoop compression, N./mm²

Fhba - Allowable hoop comp w/long compression due to moment, N./mm²

Allowable Stress Results for Case 1 : NP+EW+1.1WI+FW+BW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.174	3000	109	113	88		6895	109	6895	6895	6895
20	0.177	3000	114	118	87		6895	114	6895	6895	6895
30						108	6895		6895	6895	6895
40	0.224	5498	194	194	148	122	6895	194	6895	6895	6895
60	0.224	5498	194	194	148	122	6895	194	6895	6895	6895
70						108	6895		6895	6895	6895

Allowable Stress Results for Case 2 : NP+EW+EE+FS+BS

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.174	3000	109	113	88		6895	109	6895	6895	6895
20	0.177	3000	114	118	87		6895	114	6895	6895	6895
30						145	6895		6895	6895	6895
40	0.224	5498	194	194	148	122	6895	194	6895	6895	6895
60	0.224	5498	194	194	148	122	6895	194	6895	6895	6895
70						108	6895		6895	6895	6895

Allowable Stress Results for Case 3 : NP+OW+1.1WI+FW+BW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.174	3000	109	113	88		6895	109	6895	6895	6895
20	0.177	3000	114	118	87		6895	114	6895	6895	6895
30						108	6895		6895	6895	6895
40	0.224	5498	194	194	148	122	6895	194	6895	6895	6895
60	0.224	5498	194	194	148	122	6895	194	6895	6895	6895
70						108	6895		6895	6895	6895

Allowable Stress Results for Case 4 : NP+OW+EQ+FS+BS

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.174	3000	109	113	88		6895	109	6895	6895	6895
20	0.177	3000	114	118	87		6895	114	6895	6895	6895
30						145	6895		6895	6895	6895
40	0.224	5498	194	194	148	122	6895	194	6895	6895	6895
60	0.224	5498	194	194	148	122	6895	194	6895	6895	6895
70						108	6895		6895	6895	6895

Allowable Stress Results for Case 5 : NP+HW+HI

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.176	3000	114	117	90		6895	114	6895	6895	6895



20	0.180	3000	120	123	90	6895	120	6895	6895	6895
30						6895		6895	6895	6895
40	0.223	5498	194	194	148	122	6895	194	6895	6895
60	0.223	5498	194	194	148	122	6895	194	6895	6895
70						6895		6895	6895	6895

Allowable Stress Results for Case 6 : NP+HW+HE

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.176	3000	114	117	90		6895	114	6895	6895	6895
20	0.180	3000	120	123	90		6895	120	6895	6895	6895
30							6895		6895	6895	6895
40	0.223	5498	194	194	148	122	6895	194	6895	6895	6895
60	0.223	5498	194	194	148	122	6895	194	6895	6895	6895
70							6895		6895	6895	6895

Allowable Stress Results for Case 7 : IP+OW+1.1WI+FW+BW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.174	3000	109	113	88		6895	109	6895	6895	6895
20	0.177	3000	114	118	87		6895	114	6895	6895	6895
30						108	6895		6895	6895	6895
40	0.224	5498	194	194	148	122	6895	194	6895	6895	6895
60	0.224	5498	194	194	148	122	6895	194	6895	6895	6895
70							6895		6895	6895	6895

Allowable Stress Results for Case 8 : IP+OW+EQ+FS+BS

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.174	3000	109	113	88		6895	109	6895	6895	6895
20	0.177	3000	114	118	87		6895	114	6895	6895	6895
30						145	6895		6895	6895	6895
40	0.224	5498	194	194	148	122	6895	194	6895	6895	6895
60	0.224	5498	194	194	148	122	6895	194	6895	6895	6895
70							6895		6895	6895	6895

Allowable Stress Results for Case 9 : EP+OW+1.1WI+FW+BW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.174	3000	109	113	88		6895	109	6895	6895	6895
20	0.177	3000	114	118	87		6895	114	6895	6895	6895
30						108	6895		6895	6895	6895
40	0.224	5498	194	194	148	122	168	153	165	47	113
60	0.224	5498	194	194	148	122	141	154	139	34	116
70						145	6895		6895	6895	6895

Allowable Stress Results for Case 10 : EP+OW+EQ+FS+BS

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.174	3000	109	113	88		6895	109	6895	6895	6895
20	0.177	3000	114	118	87		6895	114	6895	6895	6895
30						145	6895		6895	6895	6895
40	0.224	5498	194	194	148	122	168	153	165	95	97
60	0.224	5498	194	194	148	122	141	154	139	61	109
70						145	6895		6895	6895	6895

Allowable Stress Results for Case 11 : HP+HW+HI

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.176	3000	114	117	90		6895	114	6895	6895	6895
20	0.180	3000	120	123	90		6895	120	6895	6895	6895
30							6895		6895	6895	6895
40	0.223	5498	194	194	148	122	6895	194	6895	6895	6895
60	0.223	5498	194	194	148	122	6895	194	6895	6895	6895
70							6895		6895	6895	6895

Allowable Stress Results for Case 12 : HP+HW+HE

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.176	3000	114	117	90		6895	114	6895	6895	6895
20	0.180	3000	120	123	90		6895	120	6895	6895	6895
30							6895		6895	6895	6895
40	0.223	5498	194	194	148	122	6895	194	6895	6895	6895
60	0.223	5498	194	194	148	122	6895	194	6895	6895	6895
70							6895		6895	6895	6895

Allowable Stress Results for Case 13 : IP+WE+EW



From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.174	3000	109	113	88		6895	109	6895	6895	6895
20	0.177	3000	114	118	87		6895	114	6895	6895	6895
30						108	6895		6895	6895	6895
40	0.224	5498	194	194	148	122	6895	194	6895	6895	6895
60	0.224	5498	194	194	148	122	6895	194	6895	6895	6895
70							6895		6895	6895	6895

Allowable Stress Results for Case 14 : IP+WF+CW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.174	3000	109	113	88		6895	109	6895	6895	6895
20	0.177	3000	114	118	87		6895	114	6895	6895	6895
30						108	6895		6895	6895	6895
40	0.224	5498	194	194	148	122	6895	194	6895	6895	6895
60	0.224	5498	194	194	148	122	6895	194	6895	6895	6895
70							6895		6895	6895	6895

Allowable Stress Results for Case 15 : IP+VO+OW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.174	3000	109	113	88		6895	109	6895	6895	6895
20	0.177	3000	114	118	87		6895	114	6895	6895	6895
30						108	6895		6895	6895	6895
40	0.224	5498	194	194	148	122	6895	194	6895	6895	6895
60	0.224	5498	194	194	148	122	6895	194	6895	6895	6895
70							6895		6895	6895	6895

Allowable Stress Results for Case 16 : IP+VE+EW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.174	3000	109	113	88		6895	109	6895	6895	6895
20	0.177	3000	114	118	87		6895	114	6895	6895	6895
30						108	6895		6895	6895	6895
40	0.224	5498	194	194	148	122	6895	194	6895	6895	6895
60	0.224	5498	194	194	148	122	6895	194	6895	6895	6895
70							6895		6895	6895	6895

Allowable Stress Results for Case 17 : NP+VO+OW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.174	3000	109	113	88		6895	109	6895	6895	6895
20	0.177	3000	114	118	87		6895	114	6895	6895	6895
30						108	6895		6895	6895	6895
40	0.224	5498	194	194	148	122	6895	194	6895	6895	6895
60	0.224	5498	194	194	148	122	6895	194	6895	6895	6895
70						108	6895		6895	6895	6895

Allowable Stress Results for Case 18 : FS+BS+IP+OW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.174	3000	109	113	88		6895	109	6895	6895	6895
20	0.177	3000	114	118	87		6895	114	6895	6895	6895
30						108	6895		6895	6895	6895
40	0.224	5498	194	194	148	122	6895	194	6895	6895	6895
60	0.224	5498	194	194	148	122	6895	194	6895	6895	6895
70							6895		6895	6895	6895

Allowable Stress Results for Case 19 : FS+BS+EP+OW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.174	3000	109	113	88		6895	109	6895	6895	6895
20	0.177	3000	114	118	87		6895	114	6895	6895	6895
30						108	6895		6895	6895	6895
40	0.224	5498	194	194	148	122	168	153	165	6895	6895
60	0.224	5498	194	194	148	122	141	154	139	6895	6895
70						108	6895		6895	6895	6895

Allowable Stress Results for Case 20 : BL+IP+OW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.174	3000	109	113	88		6895	109	6895	6895	6895
20	0.177	3000	114	118	87		6895	114	6895	6895	6895
30						108	6895		6895	6895	6895
40	0.224	5498	194	194	148	122	6895	194	6895	6895	6895
60	0.224	5498	194	194	148	122	6895	194	6895	6895	6895
70							6895		6895	6895	6895



Note:

Load cases involving Wind and Seismic are assumed to cause a net section bending moment. These load case components are: EQ, EE, WI, WE, VO, VE, VF, HE, HI, WF. Static bending moments due to the weights of nozzles etc. are considered to be local in nature and will not cause a net section bending moment. This is an important consideration when computing the unity check per section 4.4.12 equations 4.4.86, 4.4.87, 4.4.90, 4.4.91 and 4.4.92.

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Skirt Support Analysis Per EN 13445-3 Section 16.12:

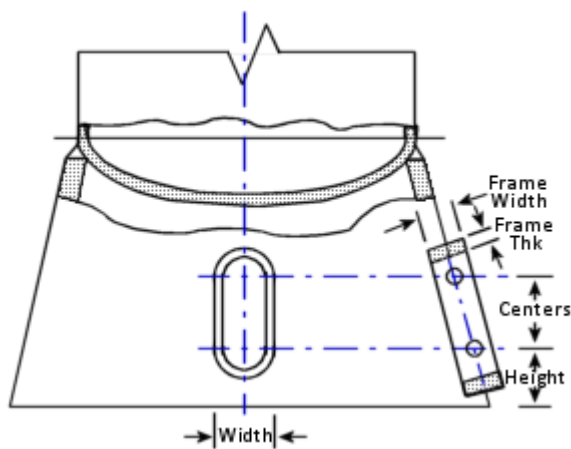
The analysis of this skirt has been conducted referencing the EN 13445-3 design code. It is important to note that the selected design code does not explicitly cover the analysis of this component. Therefore, the results provided herein should be regarded as informational only.

Item: Top Y Ring

Total vessel (empty) weight	Fg	91951.273 Kgf
Head Weight below section 2-2	DeltaFg	10595.947 Kgf
Weight of vessel contents (liquid)	Ff	9243.642 Kgf
Vessel Gross Weight	Fg + Ff	101194.914 Kgf
Skirt Mean Diameter	Dz	2609.000 mm.
Skirt Analysis Thickness (Corroded)	ez	18.500 mm.
Skirt Height	H	1000.000 mm.
Additional Axial Force at section 1-1	F1	0.000 Kgf
Moment at section 1-1	M1	67071.469 Kg-m.
Applied Moment - Bottom of Skirt	Mb	89827.375 Kg-m.
Weight at Bottom of Skirt	W	102238.828 Kgf

Structural Type B Skirt per EN 13445 Figure 16.12-2:

Type of Head:		Spherical
Head Mean Diameter at Skirt Top	Db	2626.500 mm.
Head Corroded Thickness	eb	78.500 mm.



There are no Access Openings in the Skirt - Plain Skirt Considered

Moment of Inertia	Ixx =	12901890.000 cm ⁴
Moment of Inertia	Iyy =	12901890.000 cm ⁴
Section Modulus	W =	98903048.000 mm. ³
Section Metal Area	A =	1516.337 cm ²

Analysis of the Section at weakest section Modulus [Z min]:

Radius of Gyration of Skirt at 4-4 [rg]:
 $= \sqrt{\min(Ixx, Iyy) / \text{Area}}$
 $= \sqrt{\min(12901890, 12901890) / 1516.337}$
 $= 922.421 \text{ mm.}$

Skirt Slenderness Ratio [rg/L]:
 $= rg / L$
 $= 922.421 / 1000.0$
 $= 0.922$

Correct the Moment [M4]:
 top of skirt is the critical end
 Controlling Moment [M4] = 67071.469 Kg-m.

Axial Stress at Residual (Weakest) Surface Area [Sigma]:

Downwind Side:

$$= -M / Z - W / A$$

$$= -67071.469 / .39561E+09 - 102238.828 / 1516.33691$$

$$= -8.275 \text{ N./mm}^2$$

Upwind Side:

$$= +(M) / Z - W / A$$

$$= +67071.469 / .39561E+09 - 102238.828 / 1516.33691$$

$$= -4.95 \text{ N./mm}^2$$

16.12.3.4 Checking at connection areas:

Force at the joint, 16.12-1 [Fzp]:

$$= F1 - Fg - Ff + 4 \cdot M1 / Dz$$

$$= 0.0 - 91951.273 - 9243.642 + 4 \cdot 67071.469 / 2609.0$$

$$= 1634.0 \text{ Kgf}$$

Force at the joint, 16.12-2 [Fzq]:

$$= F1 - Fg - Ff - 4 \cdot M1 / Dz$$

$$= 0.0 - 91951.273 - 9243.642 - 4 \cdot 67071.469 / 2609.0$$

$$= -204023.8 \text{ Kgf}$$

16.12.3.4.1 Membrane stresses:

Membrane stress at 1-1, 16.12-3 & 4:

Stress [σ_{1pm}]:

$$= (Fzp + \Delta Fg) / (\pi \cdot Db \cdot eb) + P \cdot Db / (4 \cdot eb)$$

$$= (1633.981 + 10595.947 + 9243.642) / (\pi \cdot 2626.5 \cdot 78.5) + 191.0 \cdot 2626.5 / (4 \cdot 78.5)$$

$$= 160.10 \text{ N./mm}^2$$

Stress [σ_{1qm}]:

$$= (Fzq + \Delta Fg + Ff) / (\pi \cdot Db \cdot eb) + P \cdot Db / (4 \cdot eb)$$

$$= (-0.2E+06 + 10595.947 + 9243.642) / (\pi \cdot 2626.5 \cdot 78.5) + 191.0 \cdot 2626.5 / (4 \cdot 78.5)$$

$$= 156.99 \text{ N./mm}^2$$

Membrane stress at 2-2, 16.12-9:

Stress [σ_{2pm} & σ_{2qm}]:

$$= (Ff + \Delta Fg) / (\pi \cdot Db \cdot eb) + P \cdot Db / (4 \cdot eb)$$

$$= (9243.642 + 10595.947) / (\pi \cdot 2626.5 \cdot 78.5) + 191.0 \cdot 2626.5 / (4 \cdot 78.5)$$

$$= 160.07 \text{ N./mm}^2$$

Membrane stress at 3-3, 16.12 12 & 13:

Stress [σ_{3pm}]:

$$= Fzp / (\pi \cdot Dz \cdot ez)$$

$$= 1633.981 / (\pi \cdot 2609.0 \cdot 18.5)$$

$$= 0.11 \text{ N./mm}^2$$

Stress [σ_{3qm}]:

$$= Fzq / (\pi \cdot Dz \cdot ez)$$

$$= -204023.797 / (\pi \cdot 2609.0 \cdot 18.5)$$

$$= -13.20 \text{ N./mm}^2$$

16.12.3.4.2 Bending stresses:

Cosine of angle, 16.12-25 [cos(γ)]:

$$= 1 - 0.5(Db + eb - Dz + ez) / (r + eb)$$

$$= 1 - 0.5(2626.5 + 78.5 - 2609.0 + 18.5) / (1274.0 + 78.5)$$

$$= 0.9577$$

$$\gamma = \arccos(\cos(\gamma)) = \arccos(0.958) = 16.73 \text{ deg}$$

Dimension [a]:

$$= 0.5 \cdot \sqrt{(eb^2 + ez^2 + 2 \cdot eb \cdot ez \cdot \cos(\gamma))}$$

$$= 0.5 \cdot \sqrt{(78.5^2 + 18.5^2 + 2 \cdot 78.5 \cdot 18.5 \cdot 0.958)}$$

$$= 48.182 \text{ mm.}$$

Bending moment at verification points p and q [Mp & Mq]:

$$\begin{aligned} M_p &= a \cdot F_{zp} = 48.18 \cdot 1634.0 = 78.73 \text{ Kg-m.} \\ M_q &= a \cdot F_{zq} = 48.18 \cdot -204023.8 = -9830.49 \text{ Kg-m.} \end{aligned}$$

Correction factor [C]:

$$\begin{aligned} &= 0.63 - 0.057 \left(\frac{e_b}{e_z} \right)^2 \\ &= 0.63 - 0.057 \left(\frac{78.5}{18.5} \right)^2 \\ &= -0.396 \end{aligned}$$

Bending stresses in sections 1-1 to 3-3 at the outer surface (o), 16.12-26 to 29:

Stress [$\sigma_{1pb(o)}$ & $\sigma_{2pb(o)}$]:

$$\begin{aligned} &= C \cdot 6 \cdot M_p / (\pi \cdot D_b \cdot e_b^2) \\ &= -0.396 \cdot 6 \cdot 78.73 / (\pi \cdot 2626.5 \cdot 78.5^2) \\ &= -0.04 \text{ N./mm}^2 \end{aligned}$$

Stress [$\sigma_{1qb(o)}$ & $\sigma_{2qb(o)}$]:

$$\begin{aligned} &= C \cdot 6 \cdot M_q / (\pi \cdot D_b \cdot e_b^2) \\ &= -0.396 \cdot 6 \cdot -9830.487 / (\pi \cdot 2626.5 \cdot 78.5^2) \\ &= 4.51 \text{ N./mm}^2 \end{aligned}$$

Stress [$\sigma_{3pb(o)}$]:

$$\begin{aligned} &= C \cdot 6 \cdot M_p / (\pi \cdot D_z \cdot e_z^2) \\ &= -0.396 \cdot 6 \cdot 78.73 / (\pi \cdot 2609.0 \cdot 18.5^2) \\ &= -0.65 \text{ N./mm}^2 \end{aligned}$$

Stress [$\sigma_{3qb(o)}$]:

$$\begin{aligned} &= C \cdot 6 \cdot M_q / (\pi \cdot D_z \cdot e_z^2) \\ &= -0.396 \cdot 6 \cdot -9830.487 / (\pi \cdot 2609.0 \cdot 18.5^2) \\ &= 81.71 \text{ N./mm}^2 \end{aligned}$$

Geometry variables, 16.12-32 to 36:

$$\begin{aligned} e_b/D_b &= 3.091/103.406 = 0.0299 \\ y &= 125 \cdot e_b/D_b = 125 \cdot 78.5/2626.5 = 3.736 \\ \alpha &= 0 \end{aligned}$$

Bending stress in sections 1-1 and 2-2 Equation 16.12-31 [$\sigma_{1b(p)}$ & $\sigma_{2b(p)}$]:

$$= 0.0 \text{ N./mm}^2 \text{ (16.12-31)}$$

16.12.3.4.3 Total stresses and strength conditions:

Structural shape B and C:

Stress [σ_{1pitot}]:

$$\begin{aligned} &= \sigma_{1pm} - \sigma_{1pb(o)} + \sigma_{1b(p)} \\ &= 160.099 - -0.036 + 0.0 \\ &= 160.14 \text{ N./mm}^2 \end{aligned}$$

Stress [σ_{1potot}]:

$$\begin{aligned} &= \sigma_{1pm} + \sigma_{1pb(o)} - \sigma_{1b(p)} \\ &= 160.099 + -0.036 - 0.0 \\ &= 160.06 \text{ N./mm}^2 \end{aligned}$$

Stress [σ_{1qitot}]:

$$\begin{aligned} &= \sigma_{1qm} - \sigma_{1qb(o)} + \sigma_{1b(p)} \\ &= 156.986 - 4.508 + 0.0 \\ &= 152.48 \text{ N./mm}^2 \end{aligned}$$

Stress [σ_{1qotot}]:

$$\begin{aligned} &= \sigma_{1qm} + \sigma_{1qb(o)} - \sigma_{1b(p)} \\ &= 156.986 + 4.508 - 0.0 \\ &= 161.49 \text{ N./mm}^2 \end{aligned}$$

Stress [σ_{2pitot}]:

$$\begin{aligned} &= \sigma_{2pm} + \sigma_{2pb(o)} + \sigma_{2b(p)} \\ &= 160.074 + -0.036 + 0.0 \\ &= 160.04 \text{ N./mm}^2 \end{aligned}$$

Stress [σ_{2potot}]:

$$\begin{aligned}
 &= \sigma_{2pm} - \sigma_{2pb(o)} - \sigma_{2b(p)} \\
 &= 160.074 - -0.036 - 0.0 \\
 &= 160.11 \text{ N./mm}^2
 \end{aligned}$$

$$\begin{aligned}
 \text{Stress } [\sigma_{2qitot}]: \\
 &= \sigma_{2qm} + \sigma_{2qb(o)} + \sigma_{2b(p)} \\
 &= 160.074 + 4.508 + 0.0 \\
 &= 164.58 \text{ N./mm}^2
 \end{aligned}$$

$$\begin{aligned}
 \text{Stress } [\sigma_{2qotot}]: \\
 &= \sigma_{2pm} - \sigma_{2pb(o)} - \sigma_{2b(p)} \\
 &= 160.074 - 4.508 - 0.0 \\
 &= 155.57 \text{ N./mm}^2
 \end{aligned}$$

$$\begin{aligned}
 \text{Stress } [\sigma_{3pitot}]: \\
 &= \sigma_{3pm} - \sigma_{3pb} \\
 &= 0.106 - -0.654 \\
 &= 0.76 \text{ N./mm}^2
 \end{aligned}$$

$$\begin{aligned}
 \text{Stress } [\sigma_{3potot}]: \\
 &= \sigma_{3pm} + \sigma_{3pb} \\
 &= 0.106 + -0.654 \\
 &= -0.55 \text{ N./mm}^2
 \end{aligned}$$

$$\begin{aligned}
 \text{Stress } [\sigma_{3qitot}]: \\
 &= \sigma_{3qm} - \sigma_{3qb} \\
 &= -13.195 - 81.713 \\
 &= -94.91 \text{ N./mm}^2
 \end{aligned}$$

$$\begin{aligned}
 \text{Stress } [\sigma_{3qotot}]: \\
 &= \sigma_{3qm} + \sigma_{3qb} \\
 &= -13.195 + 81.713 \\
 &= 68.52 \text{ N./mm}^2
 \end{aligned}$$

Allowable stresses:

$$\begin{aligned}
 \text{For } \sigma_{1pitot} \text{ and } \sigma_{1potot}, \text{ the stress must not exceed:} \\
 &= fs(3 - (1 / 1.5)) \cdot (\sigma_{1pm} / f)^2 \\
 &= 161.141(3 - (1 / 1.5)) \cdot (160.099 / 161.141)^2 \\
 &= 377.38 \text{ N./mm}^2
 \end{aligned}$$

$$\begin{aligned}
 \text{For } \sigma_{1qitot} \text{ and } \sigma_{1qotot}, \text{ the stress must not exceed:} \\
 &= fs(3 - (1 / 1.5)) \cdot (\sigma_{1qm} / f)^2 \\
 &= 161.141(3 - (1 / 1.5)) \cdot (156.986 / 161.141)^2 \\
 &= 381.47 \text{ N./mm}^2
 \end{aligned}$$

$$\begin{aligned}
 \text{For } \sigma_{2pitot} \text{ and } \sigma_{2potot}, \text{ the stress must not exceed:} \\
 &= fs(3 - (1 / 1.5)) \cdot (\sigma_{2pm} / f)^2 \\
 &= 161.141(3 - (1 / 1.5)) \cdot (160.074 / 161.141)^2 \\
 &= 377.41 \text{ N./mm}^2
 \end{aligned}$$

$$\begin{aligned}
 \text{For } \sigma_{2qitot} \text{ and } \sigma_{2qotot}, \text{ the stress must not exceed:} \\
 &= fs(3 - (1 / 1.5)) \cdot (\sigma_{2qm} / f)^2 \\
 &= 161.141(3 - (1 / 1.5)) \cdot (160.074 / 161.141)^2 \\
 &= 377.41 \text{ N./mm}^2
 \end{aligned}$$

$$\begin{aligned}
 \text{For } \sigma_{3pitot} \text{ \& } \sigma_{3potot}, \text{ the stress must not exceed:} \\
 &= fs(3 - (1 / 1.5)) \cdot (\sigma_{3pm} / f)^2 \\
 &= 161.141(3 - (1 / 1.5)) \cdot (0.106 / 161.141)^2 \\
 &= 483.42 \text{ N./mm}^2
 \end{aligned}$$

$$\begin{aligned}
 \text{For } \sigma_{3qitot} \text{ \& } \sigma_{3qotot}, \text{ the stress must not exceed:} \\
 &= fs(3 - (1 / 1.5)) \cdot (\sigma_{3qm} / f)^2 \\
 &= 161.141(3 - (1 / 1.5)) \cdot (-13.195 / 161.141)^2 \\
 &= 482.70 \text{ N./mm}^2
 \end{aligned}$$

Please note: None of the following stresses in the table below are in the region where there is an access opening.

Summary of Stresses: (N./mm²)

Stress	Actual	Allowed	Result	Equation
Stot1pi	160.14	377.38	Pass	46
Stot1po	160.06	377.38	Pass	47
Stot1qi	152.48	381.47	Pass	48
Stot1qo	161.49	381.47	Pass	49
Stot2pi	160.04	377.41	Pass	50
Stot2po	160.11	377.41	Pass	51
Stot2qi	164.58	377.41	Pass	52
Stot2qo	155.57	377.41	Pass	53
Stot3pi	0.76	483.42	Pass	54
Stot3po	-0.55	483.42	Pass	55
Stot3qi	-94.91	482.70	Pass	56
Stot3qo	68.52	482.70	Pass	57

The stresses are satisfactory in the region without openings.

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Shop/Field Installation Options :

Platform(s) installed in the Shop.
Insulation is installed in the Shop.

Note : The CG is computed from the first Element From Node

Center of Gravity of the Platforms	6340.527 mm.
Center of Gravity of the Liquid	3548.514 mm.
Center of Gravity of the Insulation	5325.082 mm.
Center of Gravity of the Nozzles	6612.053 mm.
Center of Gravity of the Added Weights (Operating)	5419.758 mm.
Center of Gravity of the Added Weights (Empty)	5419.758 mm.
Center of Gravity of Bare Shell New and Cold	5055.788 mm.
Center of Gravity of Bare Shell Corroded	5068.839 mm.
Vessel CG in the Operating Condition	5201.371 mm.
Vessel CG in the Fabricated (Shop/Empty) Condition	5348.923 mm.
Vessel CG in the Test Condition	5343.169 mm.

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Skirt Data :

Skirt Outside Diameter at Base	SOD, D6	2629.0000	mm.
Skirt Thickness	STHK	20.0000	mm.
Skirt Internal Corrosion Allowance	SCA	0.7500	mm.
Skirt External Corrosion Allowance		0.7500	mm.
Skirt Material		SA-36	

Basing Input: Type of Geometry: Continuous Top Ring W/Gussets

Thickness of Basing	TBA, ea4	28.0000	mm.
Design Temperature of the Basing		38.00	°C
Basing Material		SA-36	
Basing Ope. All. Stress	BASOPE, f4	165.00	N./mm ²
Basing Yield Stress		248.00	N./mm ²
Inside Diameter of Basing	DI, D4	2465.0000	mm.
Outside Diameter of Basing	DOU, D5	2873.0000	mm.
Nominal Diameter of Bolts	BND, d(b0)	36.0000	mm.
Bolt Corrosion Allowance	BCA	1.5000	mm.
Bolt Material		SA-36	
Bolt Operating Allowable Stress	SA, fB	149.90	N./mm ²
Number of Bolts	RN, nb	8	
Diameter of Bolt Circle	DC, DBC	2757.0000	mm.
Bolt Allowable Shear Stress		99.900	N./mm ²
Ultimate Comp. Strength of Concrete	FPC	35.0	N./mm ²
Allowable Comp. Strength of Concrete	FC	15.8	N./mm ²
Modular ratio Steel/Concrete		9.833	
Thickness of Gusset Plates	TGA, e7	13.0000	mm.
Width of Gussets at Top Plate	TWDT	122.0000	mm.
Width of Gussets at Base Plate	BWDT	122.0000	mm.
Gusset Plate Elastic Modulus	E	201734000.0	KPa.
Gusset Plate Yield Stress	SY	248.0	N./mm ²
Height of Chair	h1	262.0000	mm.
Distance between Gussets	RG, b6	102.0000	mm.
Dist. from Bolt Center to Gusset (Rg/2)	CG	51.0000	mm.
Number of Gussets per bolt	NG	2	
Thickness of Top Plate or Ring	TTA, e5	35.0000	mm.
Radial Width of the Top Plate	TOPWTH, b5	122.0000	mm.
Anchor Bolt Hole Dia. in Top Plate	dia,H	51.0000	mm.
External Corrosion Allowance	CA	1.5000	mm.
Dead Weight of Vessel	DW	96459.2	Kgf
Operating Weight of Vessel	ROW	105702.9	Kgf
Earthquake Moment on Basing	EQMOM	135718.5	Kg-m.
Wind Moment on Basing	WIMOM	46414.9	Kg-m.
Percent Bolt Preload	ppl	100.0	
Use AISC A5.2 Increase in Fc and Bolt Stress		No	
Use Allowable Weld Stress per AISC J2.5		No	
Factor for Increase of Allowables	Fact	1.0000	

Results for Brownell and Young Basing Analysis : Analyze Option

Note: This analysis is based on Neutral Axis shift method for Steel on Concrete (or a material with significantly different Young's modulus).

Calculation of Bolt Loads for Multiple Load Conditions:

Bolt Load, Earthquake Operating Case:

$$= ((4 \cdot M/DC) - W) / nb \quad \text{per Jawad \& Farr, Eq. 12.3}$$

$$= ((4 \cdot 135718/2757.0) - 105703) / 8$$

$$= 11400.1 \text{ Kgf}$$

Bolt Load, Windload Empty Case:

$$= ((4 \cdot M/DC) - W) / nb \quad \text{per Jawad \& Farr, Eq. 12.3}$$

$$= ((4 \cdot 46415/2757.0) - 96459) / 8$$

$$= -3639.9 \text{ Kgf}$$

Bolt Load, Windload Vortex Shedding Case:

$$= ((4 \cdot M/DC) - W) / nb \quad \text{per Jawad \& Farr, Eq. 12.3}$$

$$= ((4 \cdot 0/2757.0) - 105703) / 8$$

$$= -13212.9 \text{ Kgf}$$

Governing Bolt Load Condition, Earthquake + Operating Condition:

Area Available in 1 Bolt (corroded)	Abss :	6.0063	cm ²
Area Available in all the Bolts	Abss * RN :	48.0504	cm ²

Trial#	k	knew	Cc	Ct	z	j	Ft	Fc	Fs
2	0.243	0.364	1.348	2.567	0.450	0.778	2136.8	107839.7	702.3
4	0.425	0.394	1.824	2.169	0.410	0.785	7496.3	113199.2	2915.2
6	0.379	0.387	1.713	2.270	0.420	0.784	6127.4	111830.3	2277.7
8	0.383	0.381	1.723	2.261	0.419	0.784	6240.6	111943.5	2328.3
10	0.382	0.382	1.721	2.263	0.420	0.784	6212.3	111915.2	2315.6
12	0.382	0.382	1.721	2.263	0.420	0.784	6219.4	111922.2	2318.7
13	0.382	0.382	1.721	2.263	0.420	0.784	6215.8	111918.7	2317.1

The Actual Stress in a Single Bolt [Sbolt]:

$$= 2 \cdot Ft / (T1 \cdot Dc \cdot Ct)$$

$$= 2 \cdot 6215.8 / (0.555 \cdot 2757.0 \cdot 2.263)$$

$$= 35.222 \text{ N./mm}^2, \text{ Should be } \leq 149.9 \text{ N./mm}^2$$

Thickness of the Band of Bolting Steel [T1]

$$= RN \cdot \text{Bolt Area} / (\pi \cdot Dc)$$

$$= 8 \cdot 6.006 / (\pi \cdot 2757.0)$$

$$= 0.555 \text{ mm.}$$

Check the Bearing Stress in the Concrete [fc(max)]

$$= fc / [(2kd + t3) / (2kd)]$$

$$= 2.215 / [(2 \cdot 0.3821 \cdot 2757.0 + 204.0) / (2 \cdot 0.3821 \cdot 2757.0)]$$

$$= 2.429 \text{ N./mm}^2, \text{ Should be } \leq 15.8 \text{ N./mm}^2$$

Values for table 10.3, l = 122.0, b = 541.336, l/b = 0.225368

Maximum Moment per unit width [Mmax]:

$$= \max(Mx, My) = \max(382.831, 1664.045) = 1664.045 \text{ Kgf}$$

Reqd Thickness of Basing, Brownell & Young Method [T]:

$$= \sqrt{(6 \cdot Mmax / fallow)} + Ca$$

$$= \sqrt{(6 \cdot 1664.045 / 163.7)} + 1.5$$

$$= 25.958 \text{ mm.}$$

Nomenclature:

a = (Dc - Ds) / 2	Skirt Distance to Bolt Circle
P = Sa * Abss	Maximum Load on one Bolt
l	Radial Top Plate Width
lavg	Average Gusset Plate Width
g1 = Gamma 1	Constant Term f(b/l)
g2 = Gamma 2	Constant Term f(b/l)
g = Flat distance / 2	Nut 1/2 Dimension (from Tema)
Fb	Allowable Bending Stress

Values for table 10.6, l = 122.0, b = 102.0, b/l = 0.836066

As b/l (0.836) is less than 1, inverting b/l = 1.196.

Moment Term, based on geometry [Mo]:

$$= P / (4 \cdot \pi) [1.3 (\ln((2l \sin(\pi \cdot a/l)) / (\pi \cdot g))) + 1] - [(0.7 - g2) P / (4 \cdot \pi)]$$

$$= 9180.79 / (4 \cdot \pi) [1.3 (\ln((2 \cdot 122.0 \cdot \sin(\pi \cdot 64.0 / 122.0)) / (\pi \cdot 30.0))) + 1] - [(0.7 - 0.109) \cdot 9180.79 / (4 \cdot \pi)]$$

$$= 1199.5883 \text{ Kgf}$$

Required Thickness of Continuous Top Ring [Tc]:

$$= \sqrt{(6 \cdot \text{Abs}(Mo) / Fb)} + Ca$$

$$= \sqrt{(6 \cdot \text{Abs}(1199.59) / 165.33)} + 1.5$$

= 22.1624 mm.

Required Gusset Plate Thickness [tg]:

$$= P / (\text{Stress Term} \cdot l_{avg}) + Ca$$

$$= 9180.79 / (124.11 \cdot 122.0) + 1.5$$

$$= 11.025 \text{ (not less than } 9.525 + 1.5 \text{) mm.}$$

Bolt spacing [m]:

$$= \pi \cdot \text{Bolt Circle Diameter} / \text{number of Bolts}$$

$$= \pi \cdot 2757.0 / 8 = 1082.671 \text{ mm.}$$

Req. Skirt Thk. to withstand Local Bending, (Brownell and Young) [t]:

$$= 1.76 (P \cdot a / (m \cdot (h + t_{ba}) \cdot 1.5 \cdot S_{ktope}))^{2/3} \cdot r^{1/3} + Ca$$

$$= 1.76 (9181 \cdot 64.0 / (1082.67 \cdot 227.0 \cdot 230))^{2/3} \cdot 1314.51^{1/3} + Ca$$

$$= 4.211 + 1.5 = 5.711 \text{ mm.}$$

Shear Stress in a Single Bolt [tb]:

$$= \text{Shear Force} / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 22972.8 / (6.006 \cdot 8)$$

$$= 46.9 \text{ N./mm}^2. \text{ Must be less than } 99.9 \text{ N./mm}^2.$$

Summary of Basing Thickness Calculations

Required Basing Thickness (tension)	25.9584	mm.
Actual Basing Thickness as entered by user	28.0000	mm.
Required Thickness of Chair Cap	22.1624	mm.
Actual Top Ring Thickness as entered by user	35.0000	mm.
Required Gusset thickness, + CA	11.0250	mm.
Actual Gusset Thickness as entered by user	13.0000	mm.
Required Thickness of Skirt for Local Stress	5.7107	mm.
Given Thickness of Skirt	20.0000	mm.

Weld Size Calculations per Steel Plate Engineering Data - Vol. 2

Compute the Weld load at the Skirt/Base Junction [W]:

$$= \text{SkirtStress} \cdot (\text{SkirtThickness} - CA)$$

$$= 20.922 \cdot (20.0 - 1.5)$$

$$= 39.47 \text{ Kg/mm.}$$

Results for Computed Minimum Basing Weld Size [BWeld]:

$$= W / [(0.4 \cdot \text{Yield}) \cdot 2 \cdot 0.707]$$

$$= 39 / [(0.4 \cdot 230) \cdot 2 \cdot 0.707]$$

$$= 2.981 \text{ mm.}$$

Results for Computed Minimum Gusset and Top Plate to Skirt Weld Size

Vertical Plate Load [Wv]:

$$= \text{Bolt Load} / (\text{Cmwith} + 2 \cdot (Hg + Tta))$$

$$= 9180.8 / (170.0 + 2 \cdot (199.0 + 35.0))$$

$$= 14.39 \text{ Kg/mm.}$$

Horizontal Plate Load [Wh]:

$$= \text{Bolt Load} \cdot e / (\text{Cmwith} \cdot (Hg + Tta) + 0.6667 \cdot (Hg + Tta)^2)$$

$$= 9180.8 \cdot 64.0 / (170.0 \cdot (234.0) + 0.6667 \cdot (234.0)^2)$$

$$= 7.702 \text{ Kg/mm.}$$

Resultant Weld Load [Wr]:

$$= \sqrt{ Wv^2 + Wh^2 }$$

$$= \sqrt{ 14.39^2 + 7.7^2 }$$

$$= 16.322 \text{ Kg/mm.}$$

Results for Computed Min Gusset and Top Plate to Skirt Weld Size [GsWeld]:

$$= Wr / [(0.4 \cdot \text{Yield}) \cdot 2 \cdot 0.707]$$

$$= 16.32 / [(0.4 \cdot 230) \cdot 2 \cdot 0.707]$$

$$= 1.233 \text{ mm.}$$

Results for Computed Minimum Gusset to Top Plate Weld Size

Weld Load [Wv]:
= Bolt Load / (2 • TopWth)
= 9180.8/(2 • 122.0)
= 37.626 Kgf/mm.

Weld Load [Wh]:
= Bolt Load • e / (2 • Hgt • TopWth)
= 9180.8 • 64.0/(2 • 234.0 • 122.0)
= 10.291 Kgf/mm.

Resultant Weld Load [Wr]:
= $\sqrt{Wv^2 + Wh^2}$
= $\sqrt{37.63^2 + 10.29^2}$
= 39.008 Kgf/mm.

Results for Computed Min Gusset to Top Plate Weld Size [GtpWeld]:
= Wr / [(0.4 • Yield) • 2 • 0.707]
= 39.01/[(0.4 • 230) • 2 • 0.707]
= 2.946 mm.

Note: The calculated weld sizes need not exceed the component thickness framing into the weld. At the same time, the weld must meet a minimum size specification which is 3/16 in. (4.76 mm) or 1/4 in. (6.35 mm), depending on the component thickness.

Summary of Required Weld Sizes:

Required Basing to Skirt Double Fillet Weld Size	6.3500	mm.
Required Gusset to Skirt Double Fillet Weld Size	6.3500	mm.
Required Top Plate to Skirt Weld Size	6.3500	mm.
Required Gusset to Top Plate Double Fillet Weld Size	2.9462	mm.

Review notes about nozzle loadings and supports in the Warnings report.

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Input, Nozzle Desc: D13"

From: 30

Pressure for Reinforcement Calculations	P	192.419	bars
Temperature for Internal Pressure	Temp	85	°C
Design External Pressure	Pext	1.03	bars
Temperature for External Pressure	Tempex	85	°C
Parent Material		SA-516 70	+N+IT
Parent Allowable Stress at Temperature	Sv	161.14	N./mm ²
Parent Allowable Stress At Ambient	Sva	175.00	N./mm ²
Inside Diameter of Hemispherical Head	D	2545.00	mm.
Head Finished (Minimum) Thickness	t	80.0000	mm.
Head Internal Corrosion Allowance	cas	1.5000	mm.
Head External Corrosion Allowance	co	0.0000	mm.
Distance from Head Centerline	L1	0.0000	mm.
User Entered Minimum Design Metal Temperature		-40.00	°C

Type of Element Connected to the Parent : Nozzle

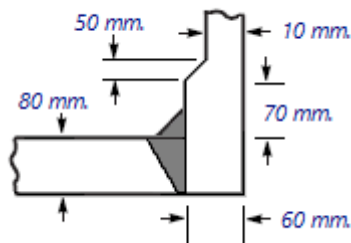
Material		SA-350 LF1	+IT
Material UNS Number		K03009	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	127.70	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		3.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	10.0000	mm.
Hub Height of Integral Nozzle	h	70.0000	mm.
Height of Beveled Transition	L'	50.0000	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		60.0000	mm.
Corrosion Allowance	can	1.5000	mm.
Outside Projection	ho	200.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	80.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.
This is a Manway, Coupling or Access Opening.			
Piping Exit Angle [North Clockwise reference]		90.0	deg
Bend Radius Multiplier		1.50	
Horizontal Run Length		2000.0	mm.
Centerline Distance to Tangent		1850.0	mm.

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: D13"

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	3.000	in.
Actual Thickness Used in Calculation	0.394	in.



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 79.2 \cdot (\exp(192.419/(127.7 \cdot 1.0)) - 1) + 1.5 + 0.0$$

$$= 6.4403 + 1.5 + 0.0 = 7.9403 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 127.7 \cdot 1.0 \cdot \ln((2 \cdot 8.5 + 79.2) / 79.2) - 0.0$$

$$= 248.302 - 0.0 = 248.3021 \text{ bars}$$

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[127.7/161.1, 1]$$

$$= 0.792$$

Thickness of Nozzle at Shell [tn]:

$$= \text{hub thickness} - \text{corrosion allowance}$$

$$= 60.0 - 1.5$$

$$= 58.500 \text{ mm.}$$

Thickness of Nozzle at Top [tn2]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 10.0 - 1.5$$

$$= 8.500 \text{ mm.}$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 2545.0/2 + 1.5$$

$$= 1274.000 \text{ mm.}$$

Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$$

$$= \min(\sqrt{1274.0 \cdot 78.5}, 2 \cdot 39.6)$$

$$= 79.200 \text{ mm.}$$

Intermediate value [Cn]:

$$= \min(((t + t_e)/t_n)^{0.35}, 1)$$

$$= \min(((78.5 + 0.0)/58.5)^{0.35}, 1)$$

$$= \min(1.108, 1)$$

$$= 1.000$$

Intermediate value [Cp]:

$$= 999999.000$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$= \min(t + t_e + F_p \cdot \sqrt{R_n \cdot t_n}, L_{pr1} + t)$$

$$= \min(78.5 + 0.0 + 1.0 \cdot \sqrt{39.6 \cdot 58.5}, 200.0 + 78.5)$$

$$= 126.631 \text{ mm.}$$

Effective Vessel Thickness [teff]:

$$= t + ((A_5 \cdot frp) / LR)$$

$$= 78.5 + ((0.0 \cdot 1.0) / 79.2)$$

$$= 78.500 \text{ mm.}$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR$$

$$= 78.5 \cdot 79.2$$

$$= 62.172 \text{ cm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= t_n \cdot LH$$

$$= 58.5 \cdot 126.631$$

$$= 74.079 \text{ cm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot \text{Leg}^2$$

$$= 0.5 \cdot 10.0^2$$

$$= 0.500 \text{ cm}^2$$

The total area contributed by A1 through A43 [AT]:

$$= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43$$

$$= 62.172 + 0.792(74.079 + 0.0) + 0.5 + 0.0 + 0.0$$

$$= 121.378 \text{ cm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n/R_n]$$

$$= 58.5 / \ln[1 + 58.5/39.6]$$

$$= 64.487 \text{ mm.}$$

Shell Radius for Force Calculation [Rxs]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$

$$= 78.5 / \ln[1 + 78.5/1274.0]$$

$$= 1312.860 \text{ mm.}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$

$$= 1.5 \cdot 161.141 \cdot 1.0$$

$$= 241.7 \text{ N./mm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot LH$$

$$= 192.419 \cdot 64.487 \cdot 126.631$$

$$= 16023.5 \text{ Kgf}$$

Force acting on the Shell [fS]:

$$= P \cdot R_{xs} \cdot (LR + t_n) / 2$$

$$= 192.419 \cdot 1312.86 \cdot (79.2 + 58.5) / 2$$

$$= 177364.3 \text{ Kgf}$$

Discontinuity Force from Internal Pressure [fY]:

$$= (P \cdot R_{xs} \cdot R_{nc}) / 2$$

$$= (192.419 \cdot 1312.86 \cdot 39.6) / 2$$

$$= 51006.7 \text{ Kgf}$$

Area Resisting Internal Pressure [Ap]:

$$= (fS + fY + fN) / P$$

$$= (177364.3 + 51006.7 + 16023.5) / 192.419$$

$$= 1245.5 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= S_{allow} / (2 \cdot A_p / AT - R_{xs} / (2 \cdot t_{eff}))$$

$$= 241.712 / (2 \cdot 1245.511 / 121.378 - 1312.86 / (2 \cdot 78.5))$$

$$= 198.8 \text{ bars}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= 2 \cdot S[t / R_{xs}]$$

$$= 2 \cdot 161.141[78.5 / 1312.86]$$

$$= 192.7 \text{ bars}$$



Maximum Allowable Working Pressure [Pmax]:

$$= \min(P_{max1}, P_{max2})$$

$$= \min(198.753, 192.691)$$

$$= 192.691 \text{ bars}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$= (f_N + f_S + f_Y) / AT$$

$$= (16023.483 + 177364.312 + 51006.73) / 121.378$$

$$= 197.461 \text{ N./mm}^2$$

General Primary Membrane Stress [σ_{Circ}]:

$$= P \cdot R_{xs} / (2 \cdot t_{eff})$$

$$= 192.419 \cdot 1312.86 / (2 \cdot 78.5)$$

$$= 160.9 \text{ N./mm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$$

$$= \max(2 \cdot 197.461 - 160.913, 160.913)$$

$$= 234.0 \text{ N./mm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	241.71	N./mm ²
Local Primary Membrane Stress	PL	234.01	N./mm ²
Maximum Allowable Working Pressure	Pmax	192.69	bars

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= (R_{nc} + t_n) / R_{nc}$$

$$= (39.6 + 58.5) / 39.6$$

$$= 2.477 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$= \pi/2 \cdot (R_n + t_n)$$

$$= \pi/2 \cdot (39.6 + 58.5)$$

$$= 154.095 \text{ mm.}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm.}$$

Required Weld Throat Dimensions :

$$t_c, L41T = 7.071, \text{ must be } \geq \min(0.7 \cdot 58.500, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$= \min(f_y \cdot k_y, 1.5 \cdot S_n(A_2 + A_3), \pi/4 \cdot P \cdot R_n^2 \cdot k_y^2)$$

$$= \min(51007 \cdot 2.48, 1.5 \cdot 127.7(74.079 + 0.0), \pi/4 \cdot 192.4 \cdot 39.6^2 \cdot 2.48^2)$$

$$= 14831.045 \text{ Kg}$$

Weld Stress Value [τ]:

$$= fwelds / (L_t(0.49 \cdot L41T + 0.6 \cdot t_{w1} + 0.49 \cdot L43T))$$

$$= 14831.045 / (154.095(0.49 \cdot 7.071 + 0.6 \cdot 78.5 + 0.49 \cdot 0.0))$$

$$= 18.667 < \text{or} = \text{to } 161.141 \text{ Weld Stress Passed}$$

Nozzle Calculations per Section 4.5: External Pressure Case:

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n/R_n]$$

$$= 58.5 / \ln[1 + 58.5/39.6]$$

$$= 64.487 \text{ mm.}$$

Shell Radius for Force Calculation [Rxs]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$

$$= 78.5 / \ln[1 + 78.5/1274.0]$$

$$= 1312.860 \text{ mm.}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S$$

$$= 1.5 \cdot 161.141$$

$$= 241.7 \text{ N./mm}^2$$



Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot LH$$

$$= 1.034 \cdot 64.487 \cdot 126.631$$

$$= 86.1 \text{ Kgf}$$

Force acting on the Shell [fS]:

$$= P \cdot R_{xs} \cdot (LR + t_n)/2$$

$$= 1.034 \cdot 1312.86 \cdot (79.2 + 58.5)/2$$

$$= 953.3 \text{ Kgf}$$

Discontinuity Force from External Pressure [fY]:

$$= (P \cdot R_{xs} \cdot R_{nc})/2$$

$$= (1.034 \cdot 1312.86 \cdot 39.6)/2$$

$$= 274.1 \text{ Kgf}$$

Area Resisting External Pressure [Ap]:

$$= (fS + fY + fN)/P$$

$$= (953.3 + 274.1 + 86.1)/1.034$$

$$= 1245.5 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= S_{allow} / (2 \cdot A_p / AT - R_{xs} / (2 \cdot t_{eff}))$$

$$= 241.712 / (2 \cdot 1245.511 / 121.378 - 1312.86 / (2 \cdot 78.5))$$

$$= 198.8 \text{ bars}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= 2 \cdot S[t/R_{xs}]$$

$$= 2 \cdot 161.141[78.5/1312.86]$$

$$= 192.7 \text{ bars}$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(P_{max1}, P_{max2})$$

$$= \min(198.753, 192.691)$$

$$= 192.691 \text{ bars}$$

Average Primary Membrane Stress [σAvg]:

$$= (fN + fS + fY) / AT$$

$$= (86.121 + 953.277 + 274.145) / 121.378$$

$$= 1.061 \text{ N./mm}^2$$

General Primary Membrane Stress [σCirc]:

$$= P \cdot R_{xs} / (2 \cdot t_{eff})$$

$$= 1.034 \cdot 1312.86 / (2 \cdot 78.5)$$

$$= 0.9 \text{ N./mm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$$

$$= \max(2 \cdot 1.061 - 0.865, 0.865)$$

$$= 1.3 \text{ N./mm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	241.71	N./mm ²
Local Primary Membrane Stress	PL	1.26	N./mm ²
Maximum Allowable External Pressure	Pmax	192.69	bars

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -29 °C

The Drop for this Nozzle is : 3.7871 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 283.7870 mm.

Input Echo, WRC107/537 Item 1, Description: D13" :



Diameter Basis for Vessel	Vbasis	ID	
Cylindrical or Spherical Vessel	Cylsph	Spherical	
Internal Corrosion Allowance	Cas	1.5000	mm.
Vessel Diameter	Dv	2545.000	mm.
Vessel Thickness	Tv	80.000	mm.
Design Temperature	T1	85.0	C
Vessel Material		SA-516 70	
Vessel UNS Number		K02700	
Vessel Cold S.I. Allowable	Smc	175.00	N./mm ²
Vessel Hot S.I. Allowable	Smh	161.14	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Attachment Type	Type	Round	
WRC107 Attachment Classification	Holsol	Hollow	
Diameter Basis for Nozzle	Nbasis	ID	
Corrosion Allowance for Nozzle	Can	1.5000	mm.
Nozzle Diameter	Dn	76.200	mm.
Nozzle Thickness	Tn	60.000	mm.
Nozzle Material		SA-350 LF1	
Nozzle UNS Number		K03009	
Nozzle Cold S.I. Allowable	SNmc	138.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh	127.70	N./mm ²
Design Internal Pressure	Dp	192.419	bars
Include Pressure Thrust		No	

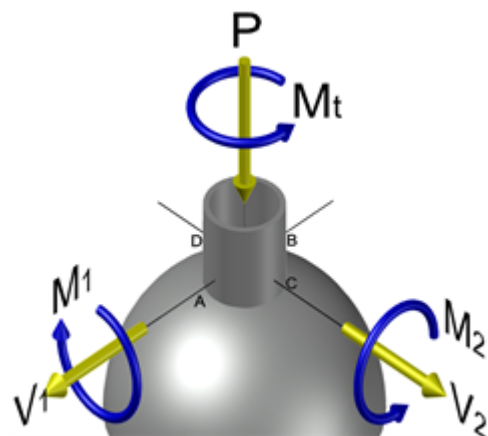
External Forces and Moments in WRC 107/537 Convention:

Radial Load (SUS)	P	-1835.6	Kgf
Longitudinal Shear (SUS)	(V1) V1	1622.5	Kgf
Circumferential Shear (SUS)	(Vc) V2	1622.5	Kgf
Circumferential Moment (SUS)	(Mc) M1	319.2	Kg-m.
Longitudinal Moment (SUS)	(Ml) M2	319.2	Kg-m.
Torsional Moment (SUS)	Mt	412.9	Kg-m.

Use Interactive Control No
 WRC107 Version Version March 1979

Include Pressure Stress Indices per Div. 2 No
 Compute Pressure Stress per WRC-368 No
 Local Loads applied at end of Nozzle/Attachment No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 196.2 + 2 \cdot 1.65 \cdot \sqrt{(1313.25 (80.0 - 1.5))}$$

$$= 1255.752 \text{ mm.}$$

WRC 107 Stress Calculation for SUSTained loads:

Radial Load	P	-1835.6	Kgf
Circumferential Shear	(VC) V2	1622.5	Kgf
Longitudinal Shear	(VL) V1	1622.5	Kgf
Circumferential Moment	(MC) M1	319.2	Kg-m.
Longitudinal Moment	(ML) M2	319.2	Kg-m.
Torsional Moment	MT	412.9	Kg-m.

Dimensionless Param: U = 0.31 TAU = 5.00 (1.18) RHO = 1.34

Dimensionless Loads for Spherical Shells at Attachment Junction:

Curves read for 1979	Figure	Value	Location
N(x) * T / P	SP 2	0.10420	(A,B,C,D)
M(x) / P	SP 2	0.11488	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / MC	SM 2	0.24851	(A,B,C,D)
M(x) * SQRT(Rm * T) / MC	SM 2	0.37073	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / ML	SM 2	0.24851	(A,B,C,D)
M(x) * SQRT(Rm * T) / ML	SM 2	0.37073	(A,B,C,D)
N(y) * T / P	SP 2	0.19165	(A,B,C,D)
M(y) / P	SP 2	0.06117	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / MC	SM 2	0.04803	(A,B,C,D)
M(y) * SQRT(Rm * T) / MC	SM 2	0.25283	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / ML	SM 2	0.04803	(A,B,C,D)
M(y) * SQRT(Rm * T) / ML	SM 2	0.25283	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (N./mm²)

Type of Stress Load		Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Memb. P		0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.3
Rad. Bend. P		2.0	-2.0	2.0	-2.0	2.0	-2.0	2.0	-2.0
Rad. Memb. MC		0.0	0.0	0.0	0.0	-0.4	-0.4	0.4	0.4
Rad. Bend. MC		0.0	0.0	0.0	0.0	-3.5	3.5	3.5	-3.5
Rad. Memb. ML		-0.4	-0.4	0.4	0.4	0.0	0.0	0.0	0.0
Rad. Bend. ML		-3.5	3.5	3.5	-3.5	0.0	0.0	0.0	0.0
Tot. Rad. Str.		-1.6	1.4	6.2	-4.8	-1.6	1.4	6.2	-4.8
Tang. Memb. P		0.6	0.6	0.6	0.6	0.6	0.6	0.6	0.6
Tang. Bend. P		1.1	-1.1	1.1	-1.1	1.1	-1.1	1.1	-1.1
Tang. Memb. MC		0.0	0.0	0.0	0.0	-0.1	-0.1	0.1	0.1
Tang. Bend. MC		0.0	0.0	0.0	0.0	-2.4	2.4	2.4	-2.4
Tang. Memb. ML		-0.1	-0.1	0.1	0.1	0.0	0.0	0.0	0.0
Tang. Bend. ML		-2.4	2.4	2.4	-2.4	0.0	0.0	0.0	0.0
Tot. Tang. Str.		-0.8	1.8	4.1	-2.8	-0.8	1.8	4.1	-2.8
Shear VC		0.7	0.7	-0.7	-0.7	0.0	0.0	0.0	0.0
Shear VL		0.0	0.0	0.0	0.0	-0.7	-0.7	0.7	0.7
Shear MT		0.9	0.9	0.9	0.9	0.9	0.9	0.9	0.9
Tot. Shear		1.5	1.5	0.2	0.2	0.2	0.2	1.5	1.5
Str. Int.		3.1	3.1	6.2	4.9	1.6	1.9	7.0	5.6



WRC 107/537 Stress Summations:

Vessel Stress Summation at Attachment Junction (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Pm (SUS)	160.9	160.9	160.9	160.9	160.9	160.9	160.9	160.9
Rad.	Pl (SUS)	-0.1	-0.1	0.7	0.7	-0.1	-0.1	0.7	0.7
Rad.	Q (SUS)	-1.5	1.5	5.5	-5.5	-1.5	1.5	5.5	-5.5
Long.	Pm (SUS)	160.9	160.9	160.9	160.9	160.9	160.9	160.9	160.9
Long.	Pl (SUS)	0.5	0.5	0.6	0.6	0.5	0.5	0.6	0.6
Long.	Q (SUS)	-1.3	1.3	3.5	-3.5	-1.3	1.3	3.5	-3.5
Shear	Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear	Pl (SUS)	0.7	0.7	-0.7	-0.7	-0.7	-0.7	0.7	0.7
Shear	Q (SUS)	0.9	0.9	0.9	0.9	0.9	0.9	0.9	0.9
Pm (SUS)		160.9	160.9	160.9	160.9	160.9	160.9	160.9	160.9
Pm+Pl (SUS)		161.8	161.8	162.2	162.2	161.8	161.8	162.2	162.2
Pm+Pl+Q (Total)		161.3	164.1	167.2	158.1	160.1	162.8	167.9	158.9

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	160.91	161.14	Passed
Pm+Pl (SUS)	162.24	241.71	Passed
Pm+Pl+Q (TOTAL)	167.93	484.02	Passed

The Pm+Pl+Q allowable above was based on the rules of ASME VIII-2 Part 4, 4.1.6.3 (a)&(b) or per ASME VIII-2, 5.5.6.1(1)&(2) Part 5 as applicable.

The Pm+Pl+Q allowable was based on a temperature range cycling from ambient to design temperature.

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Input, Nozzle Desc: L22"

From: 30

Pressure for Reinforcement Calculations	P	192.365	bars
Temperature for Internal Pressure	Temp	85	°C
Design External Pressure	Pext	1.03	bars
Temperature for External Pressure	Tempex	85	°C
Parent Material		SA-516 70	+N+IT
Parent Allowable Stress at Temperature	Sv	161.14	N./mm ²
Parent Allowable Stress At Ambient	Sva	175.00	N./mm ²
Inside Diameter of Hemispherical Head	D	2545.00	mm.
Head Finished (Minimum) Thickness	t	80.0000	mm.
Head Internal Corrosion Allowance	cas	1.5000	mm.
Head External Corrosion Allowance	co	0.0000	mm.
Distance from Head Centerline	L1	1200.0000	mm.
User Entered Minimum Design Metal Temperature		-40.00	°C

Type of Element Connected to the Parent : Nozzle

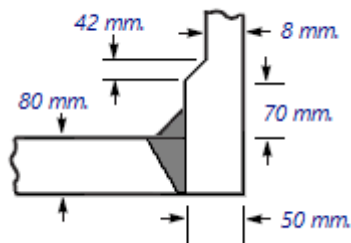
Material		SA-350 LF1	+IT
Material UNS Number		K03009	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	127.70	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	8.0000	mm.
Flange Material		SA-350 LF1	+IT
Hub Height of Integral Nozzle	h	70.0000	mm.
Height of Beveled Transition	L'	42.0000	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		50.0000	mm.
Corrosion Allowance	can	1.5000	mm.
Outside Projection	ho	200.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel Wgnv		80.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.
Flange Class		1500	
Flange Grade		GR 1.4	

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: L22"

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	2.000	in.
Actual Thickness Used in Calculation	0.315	in.



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 53.8 \cdot (\exp(192.365/(127.7 \cdot 1.0)) - 1) + 1.5 + 0.0$$

$$= 4.3736 + 1.5 + 0.0 = 5.8736 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 127.7 \cdot 1.0 \cdot \ln((2 \cdot 6.5 + 53.8) / 53.8) - 0.0$$

$$= 276.365 - 0.0 = 276.3645 \text{ bars}$$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c :	6.020 mm
Min. required shell thickness, internal p + c	79.887 mm.
Min. required shell thickness, external p + c	6.561 mm.

Nozzle neck minimum required thickness	: 6.020 mm.
--	-------------

Minimum nozzle neck thickness 8.001 mm., is greater than required thickness 6.020 mm.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[127.7/161.1, 1]$$

$$= 0.792$$

Thickness of Nozzle at Shell [tn]:

$$= \text{hub thickness} - \text{corrosion allowance}$$

$$= 50.0 - 1.5$$

$$= 48.500 \text{ mm.}$$

Thickness of Nozzle at Top [tn2]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 8.0 - 1.5$$

$$= 6.500 \text{ mm.}$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 2545.0/2 + 1.5$$

$$= 1274.000 \text{ mm.}$$

Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$$

$$= \min(\sqrt{1274.0 \cdot 78.5}, 2 \cdot 26.9)$$

$$= 53.800 \text{ mm.}$$

Intermediate value [Cn]:

$$= \min(((t + t_e)/t_n)^{0.35}, 1)$$

$$= \min(((78.5 + 0.0)/48.5)^{0.35}, 1)$$

$$= \min(1.184, 1)$$

$$= 1.000$$

Intermediate value [Cp]:

= 9999999.000

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$= \min(t + t_e + F_p \cdot \sqrt{(R_n \cdot t_n)}, L_{pr1} + t)$$

$$= \min(78.5 + 0.0 + 1.0 \cdot \sqrt{(26.9 \cdot 48.5)}, 200.0 + 78.5)$$

$$= 114.620 \text{ mm.}$$

Effective Vessel Thickness [teff]:

$$= t + ((A_5 \cdot f_{rp}) / LR)$$

$$= 78.5 + ((0.0 \cdot 1.0) / 53.8)$$

$$= 78.500 \text{ mm.}$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR$$

$$= 78.5 \cdot 53.8$$

$$= 42.233 \text{ cm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= t_n \cdot LH$$

$$= 48.5 \cdot 114.62$$

$$= 55.591 \text{ cm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot \text{Leg}^2$$

$$= 0.5 \cdot 10.0^2$$

$$= 0.500 \text{ cm}^2$$

The total area contributed by A1 through A43 [AT]:

$$= A_1 + f_{rn}(A_2 + A_3) + A_{41} + A_{42} + A_{43}$$

$$= 42.233 + 0.792(55.591 + 0.0) + 0.5 + 0.0 + 0.0$$

$$= 86.787 \text{ cm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n/R_n]$$

$$= 48.5 / \ln[1 + 48.5/26.9]$$

$$= 47.056 \text{ mm.}$$

Shell Radius for Force Calculation [Rxs]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$

$$= 78.5 / \ln[1 + 78.5/1274.0]$$

$$= 1312.860 \text{ mm.}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$

$$= 1.5 \cdot 161.141 \cdot 1.0$$

$$= 241.7 \text{ N./mm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot LH$$

$$= 192.365 \cdot 47.056 \cdot 114.62$$

$$= 10580.3 \text{ Kgf}$$

Force acting on the Shell [fS]:

$$= P \cdot R_{xs} \cdot (LR + t_n)/2$$

$$= 192.365 \cdot 1312.86 \cdot (53.8 + 48.5)/2$$

$$= 131730.3 \text{ Kgf}$$

Discontinuity Force from Internal Pressure [fY]:

$$= (P \cdot R_{xs} \cdot R_{nc})/2$$

$$= (192.365 \cdot 1312.86 \cdot 26.9)/2$$

$$= 34638.8 \text{ Kgf}$$

Area Resisting Internal Pressure [Ap]:

$$= (f_S + f_Y + f_N)/P$$

$$= (131730.3 + 34638.8 + 10580.3)/192.365$$

$$= 902.0 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:



$$= \text{Sallow} / (2 \cdot A_p / A_T - R_{xs} / (2 \cdot t_{eff}))$$

$$= 241.712 / (2 \cdot 902.043 / 86.787 - 1312.86 / (2 \cdot 78.5))$$

$$= 194.5 \text{ bars}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= 2 \cdot S[t/R_{xs}]$$

$$= 2 \cdot 161.141[78.5/1312.86]$$

$$= 192.7 \text{ bars}$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(P_{max1}, P_{max2})$$

$$= \min(194.52, 192.691)$$

$$= 192.691 \text{ bars}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$= (f_N + f_S + f_Y) / A_T$$

$$= (10580.331 + 131730.312 + 34638.758) / 86.787$$

$$= 199.950 \text{ N./mm}^2$$

General Primary Membrane Stress [σ_{Circ}]:

$$= P \cdot R_{xs} / (2 \cdot t_{eff})$$

$$= 192.365 \cdot 1312.86 / (2 \cdot 78.5)$$

$$= 160.9 \text{ N./mm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$$

$$= \max(2 \cdot 199.95 - 160.868, 160.868)$$

$$= 239.0 \text{ N./mm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	241.71	N./mm ²
Local Primary Membrane Stress	PL	239.03	N./mm ²
Maximum Allowable Working Pressure	Pmax	192.69	bars

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= (R_{nc} + t_n) / R_{nc}$$

$$= (26.9 + 48.5) / 26.9$$

$$= 2.803 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld for non-radial Nozzles [LT]:

$$= \pi/2 \cdot \sqrt{((R_{nc} + t_n)^2 + (R_n + t_n)^2) / 2}$$

$$= \pi/2 \cdot \sqrt{((26.9 + 48.5)^2 + (26.9 + 48.5)^2) / 2}$$

$$= 118.438 \text{ mm.}$$

Weld Throat Dimensions, (0.7071 • Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm.}$$

Required Weld Throat Dimensions :

$$t_c, L41T = 7.071, \text{ must be } \geq \min(0.7 \cdot 48.500, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$= \min(f_y \cdot k_y, 1.5 \cdot S_n(A_2 + A_3), \pi/4 \cdot P \cdot R_n^2 \cdot k_y^2)$$

$$= \min(34639 \cdot 2.8, 1.5 \cdot 127.7(55.591 + 0.0), \pi/4 \cdot 192.4 \cdot 26.9^2 \cdot 2.8^2)$$

$$= 8758.993 \text{ Kg}$$

Weld Stress Value [τ]:

$$= fwelds / (L_T(0.49 \cdot L41T + 0.6 \cdot t_{w1} + 0.49 \cdot L43T))$$

$$= 8758.993 / (118.438(0.49 \cdot 7.071 + 0.6 \cdot 78.5 + 0.49 \cdot 0.0))$$

$$= 14.343 < \text{or} = \text{to } 161.141 \text{ Weld Stress Passed}$$

Nozzle Calculations per Section 4.5: External Pressure Case:

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n/R_n]$$

$$= 48.5 / \ln[1 + 48.5/26.9]$$

$$= 47.056 \text{ mm.}$$

Shell Radius for Force Calculation [Rxs]:



$$\begin{aligned}
 &= t_{eff} / \ln[1 + t_{eff}/R_{eff}] \\
 &= 78.5 / \ln[1 + 78.5/1274.0] \\
 &= 1312.860 \text{ mm.}
 \end{aligned}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$\begin{aligned}
 &= 1.5 \cdot S \\
 &= 1.5 \cdot 161.141 \\
 &= 241.7 \text{ N./mm}^2
 \end{aligned}$$

Force acting on the Nozzle [fN]:

$$\begin{aligned}
 &= P \cdot R_{xn} \cdot L_H \\
 &= 1.034 \cdot 47.056 \cdot 114.62 \\
 &= 56.9 \text{ Kgf}
 \end{aligned}$$

Force acting on the Shell [fS]:

$$\begin{aligned}
 &= P \cdot R_{xs} \cdot (L_R + t_n) / 2 \\
 &= 1.034 \cdot 1312.86 \cdot (53.8 + 48.5) / 2 \\
 &= 708.2 \text{ Kgf}
 \end{aligned}$$

Discontinuity Force from External Pressure [fY]:

$$\begin{aligned}
 &= (P \cdot R_{xs} \cdot R_{nc}) / 2 \\
 &= (1.034 \cdot 1312.86 \cdot 26.9) / 2 \\
 &= 186.2 \text{ Kgf}
 \end{aligned}$$

Area Resisting External Pressure [Ap]:

$$\begin{aligned}
 &= (f_S + f_Y + f_N) / P \\
 &= (708.2 + 186.2 + 56.9) / 1.034 \\
 &= 902.0 \text{ cm}^2
 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$\begin{aligned}
 &= S_{allow} / (2 \cdot A_p / A_T - R_{xs} / (2 \cdot t_{eff})) \\
 &= 241.712 / (2 \cdot 902.044 / 86.787 - 1312.86 / (2 \cdot 78.5)) \\
 &= 194.5 \text{ bars}
 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$\begin{aligned}
 &= 2 \cdot S / [t / R_{xs}] \\
 &= 2 \cdot 161.141 [78.5 / 1312.86] \\
 &= 192.7 \text{ bars}
 \end{aligned}$$

Maximum Allowable Working Pressure [Pmax]:

$$\begin{aligned}
 &= \min(P_{max1}, P_{max2}) \\
 &= \min(194.52, 192.691) \\
 &= 192.691 \text{ bars}
 \end{aligned}$$

Average Primary Membrane Stress [σAvg]:

$$\begin{aligned}
 &= (f_N + f_S + f_Y) / A_T \\
 &= (56.882 + 708.208 + 186.225) / 86.787 \\
 &= 1.075 \text{ N./mm}^2
 \end{aligned}$$

General Primary Membrane Stress [σCirc]:

$$\begin{aligned}
 &= P \cdot R_{xs} / (2 \cdot t_{eff}) \\
 &= 1.034 \cdot 1312.86 / (2 \cdot 78.5) \\
 &= 0.9 \text{ N./mm}^2
 \end{aligned}$$

Maximum Local Primary Membrane Stress [PL]:

$$\begin{aligned}
 &= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ}) \\
 &= \max(2 \cdot 1.075 - 0.865, 0.865) \\
 &= 1.3 \text{ N./mm}^2
 \end{aligned}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	241.71	N./mm ²
Local Primary Membrane Stress	PL	1.29	N./mm ²
Maximum Allowable External Pressure	Pmax	192.69	bars

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.



Impact Test Temperature provided per Specification -29 °C

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -29 °C

Flange MDMT including temperature reduction per 3.11:

Flange MDMT with Temp reduction -34 °C

The Drop for this Nozzle is : 2.2358 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 282.2358 mm.

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Input, Nozzle Desc: L42"

From: 30

Pressure for Reinforcement Calculations	P	192.365	bars
Temperature for Internal Pressure	Temp	85	°C
Design External Pressure	Pext	1.03	bars
Temperature for External Pressure	Tempex	85	°C
Parent Material		SA-516 70	+N+IT
Parent Allowable Stress at Temperature	Sv	161.14	N./mm ²
Parent Allowable Stress At Ambient	Sva	175.00	N./mm ²
Inside Diameter of Hemispherical Head	D	2545.00	mm.
Head Finished (Minimum) Thickness	t	80.0000	mm.
Head Internal Corrosion Allowance	cas	1.5000	mm.
Head External Corrosion Allowance	co	0.0000	mm.
Distance from Head Centerline	L1	1200.0000	mm.
User Entered Minimum Design Metal Temperature		-40.00	°C

Type of Element Connected to the Parent : Nozzle

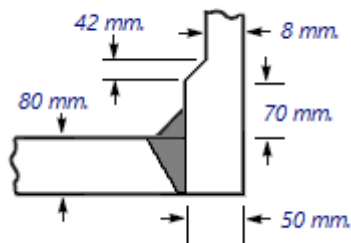
Material		SA-350 LF1	+IT
Material UNS Number		K03009	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	127.71	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	8.0000	mm.
Flange Material		SA-350 LF1	+IT
Hub Height of Integral Nozzle	h	70.0000	mm.
Height of Beveled Transition	L'	42.0000	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		50.0000	mm.
Corrosion Allowance	can	1.5000	mm.
Outside Projection	ho	200.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel Wgnv		80.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.
Flange Class		1500	
Flange Grade		GR 1.4	

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: L42"

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	2.000	in.
Actual Thickness Used in Calculation	0.315	in.



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 53.8 \cdot (\exp(192.365/(127.713 \cdot 1.0)) - 1) + 1.5 + 0.0$$

$$= 4.3731 + 1.5 + 0.0 = 5.8731 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 127.713 \cdot 1.0 \cdot \ln((2 \cdot 6.5 + 53.8) / 53.8) - 0.0$$

$$= 276.392 - 0.0 = 276.3925 \text{ bars}$$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c :	6.020 mm
Min. required shell thickness, internal p + c	79.887 mm.
Min. required shell thickness, external p + c	6.561 mm.

Nozzle neck minimum required thickness	: 6.020 mm.
--	-------------

Minimum nozzle neck thickness 8.001 mm., is greater than required thickness 6.020 mm.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[127.7/161.1, 1]$$

$$= 0.793$$

Thickness of Nozzle at Shell [tn]:

$$= \text{hub thickness} - \text{corrosion allowance}$$

$$= 50.0 - 1.5$$

$$= 48.500 \text{ mm.}$$

Thickness of Nozzle at Top [tn2]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 8.0 - 1.5$$

$$= 6.500 \text{ mm.}$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 2545.0/2 + 1.5$$

$$= 1274.000 \text{ mm.}$$

Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$$

$$= \min(\sqrt{1274.0 \cdot 78.5}, 2 \cdot 26.9)$$

$$= 53.800 \text{ mm.}$$

Intermediate value [Cn]:

$$= \min(((t + t_e)/t_n)^{0.35}, 1)$$

$$= \min(((78.5 + 0.0)/48.5)^{0.35}, 1)$$

$$= \min(1.184, 1)$$

$$= 1.000$$

Intermediate value [Cp]:

= 9999999.000

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$= \min(t + t_e + F_p \cdot \sqrt{(R_n \cdot t_n)}, L_{pr1} + t)$$

$$= \min(78.5 + 0.0 + 1.0 \cdot \sqrt{(26.9 \cdot 48.5)}, 200.0 + 78.5)$$

$$= 114.620 \text{ mm.}$$

Effective Vessel Thickness [teff]:

$$= t + ((A_5 \cdot f_{rp}) / LR)$$

$$= 78.5 + ((0.0 \cdot 1.0) / 53.8)$$

$$= 78.500 \text{ mm.}$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR$$

$$= 78.5 \cdot 53.8$$

$$= 42.233 \text{ cm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= t_n \cdot LH$$

$$= 48.5 \cdot 114.62$$

$$= 55.591 \text{ cm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot \text{Leg}^2$$

$$= 0.5 \cdot 10.0^2$$

$$= 0.500 \text{ cm}^2$$

The total area contributed by A1 through A43 [AT]:

$$= A_1 + f_{rn}(A_2 + A_3) + A_{41} + A_{42} + A_{43}$$

$$= 42.233 + 0.793(55.591 + 0.0) + 0.5 + 0.0 + 0.0$$

$$= 86.792 \text{ cm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n/R_n]$$

$$= 48.5 / \ln[1 + 48.5/26.9]$$

$$= 47.056 \text{ mm.}$$

Shell Radius for Force Calculation [Rxs]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$

$$= 78.5 / \ln[1 + 78.5/1274.0]$$

$$= 1312.860 \text{ mm.}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$

$$= 1.5 \cdot 161.141 \cdot 1.0$$

$$= 241.7 \text{ N./mm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot LH$$

$$= 192.365 \cdot 47.056 \cdot 114.62$$

$$= 10580.3 \text{ Kg}$$

Force acting on the Shell [fS]:

$$= P \cdot R_{xs} \cdot (LR + t_n)/2$$

$$= 192.365 \cdot 1312.86 \cdot (53.8 + 48.5)/2$$

$$= 131730.3 \text{ Kg}$$

Discontinuity Force from Internal Pressure [fY]:

$$= (P \cdot R_{xs} \cdot R_{nc})/2$$

$$= (192.365 \cdot 1312.86 \cdot 26.9)/2$$

$$= 34638.8 \text{ Kg}$$

Area Resisting Internal Pressure [Ap]:

$$= (f_S + f_Y + f_N)/P$$

$$= (131730.3 + 34638.8 + 10580.3)/192.365$$

$$= 902.0 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:



$$= \text{Sallow} / (2 \cdot A_p / A_T - R_{xs} / (2 \cdot t_{eff}))$$

$$= 241.712 / (2 \cdot 902.043 / 86.792 - 1312.86 / (2 \cdot 78.5))$$

$$= 194.5 \text{ bars}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= 2 \cdot S[t/R_{xs}]$$

$$= 2 \cdot 161.141[78.5/1312.86]$$

$$= 192.7 \text{ bars}$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(P_{max1}, P_{max2})$$

$$= \min(194.537, 192.691)$$

$$= 192.691 \text{ bars}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$= (f_N + f_S + f_Y) / A_T$$

$$= (10580.331 + 131730.312 + 34638.758) / 86.792$$

$$= 199.940 \text{ N./mm}^2$$

General Primary Membrane Stress [σ_{Circ}]:

$$= P \cdot R_{xs} / (2 \cdot t_{eff})$$

$$= 192.365 \cdot 1312.86 / (2 \cdot 78.5)$$

$$= 160.9 \text{ N./mm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$$

$$= \max(2 \cdot 199.94 - 160.868, 160.868)$$

$$= 239.0 \text{ N./mm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	241.71	N./mm ²
Local Primary Membrane Stress	PL	239.01	N./mm ²
Maximum Allowable Working Pressure	Pmax	192.69	bars

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= (R_{nc} + t_n) / R_{nc}$$

$$= (26.9 + 48.5) / 26.9$$

$$= 2.803 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld for non-radial Nozzles [LT]:

$$= \pi/2 \cdot \sqrt{((R_{nc} + t_n)^2 + (R_n + t_n)^2) / 2}$$

$$= \pi/2 \cdot \sqrt{((26.9 + 48.5)^2 + (26.9 + 48.5)^2) / 2}$$

$$= 118.438 \text{ mm.}$$

Weld Throat Dimensions, (0.7071 • Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm.}$$

Required Weld Throat Dimensions :

$$t_c, L41T = 7.071, \text{ must be } \geq \min(0.7 \cdot 48.500, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$= \min(f_y \cdot k_y, 1.5 \cdot S_n(A_2 + A_3), \pi/4 \cdot P \cdot R_n^2 \cdot k_y^2)$$

$$= \min(34639 \cdot 2.8, 1.5 \cdot 127.7(55.591 + 0.0), \pi/4 \cdot 192.4 \cdot 26.9^2 \cdot 2.8^2)$$

$$= 8758.993 \text{ Kg}$$

Weld Stress Value [τ]:

$$= fwelds / (L_T(0.49 \cdot L41T + 0.6 \cdot t_{w1} + 0.49 \cdot L43T))$$

$$= 8758.993 / (118.438(0.49 \cdot 7.071 + 0.6 \cdot 78.5 + 0.49 \cdot 0.0))$$

$$= 14.343 < \text{or} = \text{to } 161.141 \text{ Weld Stress Passed}$$

Nozzle Calculations per Section 4.5: External Pressure Case:

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n/R_n]$$

$$= 48.5 / \ln[1 + 48.5/26.9]$$

$$= 47.056 \text{ mm.}$$

Shell Radius for Force Calculation [Rxs]:



$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$

$$= 78.5 / \ln[1 + 78.5/1274.0]$$

$$= 1312.860 \text{ mm.}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S$$

$$= 1.5 \cdot 161.141$$

$$= 241.7 \text{ N./mm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot L_H$$

$$= 1.034 \cdot 47.056 \cdot 114.62$$

$$= 56.9 \text{ Kg}$$

Force acting on the Shell [fS]:

$$= P \cdot R_{xs} \cdot (L_R + t_n)/2$$

$$= 1.034 \cdot 1312.86 \cdot (53.8 + 48.5)/2$$

$$= 708.2 \text{ Kg}$$

Discontinuity Force from External Pressure [fY]:

$$= (P \cdot R_{xs} \cdot R_{nc})/2$$

$$= (1.034 \cdot 1312.86 \cdot 26.9)/2$$

$$= 186.2 \text{ Kg}$$

Area Resisting External Pressure [Ap]:

$$= (f_S + f_Y + f_N)/P$$

$$= (708.2 + 186.2 + 56.9)/1.034$$

$$= 902.0 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= S_{allow} / (2 \cdot A_p / A_T - R_{xs} / (2 \cdot t_{eff}))$$

$$= 241.712 / (2 \cdot 902.044 / 86.792 - 1312.86 / (2 \cdot 78.5))$$

$$= 194.5 \text{ bars}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= 2 \cdot S / [t / R_{xs}]$$

$$= 2 \cdot 161.141 [78.5 / 1312.86]$$

$$= 192.7 \text{ bars}$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(P_{max1}, P_{max2})$$

$$= \min(194.537, 192.691)$$

$$= 192.691 \text{ bars}$$

Average Primary Membrane Stress [σAvg]:

$$= (f_N + f_S + f_Y) / A_T$$

$$= (56.882 + 708.208 + 186.225) / 86.792$$

$$= 1.075 \text{ N./mm}^2$$

General Primary Membrane Stress [σCirc]:

$$= P \cdot R_{xs} / (2 \cdot t_{eff})$$

$$= 1.034 \cdot 1312.86 / (2 \cdot 78.5)$$

$$= 0.9 \text{ N./mm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$$

$$= \max(2 \cdot 1.075 - 0.865, 0.865)$$

$$= 1.3 \text{ N./mm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	241.71	N./mm ²
Local Primary Membrane Stress	PL	1.28	N./mm ²
Maximum Allowable External Pressure	Pmax	192.69	bars

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -29 °C

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -29 °C

Flange MDMT including temperature reduction per 3.11:

Flange MDMT with Temp reduction -34 °C

The Drop for this Nozzle is : 2.2358 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 282.2358 mm.

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Input, Nozzle Desc: B26"

From: 30

Pressure for Reinforcement Calculations	P	192.405	bars
Temperature for Internal Pressure	Temp	85	°C
Design External Pressure	Pext	1.03	bars
Temperature for External Pressure	Tempex	85	°C
Parent Material		SA-516 70	+N+IT
Parent Allowable Stress at Temperature	Sv	161.14	N./mm ²
Parent Allowable Stress At Ambient	Sva	175.00	N./mm ²
Inside Diameter of Hemispherical Head	D	2545.00	mm.
Head Finished (Minimum) Thickness	t	80.0000	mm.
Head Internal Corrosion Allowance	cas	1.5000	mm.
Head External Corrosion Allowance	co	0.0000	mm.
Distance from Head Centerline	L1	700.0000	mm.
User Entered Minimum Design Metal Temperature		-40.00	°C

Type of Element Connected to the Parent : Nozzle

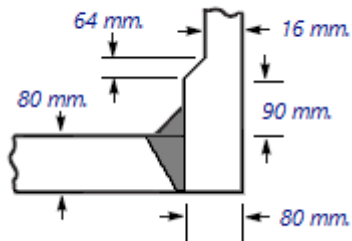
Material		SA-350 LF1	+IT
Material UNS Number		K03009	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	127.70	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		6.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.0000	mm.
Hub Height of Integral Nozzle	h	90.0000	mm.
Height of Beveled Transition	L'	64.0000	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		80.0000	mm.
Corrosion Allowance	can	1.5000	mm.
Outside Projection	ho	300.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	80.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
User Defined Nozzle/Shell Centerline Angle		45.0000	deg.
Piping Exit Angle [North Clockwise reference]		180.0	deg
Bend Radius Multiplier		1.50	
Horizontal Run Length		1650.0	mm.
Centerline Distance to Tangent		2000.0	mm.

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: B26"

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	6.000	in.
Actual Thickness Used in Calculation	0.630	in.



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 155.4 \cdot (\exp(192.405/(127.7 \cdot 1.0)) - 1) + 1.5 + 0.0$$

$$= 12.6358 + 1.5 + 0.0 = 14.1358 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 127.7 \cdot 1.0 \cdot \ln((2 \cdot 14.5 + 155.4) / 155.4) - 0.0$$

$$= 218.488 - 0.0 = 218.4883 \text{ bars}$$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c : 8.660 mm
 Min. required shell thickness, internal p + c : 79.904 mm.
 Min. required shell thickness, external p + c : 6.561 mm.

Nozzle neck minimum required thickness : 8.660 mm.

Minimum nozzle neck thickness 16.002 mm., is greater than required thickness 8.660 mm.

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 100.5,	Allowable	: 127.7 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 231.6 N./mm ²	Passed
Occasional	: 47.2,	Allowable	: 169.8 N./mm ²	Passed
Shear	: 42.2,	Allowable	: 89.4 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[127.7/161.1, 1]$$

$$= 0.792$$

Thickness of Nozzle at Shell [tn]:

$$= \text{hub thickness} - \text{corrosion allowance}$$

$$= 80.0 - 1.5$$

$$= 78.500 \text{ mm.}$$

Thickness of Nozzle at Top [tn2]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 16.0 - 1.5$$

$$= 14.500 \text{ mm.}$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 2545.0/2 + 1.5$$

$$= 1274.000 \text{ mm.}$$

Effective Length of Vessel Wall [LR]:

$$\begin{aligned}
 &= \min(\sqrt{(\text{Reff} \cdot t)}, 2 \cdot R_n) \\
 &= \min(\sqrt{(1274.0 \cdot 78.5)}, 2 \cdot 77.7) \\
 &= 155.400 \text{ mm.}
 \end{aligned}$$

Intermediate value [Cn]:

$$\begin{aligned}
 &= \min((t + t_e)/t_n)^{0.35}, 1) \\
 &= \min((78.5 + 0.0)/78.5)^{0.35}, 1) \\
 &= \min(1.0, 1) \\
 &= 1.000
 \end{aligned}$$

Intermediate value [Cp]:

$$= 9999999.000$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$\begin{aligned}
 &= \min(t + t_e + F_p \cdot \sqrt{(R_n \cdot t_n)}, L_{pr1} + t) \\
 &= \min(78.5 + 0.0 + 1.0 \cdot \sqrt{(77.7 \cdot 78.5)}, 300.0 + 78.5) \\
 &= 156.599 \text{ mm.}
 \end{aligned}$$

Effective Vessel Thickness [teff]:

$$\begin{aligned}
 &= t + ((A_5 \cdot \text{frp}) / LR) \\
 &= 78.5 + ((0.0 \cdot 1.0) / 155.4) \\
 &= 78.500 \text{ mm.}
 \end{aligned}$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$\begin{aligned}
 &= t \cdot LR \\
 &= 78.5 \cdot 155.4 \\
 &= 121.989 \text{ cm}^2
 \end{aligned}$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$\begin{aligned}
 &= t_n \cdot LH \\
 &= 78.5 \cdot 156.599 \\
 &= 122.930 \text{ cm}^2
 \end{aligned}$$

Area Contributed by the Outside Fillet Weld [A41]:

$$\begin{aligned}
 &= 0.5 \cdot \text{Leg}^2 \\
 &= 0.5 \cdot 10.0^2 \\
 &= 0.500 \text{ cm}^2
 \end{aligned}$$

The total area contributed by A1 through A43 [AT]:

$$\begin{aligned}
 &= A_1 + \text{frn}(A_2 + A_3) + A_{41} + A_{42} + A_{43} \\
 &= 121.989 + 0.792(122.93 + 0.0) + 0.5 + 0.0 + 0.0 \\
 &= 219.908 \text{ cm}^2
 \end{aligned}$$

Nozzle Radius for Force Calculation [Rxn]:

$$\begin{aligned}
 &= t_n / \ln[1 + t_n/R_n] \\
 &= 78.5 / \ln[1 + 78.5/77.7] \\
 &= 112.419 \text{ mm.}
 \end{aligned}$$

Shell Radius for Force Calculation [Rxs]:

$$\begin{aligned}
 &= \text{teff} / \ln[1 + \text{teff}/\text{Reff}] \\
 &= 78.5 / \ln[1 + 78.5/1274.0] \\
 &= 1312.860 \text{ mm.}
 \end{aligned}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$\begin{aligned}
 &= 1.5 \cdot S \cdot E \\
 &= 1.5 \cdot 161.141 \cdot 1.0 \\
 &= 241.7 \text{ N./mm}^2
 \end{aligned}$$

Force acting on the Nozzle [fN]:

$$\begin{aligned}
 &= P \cdot R_{xn} \cdot LH \\
 &= 192.405 \cdot 112.419 \cdot 156.599 \\
 &= 34541.5 \text{ Kgf}
 \end{aligned}$$

Force acting on the Shell [fS]:

$$\begin{aligned}
 &= P \cdot R_{xs} \cdot (LR + t_n)/2 \\
 &= 192.405 \cdot 1312.86 \cdot (155.4 + 78.5)/2 \\
 &= 301253.7 \text{ Kgf}
 \end{aligned}$$



Discontinuity Force from Internal Pressure [fY]:

$$= (P \cdot Rxs \cdot Rnc) / 2$$

$$= (192.405 \cdot 1312.86 \cdot 77.7) / 2$$

$$= 100074.4 \text{ Kg}$$

Area Resisting Internal Pressure [Ap]:

$$= (fS + fY + fN) / P$$

$$= (301253.7 + 100074.4 + 34541.5) / 192.405$$

$$= 2221.5 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= Sallow / (2 \cdot Ap / AT - Rxs / (2 \cdot teff))$$

$$= 241.712 / (2 \cdot 2221.482 / 219.908 - 1312.86 / (2 \cdot 78.5))$$

$$= 204.1 \text{ bars}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= 2 \cdot S[t/Rxs]$$

$$= 2 \cdot 161.141[78.5/1312.86]$$

$$= 192.7 \text{ bars}$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(Pmax1, Pmax2)$$

$$= \min(204.109, 192.691)$$

$$= 192.691 \text{ bars}$$

Average Primary Membrane Stress [σAvg]:

$$= (fN + fS + fY) / AT$$

$$= (34541.523 + 301253.656 + 100074.43) / 219.908$$

$$= 194.377 \text{ N./mm}^2$$

General Primary Membrane Stress [σCirc]:

$$= P \cdot Rxs / (2 \cdot teff)$$

$$= 192.405 \cdot 1312.86 / (2 \cdot 78.5)$$

$$= 160.9 \text{ N./mm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma Avg - \sigma Circ, \sigma Circ)$$

$$= \max(2 \cdot 194.377 - 160.902, 160.902)$$

$$= 227.9 \text{ N./mm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	241.71	N./mm ²
Local Primary Membrane Stress	PL	227.85	N./mm ²
Maximum Allowable Working Pressure	Pmax	192.69	bars

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= (Rnc + tn) / Rnc$$

$$= (77.7 + 78.5) / 77.7$$

$$= 2.010 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld for non-radial Nozzles [LT]:

$$= \pi / 2 \cdot \sqrt{((Rnc + Tn)^2 + (Rn + Tn)^2) / 2}$$

$$= \pi / 2 \cdot \sqrt{((77.7 + 78.5)^2 + (77.7 + 78.5)^2) / 2}$$

$$= 245.358 \text{ mm.}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm.}$$

Required Weld Throat Dimensions :

$$tc, L41T = 7.071, \text{ must be } \geq \min(0.7 \cdot 78.500, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$= \min(fy \cdot ky, 1.5 \cdot Sn(A2 + A3), \pi / 4 \cdot P \cdot Rn^2 \cdot ky^2)$$

$$= \min(100074 \cdot 2.01, 1.5 \cdot 127.7(122.93 + 0.0), \pi / 4 \cdot 192.4 \cdot 77.7^2 \cdot 2.01^2)$$

$$= 37598.078 \text{ Kg}$$

Weld Stress Value [τ]:

$$= fwelds / (Lt(0.49 \cdot L41T + 0.6 \cdot tw1 + 0.49 \cdot L43T))$$



= 37598.078/(245.358(0.49•7.071 + 0.6•78.5 + 0.49•0.0))
 = 29.720 < or = to 161.141 Weld Stress Passed

Nozzle Calculations per Section 4.5: External Pressure Case:

Nozzle Radius for Force Calculation [Rxn]:

= $t_n / \ln[1 + t_n/R_n]$
 = 78.5/ $\ln[1 + 78.5/77.7]$
 = 112.419 mm.

Shell Radius for Force Calculation [Rxs]:

= $t_{eff} / \ln[1 + t_{eff}/R_{eff}]$
 = 78.5/ $\ln[1 + 78.5/1274.0]$
 = 1312.860 mm.

Allowable Local Primary Membrane Stress [Sallow]:

= 1.5 • S
 = 1.5 • 161.141
 = 241.7 N./mm²

Force acting on the Nozzle [fN]:

= P • Rxn • LH
 = 1.034 • 112.419 • 156.599
 = 185.7 Kgf

Force acting on the Shell [fS]:

= P • Rxs • (LR + t_n)/2
 = 1.034 • 1312.86 • (155.4 + 78.5)/2
 = 1619.3 Kgf

Discontinuity Force from External Pressure [fY]:

= (P • Rxs • Rnc)/2
 = (1.034 • 1312.86 • 77.7)/2
 = 537.9 Kgf

Area Resisting External Pressure [Ap]:

= (fS + fY + fN)/P
 = (1619.3 + 537.9 + 185.7)/1.034
 = 2221.5 cm²

Maximum Allowable Working Pressure Candidate [Pmax1]:

= Sallow / (2 • Ap/AT - Rxs/(2 • t_{eff}))
 = 241.712/(2 • 2221.482/219.908 - 1312.86/(2•78.5))
 = 204.1 bars

Maximum Allowable Working Pressure Candidate [Pmax2]:

= 2 • S[t/Rxs]
 = 2 • 161.141[78.5/1312.86]
 = 192.7 bars

Maximum Allowable Working Pressure [Pmax]:

= min(Pmax1, Pmax2)
 = min(204.109, 192.691)
 = 192.691 bars

Average Primary Membrane Stress [σAvg]:

= (fN + fS + fY) / AT
 = (185.663 + 1619.256 + 537.906)/219.908
 = 1.045 N./mm²

General Primary Membrane Stress [σCirc]:

= P • Rxs / (2 • t_{eff})
 = 1.034 • 1312.86/(2 • 78.5)
 = 0.9 N./mm²

Maximum Local Primary Membrane Stress [PL]:

= max(2 • σAvg - σCirc, σCirc)
 = max(2 • 1.045 - 0.865, 0.865)
 = 1.2 N./mm²



Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	241.71	N./mm ²
Local Primary Membrane Stress	PL	1.22	N./mm ²
Maximum Allowable External Pressure	Pmax	192.69	bars

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -29 °C

The Drop for this Nozzle is : 9.6233 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 389.6232 mm.

Warning:

The equations for nozzle local stress analysis pertain to radial nozzles only and this nozzle is not specified as radial. Please check the specific analysis code for all assumptions and limitations.

Input Echo, WRC107/537 Item 1, Description: B26" :

Diameter Basis for Vessel	Vbasis	ID	
Cylindrical or Spherical Vessel	Cylsph	Spherical	
Internal Corrosion Allowance	Cas	1.5000	mm.
Vessel Diameter	Dv	2545.000	mm.
Vessel Thickness	Tv	80.000	mm.
Design Temperature	T1	85.0	C
Vessel Material		SA-516 70	
Vessel UNS Number		K02700	
Vessel Cold S.I. Allowable	Smc	175.00	N./mm ²
Vessel Hot S.I. Allowable	Smh	161.14	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Attachment Type	Type	Round	
WRC107 Attachment Classification	Holsol	Hollow	
Diameter Basis for Nozzle	Nbasis	ID	
Corrosion Allowance for Nozzle	Can	1.5000	mm.
Nozzle Diameter	Dn	152.400	mm.
Nozzle Thickness	Tn	80.000	mm.
Nozzle Material		SA-350 LF1	
Nozzle UNS Number		K03009	
Nozzle Cold S.I. Allowable	SNmc	138.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh	127.70	N./mm ²
Design Internal Pressure	Dp	192.405	bars
Include Pressure Thrust		No	

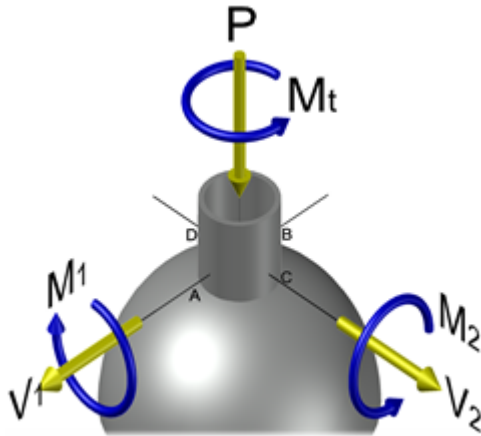
External Forces and Moments in WRC 107/537 Convention:

Radial Load (SUS)	P	-3671.2	Kgf
Longitudinal Shear (SUS)	(V1) V1	3245.0	Kgf
Circumferential Shear (SUS)	(Vc) V2	3245.0	Kgf
Circumferential Moment (SUS)	(Mc) M1	1276.7	Kg-m.
Longitudinal Moment (SUS)	(Ml) M2	1276.7	Kg-m.
Torsional Moment (SUS)	Mt	1651.4	Kg-m.

Use Interactive Control	No
WRC107 Version	Version March 1979

Include Pressure Stress Indices per Div. 2	No
Compute Pressure Stress per WRC-368	No
Local Loads applied at end of Nozzle/Attachment	No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean} (t - ca))}$$

$$= 312.4 + 2 \cdot 1.65 \cdot \sqrt{(1313.25 (80.0 - 1.5))}$$

$$= 1371.953 \text{ mm.}$$

WRC 107 Stress Calculation for SUSained loads:

Radial Load	P	-3671.2	Kgf
Circumferential Shear	(VC) V2	3245.0	Kgf
Longitudinal Shear	(VL) V1	3245.0	Kgf
Circumferential Moment	(MC) M1	1276.7	Kg-m.
Longitudinal Moment	(ML) M2	1276.7	Kg-m.
Torsional Moment	MT	1651.4	Kg-m.

Dimensionless Param: U = 0.49 TAU = 5.00 (1.49) RHO = 1.00

Dimensionless Loads for Spherical Shells at Attachment Junction:

Curves read for 1979	Figure	Value	Location
$N(x) * T / P$	SP 2	0.07380	(A,B,C,D)
$M(x) / P$	SP 2	0.08962	(A,B,C,D)
$N(x) * T * \text{SQRT}(Rm * T) / MC$	SM 2	0.14038	(A,B,C,D)
$M(x) * \text{SQRT}(Rm * T) / MC$	SM 2	0.23095	(A,B,C,D)
$N(x) * T * \text{SQRT}(Rm * T) / ML$	SM 2	0.14038	(A,B,C,D)
$M(x) * \text{SQRT}(Rm * T) / ML$	SM 2	0.23095	(A,B,C,D)
$N(y) * T / P$	SP 2	0.13991	(A,B,C,D)
$M(y) / P$	SP 2	0.03380	(A,B,C,D)
$N(y) * T * \text{SQRT}(Rm * T) / MC$	SM 2	0.08707	(A,B,C,D)
$M(y) * \text{SQRT}(Rm * T) / MC$	SM 2	0.10400	(A,B,C,D)
$N(y) * T * \text{SQRT}(Rm * T) / ML$	SM 2	0.08707	(A,B,C,D)
$M(y) * \text{SQRT}(Rm * T) / ML$	SM 2	0.10400	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Memb. P	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.4
Rad.	Bend. P	3.1	-3.1	3.1	-3.1	3.1	-3.1	3.1	-3.1
Rad.	Memb. MC	0.0	0.0	0.0	0.0	-0.9	-0.9	0.9	0.9



Rad. Bend. MC	0.0	0.0	0.0	0.0	-8.8	8.8	8.8	-8.8
Rad. Memb. ML	-0.9	-0.9	0.9	0.9	0.0	0.0	0.0	0.0
Rad. Bend. ML	-8.8	8.8	8.8	-8.8	0.0	0.0	0.0	0.0
Tot. Rad. Str.	-6.1	5.2	13.2	-10.6	-6.1	5.2	13.2	-10.6
Tang. Memb. P	0.8	0.8	0.8	0.8	0.8	0.8	0.8	0.8
Tang. Bend. P	1.2	-1.2	1.2	-1.2	1.2	-1.2	1.2	-1.2
Tang. Memb. MC	0.0	0.0	0.0	0.0	-0.6	-0.6	0.6	0.6
Tang. Bend. MC	0.0	0.0	0.0	0.0	-3.9	3.9	3.9	-3.9
Tang. Memb. ML	-0.6	-0.6	0.6	0.6	0.0	0.0	0.0	0.0
Tang. Bend. ML	-3.9	3.9	3.9	-3.9	0.0	0.0	0.0	0.0
Tot. Tang. Str.	-2.5	3.0	6.5	-3.8	-2.5	3.0	6.5	-3.8
Shear VC	0.8	0.8	-0.8	-0.8	0.0	0.0	0.0	0.0
Shear VL	0.0	0.0	0.0	0.0	-0.8	-0.8	0.8	0.8
Shear MT	1.3	1.3	1.3	1.3	1.3	1.3	1.3	1.3
Tot. Shear	2.2	2.2	0.5	0.5	0.5	0.5	2.2	2.2
Str. Int.	7.1	6.5	13.3	10.6	6.2	5.3	13.9	11.2

WRC 107/537 Stress Summations:**Vessel Stress Summation at Attachment Junction (N./mm²)**

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Pm (SUS)		160.9	160.9	160.9	160.9	160.9	160.9	160.9	160.9
Rad. Pl (SUS)		-0.5	-0.5	1.3	1.3	-0.5	-0.5	1.3	1.3
Rad. Q (SUS)		-5.6	5.6	11.9	-11.9	-5.6	5.6	11.9	-11.9
Long. Pm (SUS)		160.9	160.9	160.9	160.9	160.9	160.9	160.9	160.9
Long. Pl (SUS)		0.3	0.3	1.4	1.4	0.3	0.3	1.4	1.4
Long. Q (SUS)		-2.8	2.8	5.1	-5.1	-2.8	2.8	5.1	-5.1
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)		0.8	0.8	-0.8	-0.8	-0.8	-0.8	0.8	0.8
Shear Q (SUS)		1.3	1.3	1.3	1.3	1.3	1.3	1.3	1.3
Pm (SUS)		160.9	160.9	160.9	160.9	160.9	160.9	160.9	160.9
Pm+Pl (SUS)		161.7	161.7	163.1	163.1	161.7	161.7	163.1	163.1
Pm+Pl+Q (Total)		159.4	167.4	174.2	157.2	158.5	166.2	174.8	157.8

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	160.90	161.14	Passed
Pm+Pl (SUS)	163.07	241.71	Passed
Pm+Pl+Q (TOTAL)	174.77	484.02	Passed

The Pm+Pl+Q allowable above was based on the rules of ASME VIII-2 Part 4, 4.1.6.3 (a)&(b) or per ASME VIII-2, 5.5.6.1(1)&(2) Part 5 as applicable.

The Pm+Pl+Q allowable was based on a temperature range cycling from ambient to design temperature.



Input, Nozzle Desc: L62"

From: 30

Pressure for Reinforcement Calculations	P	192.365	bars
Temperature for Internal Pressure	Temp	85	°C
Design External Pressure	Pext	1.03	bars
Temperature for External Pressure	Tempex	85	°C
Parent Material		SA-516 70	+N+IT
Parent Allowable Stress at Temperature	Sv	161.14	N./mm ²
Parent Allowable Stress At Ambient	Sva	175.00	N./mm ²
Inside Diameter of Hemispherical Head	D	2545.00	mm.
Head Finished (Minimum) Thickness	t	80.0000	mm.
Head Internal Corrosion Allowance	cas	1.5000	mm.
Head External Corrosion Allowance	co	0.0000	mm.
Distance from Head Centerline	L1	1200.0000	mm.
User Entered Minimum Design Metal Temperature		-40.00	°C

Type of Element Connected to the Parent : Nozzle

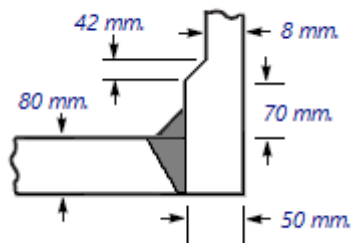
Material		SA-350 LF1	+IT
Material UNS Number		K03009	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	127.71	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	8.0000	mm.
Flange Material		SA-350 LF1	+IT
Hub Height of Integral Nozzle	h	70.0000	mm.
Height of Beveled Transition	L'	42.0000	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		50.0000	mm.
Corrosion Allowance	can	1.5000	mm.
Outside Projection	ho	200.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel Wgnv		80.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.
Flange Class		1500	
Flange Grade		GR 1.4	

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: L62"

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	2.000	in.
Actual Thickness Used in Calculation	0.315	in.



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 53.8 \cdot (\exp(192.365/(127.713 \cdot 1.0)) - 1) + 1.5 + 0.0$$

$$= 4.3731 + 1.5 + 0.0 = 5.8731 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 127.713 \cdot 1.0 \cdot \ln((2 \cdot 6.5 + 53.8) / 53.8) - 0.0$$

$$= 276.392 - 0.0 = 276.3925 \text{ bars}$$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c :	6.020 mm
Min. required shell thickness, internal p + c	79.887 mm.
Min. required shell thickness, external p + c	6.561 mm.

Nozzle neck minimum required thickness	: 6.020 mm.
--	-------------

Minimum nozzle neck thickness 8.001 mm., is greater than required thickness 6.020 mm.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[127.7/161.1, 1]$$

$$= 0.793$$

Thickness of Nozzle at Shell [tn]:

$$= \text{hub thickness} - \text{corrosion allowance}$$

$$= 50.0 - 1.5$$

$$= 48.500 \text{ mm.}$$

Thickness of Nozzle at Top [tn2]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 8.0 - 1.5$$

$$= 6.500 \text{ mm.}$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 2545.0/2 + 1.5$$

$$= 1274.000 \text{ mm.}$$

Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$$

$$= \min(\sqrt{1274.0 \cdot 78.5}, 2 \cdot 26.9)$$

$$= 53.800 \text{ mm.}$$

Intermediate value [Cn]:

$$= \min(((t + t_e)/t_n)^{0.35}, 1)$$

$$= \min(((78.5 + 0.0)/48.5)^{0.35}, 1)$$

$$= \min(1.184, 1)$$

$$= 1.000$$

Intermediate value [Cp]:

= 9999999.000

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$= \min(t + t_e + F_p \cdot \sqrt{(R_n \cdot t_n)}, L_{pr1} + t)$$

$$= \min(78.5 + 0.0 + 1.0 \cdot \sqrt{(26.9 \cdot 48.5)}, 200.0 + 78.5)$$

$$= 114.620 \text{ mm.}$$

Effective Vessel Thickness [teff]:

$$= t + ((A_5 \cdot f_{rp}) / LR)$$

$$= 78.5 + ((0.0 \cdot 1.0) / 53.8)$$

$$= 78.500 \text{ mm.}$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR$$

$$= 78.5 \cdot 53.8$$

$$= 42.233 \text{ cm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= t_n \cdot LH$$

$$= 48.5 \cdot 114.62$$

$$= 55.591 \text{ cm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot \text{Leg}^2$$

$$= 0.5 \cdot 10.0^2$$

$$= 0.500 \text{ cm}^2$$

The total area contributed by A1 through A43 [AT]:

$$= A_1 + f_{rn}(A_2 + A_3) + A_{41} + A_{42} + A_{43}$$

$$= 42.233 + 0.793(55.591 + 0.0) + 0.5 + 0.0 + 0.0$$

$$= 86.792 \text{ cm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n/R_n]$$

$$= 48.5 / \ln[1 + 48.5/26.9]$$

$$= 47.056 \text{ mm.}$$

Shell Radius for Force Calculation [Rxs]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$

$$= 78.5 / \ln[1 + 78.5/1274.0]$$

$$= 1312.860 \text{ mm.}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$

$$= 1.5 \cdot 161.141 \cdot 1.0$$

$$= 241.7 \text{ N./mm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot LH$$

$$= 192.365 \cdot 47.056 \cdot 114.62$$

$$= 10580.3 \text{ Kg}$$

Force acting on the Shell [fS]:

$$= P \cdot R_{xs} \cdot (LR + t_n)/2$$

$$= 192.365 \cdot 1312.86 \cdot (53.8 + 48.5)/2$$

$$= 131730.3 \text{ Kg}$$

Discontinuity Force from Internal Pressure [fY]:

$$= (P \cdot R_{xs} \cdot R_{nc})/2$$

$$= (192.365 \cdot 1312.86 \cdot 26.9)/2$$

$$= 34638.8 \text{ Kg}$$

Area Resisting Internal Pressure [Ap]:

$$= (f_S + f_Y + f_N)/P$$

$$= (131730.3 + 34638.8 + 10580.3)/192.365$$

$$= 902.0 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:



$$= \text{Sallow} / (2 \cdot \text{Ap}/\text{AT} - \text{Rxs}/(2 \cdot \text{teff}))$$

$$= 241.712 / (2 \cdot 902.043/86.792 - 1312.86/(2 \cdot 78.5))$$

$$= 194.5 \text{ bars}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= 2 \cdot \text{S}[\text{t}/\text{Rxs}]$$

$$= 2 \cdot 161.141[78.5/1312.86]$$

$$= 192.7 \text{ bars}$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(\text{Pmax1}, \text{Pmax2})$$

$$= \min(194.537, 192.691)$$

$$= 192.691 \text{ bars}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$= (\text{fN} + \text{fS} + \text{fY}) / \text{AT}$$

$$= (10580.331 + 131730.312 + 34638.758) / 86.792$$

$$= 199.940 \text{ N./mm}^2$$

General Primary Membrane Stress [σ_{Circ}]:

$$= \text{P} \cdot \text{Rxs} / (2 \cdot \text{teff})$$

$$= 192.365 \cdot 1312.86 / (2 \cdot 78.5)$$

$$= 160.9 \text{ N./mm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{\text{Avg}} - \sigma_{\text{Circ}}, \sigma_{\text{Circ}})$$

$$= \max(2 \cdot 199.94 - 160.868, 160.868)$$

$$= 239.0 \text{ N./mm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	241.71	N./mm ²
Local Primary Membrane Stress	PL	239.01	N./mm ²
Maximum Allowable Working Pressure	Pmax	192.69	bars

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= (\text{Rnc} + \text{tn}) / \text{Rnc}$$

$$= (26.9 + 48.5) / 26.9$$

$$= 2.803 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld for non-radial Nozzles [LT]:

$$= \pi/2 \cdot \sqrt{(((\text{Rnc} + \text{Tn})^2 + (\text{Rn} + \text{Tn})^2)/2)}$$

$$= \pi/2 \cdot \sqrt{(((26.9 + 48.5)^2 + (26.9 + 48.5)^2)/2)}$$

$$= 118.438 \text{ mm.}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm.}$$

Required Weld Throat Dimensions :

$$\text{tc}, \text{L41T} = 7.071, \text{ must be } \geq \min(0.7 \cdot 48.500, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$= \min(\text{fy} \cdot \text{ky}, 1.5 \cdot \text{Sn}(\text{A2} + \text{A3}), \pi/4 \cdot \text{P} \cdot \text{Rn}^2 \cdot \text{ky}^2)$$

$$= \min(34639 \cdot 2.8, 1.5 \cdot 127.7(55.591 + 0.0), \pi/4 \cdot 192.4 \cdot 26.9^2 \cdot 2.8^2)$$

$$= 8758.993 \text{ Kg}$$

Weld Stress Value [τ]:

$$= \text{fwelds} / (\text{LT}(0.49 \cdot \text{L41T} + 0.6 \cdot \text{tw1} + 0.49 \cdot \text{L43T}))$$

$$= 8758.993 / (118.438(0.49 \cdot 7.071 + 0.6 \cdot 78.5 + 0.49 \cdot 0.0))$$

$$= 14.343 < \text{or} = \text{to } 161.141 \text{ Weld Stress Passed}$$

Nozzle Calculations per Section 4.5: External Pressure Case:

Nozzle Radius for Force Calculation [Rxn]:

$$= \text{tn} / \ln[1 + \text{tn}/\text{Rn}]$$

$$= 48.5 / \ln[1 + 48.5/26.9]$$

$$= 47.056 \text{ mm.}$$

Shell Radius for Force Calculation [Rxs]:

$$\begin{aligned}
 &= t_{eff} / \ln[1 + t_{eff}/R_{eff}] \\
 &= 78.5 / \ln[1 + 78.5/1274.0] \\
 &= 1312.860 \text{ mm.}
 \end{aligned}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$\begin{aligned}
 &= 1.5 \cdot S \\
 &= 1.5 \cdot 161.141 \\
 &= 241.7 \text{ N./mm}^2
 \end{aligned}$$

Force acting on the Nozzle [fN]:

$$\begin{aligned}
 &= P \cdot R_{xn} \cdot L_H \\
 &= 1.034 \cdot 47.056 \cdot 114.62 \\
 &= 56.9 \text{ Kgf}
 \end{aligned}$$

Force acting on the Shell [fS]:

$$\begin{aligned}
 &= P \cdot R_{xs} \cdot (L_R + t_n)/2 \\
 &= 1.034 \cdot 1312.86 \cdot (53.8 + 48.5)/2 \\
 &= 708.2 \text{ Kgf}
 \end{aligned}$$

Discontinuity Force from External Pressure [fY]:

$$\begin{aligned}
 &= (P \cdot R_{xs} \cdot R_{nc})/2 \\
 &= (1.034 \cdot 1312.86 \cdot 26.9)/2 \\
 &= 186.2 \text{ Kgf}
 \end{aligned}$$

Area Resisting External Pressure [Ap]:

$$\begin{aligned}
 &= (f_S + f_Y + f_N)/P \\
 &= (708.2 + 186.2 + 56.9)/1.034 \\
 &= 902.0 \text{ cm}^2
 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$\begin{aligned}
 &= S_{allow} / (2 \cdot A_p / A_T - R_{xs} / (2 \cdot t_{eff})) \\
 &= 241.712 / (2 \cdot 902.044 / 86.792 - 1312.86 / (2 \cdot 78.5)) \\
 &= 194.5 \text{ bars}
 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$\begin{aligned}
 &= 2 \cdot S / [t / R_{xs}] \\
 &= 2 \cdot 161.141 [78.5 / 1312.86] \\
 &= 192.7 \text{ bars}
 \end{aligned}$$

Maximum Allowable Working Pressure [Pmax]:

$$\begin{aligned}
 &= \min(P_{max1}, P_{max2}) \\
 &= \min(194.537, 192.691) \\
 &= 192.691 \text{ bars}
 \end{aligned}$$

Average Primary Membrane Stress [σAvg]:

$$\begin{aligned}
 &= (f_N + f_S + f_Y) / A_T \\
 &= (56.882 + 708.208 + 186.225) / 86.792 \\
 &= 1.075 \text{ N./mm}^2
 \end{aligned}$$

General Primary Membrane Stress [σCirc]:

$$\begin{aligned}
 &= P \cdot R_{xs} / (2 \cdot t_{eff}) \\
 &= 1.034 \cdot 1312.86 / (2 \cdot 78.5) \\
 &= 0.9 \text{ N./mm}^2
 \end{aligned}$$

Maximum Local Primary Membrane Stress [PL]:

$$\begin{aligned}
 &= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ}) \\
 &= \max(2 \cdot 1.075 - 0.865, 0.865) \\
 &= 1.3 \text{ N./mm}^2
 \end{aligned}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	241.71	N./mm ²
Local Primary Membrane Stress	PL	1.28	N./mm ²
Maximum Allowable External Pressure	Pmax	192.69	bars

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -29 °C

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -29 °C

Flange MDMT including temperature reduction per 3.11:

Flange MDMT with Temp reduction -34 °C

The Drop for this Nozzle is : 2.2358 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 282.2358 mm.

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Input, Nozzle Desc: M124"
From: 60

Pressure for Reinforcement Calculations	P	192.204	bars
Temperature for Internal Pressure	Temp	85	°C
Design External Pressure	Pext	1.03	bars
Temperature for External Pressure	Tempex	85	°C
Parent Material		SA-516 70	+N+IT
Parent Allowable Stress at Temperature	Sv	161.10	N./mm ²
Parent Allowable Stress At Ambient	Sva	175.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	2450.00	mm.
Design Length of Section	L	5498.3330	mm.
Shell Finished (Minimum) Thickness	t	157.0000	mm.
Shell Internal Corrosion Allowance	cas	1.5000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		3750.00	mm.
User Entered Minimum Design Metal Temperature		-40.00	°C

Type of Element Connected to the Parent : Nozzle

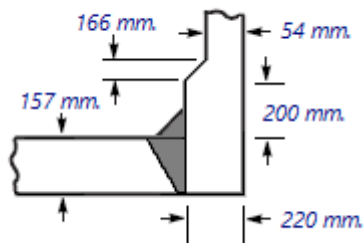
Material		SA-350 LF1	+IT
Material UNS Number		K03009	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	127.70	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		24.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	54.0000	mm.
Flange Material		SA-350 LF1	+IT
Hub Height of Integral Nozzle	h	200.0000	mm.
Height of Beveled Transition	L'	166.0000	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		220.0000	mm.
Corrosion Allowance	can	1.5000	mm.
Outside Projection	ho	300.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	157.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.
This is a Manway, Coupling or Access Opening.			
Flange Class		1500	
Flange Grade		GR 1.4	

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: M124"

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	24.000	in.
Actual Thickness Used in Calculation	2.126	in.



Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 612.6 \cdot (\exp(192.204/(127.7 \cdot 1.0)) - 1) + 1.5 + 0.0$$

$$= 49.7553 + 1.5 + 0.0 = 51.2553 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 127.7 \cdot 1.0 \cdot \ln((2 \cdot 52.5 + 612.6) / 612.6) - 0.0$$

$$= 202.01 - 0.0 = 202.0098 \text{ bars}$$

Please check Code paragraph 4.5.17.1 regarding openings in compression. A Part 5 analysis may be required.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[127.7/161.1, 1]$$

$$= 0.793$$

Thickness of Nozzle at Shell [tn]:

$$= \text{hub thickness} - \text{corrosion allowance}$$

$$= 220.0 - 1.5$$

$$= 218.500 \text{ mm.}$$

Thickness of Nozzle at Top [tn2]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 54.0 - 1.5$$

$$= 52.500 \text{ mm.}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D/t$$

$$= 2453.0/155.5$$

$$= 15.775 \text{ must be less than 400.}$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 2450.0/2 + 1.5$$

$$= 1226.500 \text{ mm.}$$

Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$$

$$= \min(\sqrt{1226.5 \cdot 155.5}, 2 \cdot 306.3)$$

$$= 436.716 \text{ mm.}$$

Thickness Limit Candidate [LH1]:

$$= \min[1.5t, t_e] + \sqrt{R_n \cdot t_n}$$

$$= \min[1.5 \cdot 155.5, 0.0] + \sqrt{306.3 \cdot 218.5}$$

$$= 258.702 \text{ mm.}$$

Thickness Limit Candidate [LH2]:

$$= L_{pr1}$$

$$= 300.0$$

$$= 300.000 \text{ mm.}$$

Thickness Limit Candidate [LH3]:

$$= 8(t + t_e)$$

$$= 8(155.5 + 0.0)$$

$$= 1244.000 \text{ mm.}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$= \min[LH1, LH2, LH3] + t \text{ for set-in nozzles}$$

$$= \min[258.702, 300.0, 1244.0] + 155.5$$

$$= 414.202 \text{ mm.}$$

Effective Vessel Thickness [teff]:

$$= t + ((A5 \cdot frp) / LR)$$

$$= 155.5 + ((0.0 \cdot 1.0) / 436.716)$$

$$= 155.500 \text{ mm.}$$

Parameter [λ]:

$$= \min(12, (2 \cdot R_n + t_n) / (\sqrt{(D_i + t_{eff}) \cdot t_{eff}}))$$

$$= \min(12, (2 \cdot 306.3 + 218.5) / (\sqrt{(2453.0 + 155.5) \cdot 155.5}))$$

$$= 1.305$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1)$$

$$= 155.5 \cdot 436.716 \cdot \max((1.305/5)^{0.85}, 1)$$

$$= 679.093 \text{ cm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= A2A + A2B$$

$$= 776.768 + 111.034$$

$$= 887.801 \text{ cm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot \text{Leg}^2$$

$$= 0.5 \cdot 10.0^2$$

$$= 0.500 \text{ cm}^2$$

The total area contributed by A1 through A43 [AT]:

$$= A1 + frn(A2 + A3) + A41 + A42 + A43$$

$$= 679.093 + 0.793(887.801 + 0.0) + 0.5 + 0.0 + 0.0$$

$$= 1383.331 \text{ cm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n/R_n]$$

$$= 218.5 / \ln[1 + 218.5/306.3]$$

$$= 405.793 \text{ mm.}$$

Shell Radius for Force Calculation [Rxs]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$

$$= 155.5 / \ln[1 + 155.5/1226.5]$$

$$= 1302.704 \text{ mm.}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$

$$= 1.5 \cdot 161.1 \cdot 1.0$$

$$= 241.7 \text{ N./mm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot LH$$

$$= 192.204 \cdot 405.793 \cdot 414.202$$

$$= 329439.5 \text{ Kg}$$

Force acting on the Shell [fS]:

$$= P \cdot R_{xs}(LR + t_n)$$

$$= 192.204 \cdot 1302.704 (436.716 + 218.5)$$

$$= 1672975.9 \text{ Kgf}$$

Discontinuity Force from Internal Pressure [fY]:

$$= P \cdot Rxs \cdot Rnc$$

$$= 192.204 \cdot 1302.704 \cdot 306.3$$

$$= 782082.0 \text{ Kgf}$$

Area Resisting Internal Pressure [Ap]:

$$= (fS + fY + fN) / P$$

$$= (1672976 + 782082 + 329439.5) / 192.204$$

$$= 14206.5 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= Sallow / (2 \cdot Ap / AT - Rxs / teff)$$

$$= 241.65 / (2 \cdot 14206.502 / 1383.331 - 1302.704 / 155.5)$$

$$= 198.7 \text{ bars}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= S[t/Rxs]$$

$$= 161.1 [155.5 / 1302.704]$$

$$= 192.3 \text{ bars}$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(Pmax1, Pmax2)$$

$$= \min(198.68, 192.289)$$

$$= 192.289 \text{ bars}$$

Average Primary Membrane Stress [σAvg]:

$$= (fN + fS + fY) / AT$$

$$= (329439.531 + 1672976 + 782082) / 1383.331$$

$$= 197.401 \text{ N./mm}^2$$

General Primary Membrane Stress [σCirc]:

$$= P \cdot Rxs / teff$$

$$= 192.204 \cdot 1302.704 / 155.5$$

$$= 161.0 \text{ N./mm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma Avg - \sigma Circ, \sigma Circ)$$

$$= \max(2 \cdot 197.401 - 161.029, 161.029)$$

$$= 233.8 \text{ N./mm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	241.65	N./mm ²
Local Primary Membrane Stress	PL	233.77	N./mm ²
Maximum Allowable Working Pressure	Pmax	192.29	bars

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= (Rnc + tn) / Rnc$$

$$= (306.3 + 218.5) / 306.3$$

$$= 1.713 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$= \pi / 2 \cdot (Rn + tn)$$

$$= \pi / 2 \cdot (306.3 + 218.5)$$

$$= 824.354 \text{ mm.}$$

Weld Throat Dimensions, (0.7071 • Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm.}$$

Required Weld Throat Dimensions :

$$tc, L41T = 7.071, \text{ must be } \geq \min(0.7 \cdot 218.500, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$= \min(fy \cdot ky, 1.5 \cdot Sn(A2 + A3), \pi / 4 \cdot P \cdot Rn^2 \cdot ky^2)$$

$$= \min(782082 \cdot 1.71, 1.5 \cdot 127.7(887.801 + 0.0), \pi / 4 \cdot 192.2 \cdot 306.3^2 \cdot 1.71^2)$$

$$= 423971.969 \text{ Kgf}$$



Weld Stress Value [τ]:
 $= \text{fwelds} / (L_t(0.49 \cdot L_{41T} + 0.6 \cdot t_{w1} + 0.49 \cdot L_{43T}))$
 $= 423971.969 / (824.354(0.49 \cdot 7.071 + 0.6 \cdot 155.5 + 0.49 \cdot 0.0))$
 $= 52.124 < \text{or} = \text{to } 161.100 \text{ Weld Stress Passed}$

Nozzle/Structural Discontinuity Proximity Warning:

This nozzle's diameter limit intersects a major structural discontinuity. Reduce the diameter limit in the nozzle dialog so that there is no interference or move the nozzle farther away from the intersecting weld.

Please check Code paragraph 4.5.17.1 regarding openings in compression. A Part 5 analysis may be required.

Nozzle Calculations per Section 4.5: External Pressure Case:

Nozzle Radius for Force Calculation [R_{xn}]:
 $= t_n / \ln[1 + t_n/R_n]$
 $= 218.5 / \ln[1 + 218.5/306.3]$
 $= 405.793 \text{ mm.}$

Shell Radius for Force Calculation [R_{xs}]:
 $= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$
 $= 155.5 / \ln[1 + 155.5/1226.5]$
 $= 1302.704 \text{ mm.}$

Allowable Local Primary Membrane Stress [S_{allow}]:
 $= 1.5 \cdot S$
 $= 1.5 \cdot 161.1$
 $= 241.7 \text{ N./mm}^2$

Force acting on the Nozzle [f_N]:
 $= P \cdot R_{xn} \cdot L_H$
 $= 1.034 \cdot 405.793 \cdot 414.202$
 $= 1772.6 \text{ Kg}$

Force acting on the Shell [f_S]:
 $= P \cdot R_{xs} (L_R + t_n)$
 $= 1.034 \cdot 1302.704 (436.716 + 218.5)$
 $= 9001.7 \text{ Kg}$

Discontinuity Force from External Pressure [f_Y]:
 $= P \cdot R_{xs} \cdot R_{nc}$
 $= 1.034 \cdot 1302.704 \cdot 306.3$
 $= 4208.1 \text{ Kg}$

Area Resisting External Pressure [A_p]:
 $= (f_S + f_Y + f_N) / P$
 $= (9001.7 + 4208.1 + 1772.6) / 1.034$
 $= 14206.5 \text{ cm}^2$

Maximum Allowable Working Pressure Candidate [P_{max1}]:
 $= S_{allow} / (2 \cdot A_p / A_T - R_{xs} / t_{eff})$
 $= 241.65 / (2 \cdot 14206.504 / 1383.331 - 1302.704 / 155.5)$
 $= 198.7 \text{ bars}$

Maximum Allowable Working Pressure Candidate [P_{max2}]:
 $= S[t/R_{xs}]$
 $= 161.1[155.5/1302.704]$
 $= 192.3 \text{ bars}$

Maximum Allowable Working Pressure [P_{max}]:
 $= \min(P_{max1}, P_{max2})$
 $= \min(198.68, 192.289)$
 $= 192.289 \text{ bars}$

Average Primary Membrane Stress [σ_{Avg}]:
 $= (f_N + f_S + f_Y) / A_T$
 $= (1772.609 + 9001.749 + 4208.134) / 1383.331$



= 1.062 N./mm²

General Primary Membrane Stress [σ_{Circ}]:

= $P \cdot R_{xs} / t_{eff}$

= $1.034 \cdot 1302.704 / 155.5$

= 0.9 N./mm²

Maximum Local Primary Membrane Stress [PL]:

= $\max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$

= $\max(2 \cdot 1.062 - 0.866, 0.866)$

= 1.3 N./mm²

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	241.65	N./mm ²
Local Primary Membrane Stress	PL	1.26	N./mm ²
Maximum Allowable External Pressure	Pmax	192.29	bars

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -29 °C

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -29 °C

Flange MDMT including temperature reduction per 3.11:

Flange MDMT with Temp reduction -34 °C

The Drop for this Nozzle is : 118.1079 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 575.1079 mm.

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Input, Nozzle Desc: A120"

From: 60

Pressure for Reinforcement Calculations	P	192.204	bars
Temperature for Internal Pressure	Temp	85	°C
Design External Pressure	Pext	1.03	bars
Temperature for External Pressure	Tempex	85	°C
Parent Material		SA-516 70	+N+IT
Parent Allowable Stress at Temperature	Sv	161.10	N./mm ²
Parent Allowable Stress At Ambient	Sva	175.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	2450.00	mm.
Design Length of Section	L	5498.3330	mm.
Shell Finished (Minimum) Thickness	t	157.0000	mm.
Shell Internal Corrosion Allowance	cas	1.5000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		2875.00	mm.
User Entered Minimum Design Metal Temperature		-40.00	°C

Type of Element Connected to the Parent : Nozzle

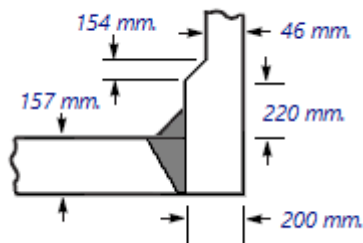
Material		SA-350 LF1	+IT
Material UNS Number		K03009	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	127.70	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		20.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	46.0000	mm.
Hub Height of Integral Nozzle	h	220.0000	mm.
Height of Beveled Transition	L'	154.0000	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		200.0000	mm.
Corrosion Allowance	can	1.5000	mm.
Outside Projection	ho	250.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	157.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: A120"

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	20.000	in.
Actual Thickness Used in Calculation	1.811	in.



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 511.0 \cdot (\exp(192.204/(127.7 \cdot 1.0)) - 1) + 1.5 + 0.0$$

$$= 41.5034 + 1.5 + 0.0 = 43.0034 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 127.7 \cdot 1.0 \cdot \ln((2 \cdot 44.5 + 511.0) / 511.0) - 0.0$$

$$= 205.023 - 0.0 = 205.0232 \text{ bars}$$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c : 9.840 mm
 Min. required shell thickness, internal p + c : 157.123 mm.
 Min. required shell thickness, external p + c : 14.107 mm.

Nozzle neck minimum required thickness : 9.840 mm.

Minimum nozzle neck thickness 45.999 mm., is greater than required thickness 9.840 mm.

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 68.8,	Allowable	: 127.7 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 263.3 N./mm ²	Passed
Occasional	: 50.8,	Allowable	: 169.8 N./mm ²	Passed
Shear	: 14.1,	Allowable	: 89.4 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[127.7/161.1, 1]$$

$$= 0.793$$

Thickness of Nozzle at Shell [tn]:

$$= \text{hub thickness} - \text{corrosion allowance}$$

$$= 200.0 - 1.5$$

$$= 198.500 \text{ mm.}$$

Thickness of Nozzle at Top [tn2]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 46.0 - 1.5$$

$$= 44.500 \text{ mm.}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D_i/t$$

$$= 2453.0/155.5$$

$$= 15.775 \text{ must be less than } 400.$$

Effective Pressure Radius [Reff]:



$$\begin{aligned}
 &= Di/2 + \text{corrosion allowance} \\
 &= 2450.0/2 + 1.5 \\
 &= 1226.500 \text{ mm.}
 \end{aligned}$$

$$\begin{aligned}
 &\text{Effective Length of Vessel Wall [LR]:} \\
 &= \min(\sqrt{Reff \cdot t}, 2 \cdot Rn) \\
 &= \min(\sqrt{1226.5 \cdot 155.5}, 2 \cdot 255.5) \\
 &= 436.716 \text{ mm.}
 \end{aligned}$$

$$\begin{aligned}
 &\text{Thickness Limit Candidate [LH1]:} \\
 &= \min[1.5t, te] + \sqrt{Rn \cdot tn} \\
 &= \min[1.5 \cdot 155.5, 0.0] + \sqrt{255.5 \cdot 198.5} \\
 &= 225.204 \text{ mm.}
 \end{aligned}$$

$$\begin{aligned}
 &\text{Thickness Limit Candidate [LH2]:} \\
 &= Lpr1 \\
 &= 250.0 \\
 &= 250.000 \text{ mm.}
 \end{aligned}$$

$$\begin{aligned}
 &\text{Thickness Limit Candidate [LH3]:} \\
 &= 8(t + te) \\
 &= 8(155.5 + 0.0) \\
 &= 1244.000 \text{ mm.}
 \end{aligned}$$

$$\begin{aligned}
 &\text{Effective Nozzle Wall Length Outside the Vessel [LH]:} \\
 &= \min[LH1, LH2, LH3] + t \quad \text{for set-in nozzles} \\
 &= \min[225.204, 250.0, 1244.0] + 155.5 \\
 &= 380.704 \text{ mm.}
 \end{aligned}$$

$$\begin{aligned}
 &\text{Effective Vessel Thickness [teff]:} \\
 &= t + ((A5 \cdot frp) / LR) \\
 &= 155.5 + ((0.0 \cdot 1.0) / 436.716) \\
 &= 155.500 \text{ mm.}
 \end{aligned}$$

$$\begin{aligned}
 &\text{Parameter } [\lambda]: \\
 &= \min(12, (2 \cdot Rn + tn) / (\sqrt{(Di + teff) \cdot teff})) \\
 &= \min(12, (2 \cdot 255.5 + 198.5) / (\sqrt{(2453.0 + 155.5) \cdot 155.5})) \\
 &= 1.114
 \end{aligned}$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

$$\begin{aligned}
 &\text{Area Contributed by the Vessel Wall [A1]:} \\
 &= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1) \\
 &= 155.5 \cdot 436.716 \cdot \max((1.114/5)^{0.85}, 1) \\
 &= 679.093 \text{ cm}^2
 \end{aligned}$$

$$\begin{aligned}
 &\text{Area Contributed by the Nozzle Outside the Vessel Wall [A2]:} \\
 &= A2A + A2B \\
 &= 745.367 + 10.194 \\
 &= 755.562 \text{ cm}^2
 \end{aligned}$$

$$\begin{aligned}
 &\text{Area Contributed by the Outside Fillet Weld [A41]:} \\
 &= 0.5 \cdot Leg^2 \\
 &= 0.5 \cdot 10.0^2 \\
 &= 0.500 \text{ cm}^2
 \end{aligned}$$

$$\begin{aligned}
 &\text{The total area contributed by A1 through A43 [AT]:} \\
 &= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43 \\
 &= 679.093 + 0.793(755.562 + 0.0) + 0.5 + 0.0 + 0.0 \\
 &= 1278.508 \text{ cm}^2
 \end{aligned}$$

$$\begin{aligned}
 &\text{Nozzle Radius for Force Calculation [Rxn]:} \\
 &= tn / \ln[1 + tn/Rn] \\
 &= 198.5 / \ln[1 + 198.5/255.5] \\
 &= 345.293 \text{ mm.}
 \end{aligned}$$

$$\begin{aligned}
 &\text{Shell Radius for Force Calculation [Rxs]:} \\
 &= teff / \ln[1 + teff/Reff] \\
 &= 155.5 / \ln[1 + 155.5/1226.5] \\
 &= 1302.704 \text{ mm.}
 \end{aligned}$$



Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$

$$= 1.5 \cdot 161.1 \cdot 1.0$$

$$= 241.7 \text{ N./mm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot L_H$$

$$= 192.204 \cdot 345.293 \cdot 380.704$$

$$= 257652.3 \text{ Kg}$$

Force acting on the Shell [fS]:

$$= P \cdot R_{xs} (L_R + t_n)$$

$$= 192.204 \cdot 1302.704 (436.716 + 198.5)$$

$$= 1621909.5 \text{ Kg}$$

Discontinuity Force from Internal Pressure [fY]:

$$= P \cdot R_{xs} \cdot R_{nc}$$

$$= 192.204 \cdot 1302.704 \cdot 255.5$$

$$= 652373.4 \text{ Kg}$$

Area Resisting Internal Pressure [Ap]:

$$= (f_S + f_Y + f_N) / P$$

$$= (1621910 + 652373 + 257652.3) / 192.204$$

$$= 12917.9 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= Sallow / (2 \cdot A_p / A_T - R_{xs} / t_{eff})$$

$$= 241.65 / (2 \cdot 12917.931 / 1278.508 - 1302.704 / 155.5)$$

$$= 204.3 \text{ bars}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= S / [t / R_{xs}]$$

$$= 161.1 [155.5 / 1302.704]$$

$$= 192.3 \text{ bars}$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(P_{max1}, P_{max2})$$

$$= \min(204.252, 192.289)$$

$$= 192.289 \text{ bars}$$

Average Primary Membrane Stress [σAvg]:

$$= (f_N + f_S + f_Y) / A_T$$

$$= (257652.328 + 1621910 + 652373) / 1278.508$$

$$= 194.213 \text{ N./mm}^2$$

General Primary Membrane Stress [σCirc]:

$$= P \cdot R_{xs} / t_{eff}$$

$$= 192.204 \cdot 1302.704 / 155.5$$

$$= 161.0 \text{ N./mm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$$

$$= \max(2 \cdot 194.213 - 161.029, 161.029)$$

$$= 227.4 \text{ N./mm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	241.65	N./mm ²
Local Primary Membrane Stress	PL	227.40	N./mm ²
Maximum Allowable Working Pressure	Pmax	192.29	bars

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= (R_{nc} + t_n) / R_{nc}$$

$$= (255.5 + 198.5) / 255.5$$

$$= 1.777 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$= \pi / 2 \cdot (R_n + t_n)$$



$$= \pi/2 \cdot (255.5 + 198.5) \\ = 713.142 \text{ mm.}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:
= 7.071, 0.000, 0.000, mm.

Required Weld Throat Dimensions :
tc, L41T = 7.071, must be $\geq \min(0.7 \cdot 198.500, 6.000)$ Passed

Weld Load Value [fwelds]:
= $\min(f_y \cdot k_y, 1.5 \cdot S_n(A_2 + A_3), \pi/4 \cdot P \cdot R_n^2 \cdot k_y^2)$
= $\min(652373 \cdot 1.78, 1.5 \cdot 127.7(755.562 + 0.0), \pi/4 \cdot 192.2 \cdot 255.5^2 \cdot 1.78^2)$
= 317293.531 Kgf

Weld Stress Value [τ]:
= $fwelds / (L_t(0.49 \cdot L_{41T} + 0.6 \cdot t_{w1} + 0.49 \cdot L_{43T}))$
= $317293.531 / (713.142(0.49 \cdot 7.071 + 0.6 \cdot 155.5 + 0.49 \cdot 0.0))$
= 45.092 < or = to 161.100 Weld Stress Passed

Nozzle Calculations per Section 4.5: External Pressure Case:

Nozzle Radius for Force Calculation [Rxn]:
= $t_n / \ln[1 + t_n/R_n]$
= $198.5 / \ln[1 + 198.5/255.5]$
= 345.293 mm.

Shell Radius for Force Calculation [Rxs]:
= $t_{eff} / \ln[1 + t_{eff}/R_{eff}]$
= $155.5 / \ln[1 + 155.5/1226.5]$
= 1302.704 mm.

Allowable Local Primary Membrane Stress [Sallow]:
= $1.5 \cdot S$
= $1.5 \cdot 161.1$
= 241.7 N./mm²

Force acting on the Nozzle [fN]:
= $P \cdot R_{xn} \cdot L_H$
= $1.034 \cdot 345.293 \cdot 380.704$
= 1386.3 Kgf

Force acting on the Shell [fS]:
= $P \cdot R_{xs}(L_R + t_n)$
= $1.034 \cdot 1302.704(436.716 + 198.5)$
= 8727.0 Kgf

Discontinuity Force from External Pressure [fY]:
= $P \cdot R_{xs} \cdot R_{nc}$
= $1.034 \cdot 1302.704 \cdot 255.5$
= 3510.2 Kgf

Area Resisting External Pressure [Ap]:
= $(f_S + f_Y + f_N)/P$
= $(8727.0 + 3510.2 + 1386.3)/1.034$
= 12917.9 cm²

Maximum Allowable Working Pressure Candidate [Pmax1]:
= $S_{allow} / (2 \cdot A_p / A_T - R_{xs} / t_{eff})$
= $241.65 / (2 \cdot 12917.932 / 1278.508 - 1302.704 / 155.5)$
= 204.3 bars

Maximum Allowable Working Pressure Candidate [Pmax2]:
= $S / (t / R_{xs})$
= $161.1(155.5 / 1302.704)$
= 192.3 bars

Maximum Allowable Working Pressure [Pmax]:
= $\min(P_{max1}, P_{max2})$
= $\min(204.252, 192.289)$
= 192.289 bars

Average Primary Membrane Stress [σ_{Avg}]:
 $= (fN + fS + fY) / AT$
 $= (1386.345 + 8726.978 + 3510.213) / 1278.508$
 $= 1.045 \text{ N./mm}^2$

General Primary Membrane Stress [σ_{Circ}]:
 $= P \cdot Rxs / teff$
 $= 1.034 \cdot 1302.704 / 155.5$
 $= 0.9 \text{ N./mm}^2$

Maximum Local Primary Membrane Stress [PL]:
 $= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$
 $= \max(2 \cdot 1.045 - 0.866, 0.866)$
 $= 1.2 \text{ N./mm}^2$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	241.65	N./mm ²
Local Primary Membrane Stress	PL	1.22	N./mm ²
Maximum Allowable External Pressure	Pmax	192.29	bars

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -29 °C

The Drop for this Nozzle is : 87.2351 mm.
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 494.2351 mm.

Input Echo, WRC297 Item 1, Description: A120" :

Diameter Basis for Cylindrical Shell	ID		
Shell Internal Corrosion Allowance	1.5000	mm.	
Shell External Corrosion Allowance	0.0000	mm.	
Shell Diameter	2450.000	mm.	
Shell Thickness	157.0000	mm.	
Shell Stress Concentration Factor	1.000		
Vessel Material	SA-516 70		
Vessel Cold S.I. Allowable	Smc	175.00	N./mm ²
Vessel Hot S.I. Allowable	Smh	161.10	N./mm ²
Diameter Basis for Nozzle	ID		
Nozzle Corrosion Allowance	1.5000	mm.	
Nozzle Diameter	508.000	mm.	
Nozzle Thickness	200.0000	mm.	
Nozzle Stress Concentration Factor	1.000		
Nozzle Material	SA-350 LF1		
Nozzle Cold S.I. Allowable	SNmc	138.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh	127.70	N./mm ²

Note:

*External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSained loads.*

Design Internal Pressure	Dp	192.20	bars
Radial Load	P	-12237.40	Kgf
Circumferential Shear	Vc	9178.05	Kgf
Longitudinal Shear	Vl	12237.40	Kgf
Circumferential Moment	Mc	12232.69	Kg-m.
Longitudinal Moment	Ml	15922.85	Kg-m.
Torsional Moment	Mt	18349.04	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

*Warning - The ratio, Dv/Tv (17.605) is < 20
 check the limitations of WRC bulletin 297.
 Warning - The ratio, Dn/Tn (4.540) is < 10
 check the limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$
 $= 908.0 + 2 \cdot 1.65 \cdot \sqrt{(1304.25(157.0 - 1.5))}$
 $= 2394.139 \text{ mm.}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 2608.500 mm.
 Nozzle Outside Diameter (d) = 908.000 mm.
 Vessel Thickness used (T) = 155.500 mm.
 Nozzle Thickness used (t) = 198.500 mm.
 T / t = 0.783
 d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.426$$

Nr/P = 0.096
 Mr/P = 0.161
 M0/P = 0.051
 N0/P = 0.112
 MrD/Mc = 0.280
 NrDL/Mc = 0.124
 M0d/Mc = 0.096
 N0DL/Mc = 0.052
 MrD/M1 = 0.153
 NrDL/M1 = 0.087
 M0D/M1 = 0.053
 N0DL/M1 = 0.134

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	0	0	0	0
Outplane Bending (P)	1	-1	1	-1
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	-2	2	2	-2
Normal Pressure Stress	142	161	142	161
Outplane Stress Summary	141	162	147	159

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	4	-4	4	-4
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	-6	6	6	-6
Inplane Pressure Stress	71	71	71	71
Inplane Stress Summary	69	72	83	61

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to	Au Top	Al Top	Bu Bottom	Bl Bottom
---	-----------	-----------	--------------	--------------



longitudinal plane)	Outside	Inside	Outside	Inside
---------------------	---------	--------	---------	--------

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	0	0	0	0

Shear Stress Summary	1	1	0	0
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
---	----------------------	---------------------	-------------------------	------------------------

Two * Max Shear Stress	141	162	147	159
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Outplane Membrane (P)	0	0	0	0
Outplane Bending (P)	1	-1	1	-1
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	-3	3	3	-3
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	71	71	71	71

Outplane Stress Summary	69	73	76	67
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	4	-4	4	-4
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	-9	9	9	-9
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	142	161	142	161

Inplane Stress Summary	137	165	157	148
------------------------	-----	-----	-----	-----

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	0	0	0	0

Shear Stress Summary	0	0	1	1
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Two * Max Shear Stress	137	165	157	148
------------------------	-----	-----	-----	-----

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
---	----------------------	---------------------	-------------------------	------------------------

Hoop Membrane (P)	0	0	0	0
Hoop Bending (P)	0	0	0	0



Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	21	44	21	44

Hoop Stress Summary	20	44	22	45
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to	Top	Top	Bottom	Bottom
pipe cross-section)	Outside	Inside	Outside	Inside

Axial Membrane (P)	0	0	0	0
Axial Bending (P)	2	-2	2	-2
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-2	-2	2	2
Axial Bending (ML)	-2	2	2	-2
Axial Pressure Stress	10	10	10	10

Axial Stress Summary	7	9	18	8
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0

Shear Stress Summary	1	1	0	0
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities	Outside	Inside	Outside	Inside

Two * Max Shear Stress	21	44	22	45
------------------------	----	----	----	----

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses in the	Left	Left	Right	Right
hoop direction)	Outside	Inside	Outside	Inside

Hoop Membrane (P)	0	0	0	0
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	21	44	21	44

Hoop Stress Summary	21	44	22	45
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to	Left	Left	Right	Right
pipe cross-section)	Outside	Inside	Outside	Inside

Axial Membrane (P)	0	0	0	0
Axial Bending (P)	2	-2	2	-2
Axial Membrane (Mc)	-1	-1	1	1
Axial Bending (Mc)	-4	4	4	-4
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	10	10	10	10

Axial Stress Summary	6	11	19	6
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0
Shear Stress Summary	0	0	1	1

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	21	44	22	45

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	142	161	142	161	142	161	142	161
Circ. Pl (SUS)	0	0	0	0	0	0	0	0
Circ. Q (SUS)	-1	1	3	-3	-5	5	13	-13
Long. Pm (SUS)	71	71	71	71	71	71	71	71
Long. Pl (SUS)	0	0	0	0	0	0	0	0
Long. Q (SUS)	-2	2	10	-10	-2	2	4	-4
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	142.0	161.0	142.0	161.0	142.0	161.0	142.0	161.0
Pm+Pl (SUS)	142.0	161.0	142.0	161.0	142.0	161.0	142.0	161.0
Pm+Pl+Q (Total)	141.0	162.0	145.0	158.0	137.0	166.0	155.0	148.0

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	161.00	161.10	Passed
Pm+Pl (SUS)	161.00	241.65	Passed
Pm+Pl+Q (TOTAL)	166.00	483.30	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	21	44	21	44	21	44	21	44
Circ. Pl (SUS)	0	0	0	0	0	0	0	0
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	10	10	10	10	10	10	10	10
Long. Pl (SUS)	-2	-2	2	2	-1	-1	1	1
Long. Q (SUS)	0	0	4	-4	-2	2	6	-6
Shear Pm (SUS)	0	0	0	0	0	0	0	0



Shear P1 (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	21.0	44.0	21.0	44.0	21.0	44.0	21.0	44.0
Pm+P1 (SUS)	21.0	44.0	21.0	44.0	21.0	44.0	21.0	44.0
Pm+P1+Q (Total)	21.0	44.0	21.0	44.0	21.0	44.0	21.0	44.0

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	44.00	127.70	Passed
Pm+P1 (SUS)	44.00	191.55	Passed
Pm+P1+Q (TOTAL)	44.00	383.10	Passed

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.

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Input, Nozzle Desc: L12"

From: 60

Pressure for Reinforcement Calculations	P	192.204	bars
Temperature for Internal Pressure	Temp	85	°C
Design External Pressure	Pext	1.03	bars
Temperature for External Pressure	Tempex	85	°C
Parent Material		SA-516 70	+N+IT
Parent Allowable Stress at Temperature	Sv	161.10	N./mm ²
Parent Allowable Stress At Ambient	Sva	175.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	2450.00	mm.
Design Length of Section	L	5498.3330	mm.
Shell Finished (Minimum) Thickness	t	157.0000	mm.
Shell Internal Corrosion Allowance	cas	1.5000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		2200.00	mm.
User Entered Minimum Design Metal Temperature		-40.00	°C

Type of Element Connected to the Parent : Nozzle

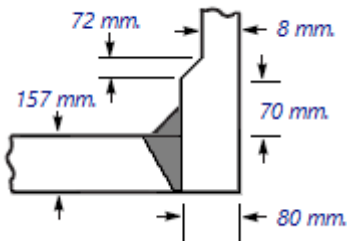
Material		SA-350 LF1	+IT
Material UNS Number		K03009	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	127.70	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	8.0000	mm.
Flange Material		SA-350 LF1	+IT
Hub Height of Integral Nozzle	h	70.0000	mm.
Height of Beveled Transition	L'	72.0000	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		80.0000	mm.
Corrosion Allowance	can	1.5000	mm.
Outside Projection	ho	200.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	157.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.
Flange Class		1500	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: L12"

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	2.000	in.
Actual Thickness Used in Calculation	0.315	in.



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 53.8 \cdot (\exp(192.204/(127.7 \cdot 1.0)) - 1) + 1.5 + 0.0$$

$$= 4.3696 + 1.5 + 0.0 = 5.8696 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 127.7 \cdot 1.0 \cdot \ln((2 \cdot 6.5 + 53.8) / 53.8) - 0.0$$

$$= 276.365 - 0.0 = 276.3645 \text{ bars}$$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c :	6.020 mm
Min. required shell thickness, internal p + c	157.123 mm.
Min. required shell thickness, external p + c	14.107 mm.

Nozzle neck minimum required thickness	: 6.020 mm.
--	-------------

Minimum nozzle neck thickness 8.001 mm., is greater than required thickness 6.020 mm.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[127.7/161.1, 1]$$

$$= 0.793$$

Thickness of Nozzle at Shell [tn]:

$$= \text{hub thickness} - \text{corrosion allowance}$$

$$= 80.0 - 1.5$$

$$= 78.500 \text{ mm.}$$

Thickness of Nozzle at Top [tn2]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 8.0 - 1.5$$

$$= 6.500 \text{ mm.}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D_i/t$$

$$= 2453.0/155.5$$

$$= 15.775 \text{ must be less than 400.}$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 2450.0/2 + 1.5$$

$$= 1226.500 \text{ mm.}$$

Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$$

$$= \min(\sqrt{1226.5 \cdot 155.5}, 2 \cdot 26.9)$$

$$= 53.800 \text{ mm.}$$

Thickness Limit Candidate [LH1]:

$$= \min[1.5t, t_e] + \sqrt{R_n \cdot t_n}$$



$$= \min[1.5 \cdot 155.5, 0.0] + \sqrt{(26.9 \cdot 78.5)}$$

$$= 45.953 \text{ mm.}$$

Thickness Limit Candidate [LH2]:

$$= L_{pr1}$$

$$= 200.0$$

$$= 200.000 \text{ mm.}$$

Thickness Limit Candidate [LH3]:

$$= 8(t + t_e)$$

$$= 8(155.5 + 0.0)$$

$$= 1244.000 \text{ mm.}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$= \min[LH1, LH2, LH3] + t \text{ for set-in nozzles}$$

$$= \min[45.953, 200.0, 1244.0] + 155.5$$

$$= 201.453 \text{ mm.}$$

Effective Vessel Thickness [teff]:

$$= t + ((A5 \cdot frp) / LR)$$

$$= 155.5 + ((0.0 \cdot 1.0) / 53.8)$$

$$= 155.500 \text{ mm.}$$

Parameter [λ]:

$$= \min(12, (2 \cdot R_n + t_n) / (\sqrt{(D_i + t_{eff}) \cdot t_{eff}}))$$

$$= \min(12, (2 \cdot 26.9 + 78.5) / (\sqrt{(2453.0 + 155.5) \cdot 155.5}))$$

$$= 0.208$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1)$$

$$= 155.5 \cdot 53.8 \cdot \max((0.208/5)^{0.85}, 1)$$

$$= 83.659 \text{ cm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= t_n \cdot LH$$

$$= 78.5 \cdot 201.453$$

$$= 158.140 \text{ cm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot \text{Leg}^2$$

$$= 0.5 \cdot 10.0^2$$

$$= 0.500 \text{ cm}^2$$

The total area contributed by A1 through A43 [AT]:

$$= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43$$

$$= 83.659 + 0.793(158.14 + 0.0) + 0.5 + 0.0 + 0.0$$

$$= 209.513 \text{ cm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n/R_n]$$

$$= 78.5 / \ln[1 + 78.5/26.9]$$

$$= 57.482 \text{ mm.}$$

Shell Radius for Force Calculation [Rxs]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$

$$= 155.5 / \ln[1 + 155.5/1226.5]$$

$$= 1302.704 \text{ mm.}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$

$$= 1.5 \cdot 161.1 \cdot 1.0$$

$$= 241.7 \text{ N/mm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot LH$$

$$= 192.204 \cdot 57.482 \cdot 201.453$$

$$= 22696.9 \text{ Kg}$$

Force acting on the Shell [fS]:

$$= P \cdot Rxs(LR + tn)$$

$$= 192.204 \cdot 1302.704(53.8 + 78.5)$$

$$= 337804.3 \text{ Kgf}$$

Discontinuity Force from Internal Pressure [fY]:

$$= P \cdot Rxs \cdot Rnc$$

$$= 192.204 \cdot 1302.704 \cdot 26.9$$

$$= 68684.3 \text{ Kgf}$$

Area Resisting Internal Pressure [Ap]:

$$= (fS + fY + fN)/P$$

$$= (337804.3 + 68684.3 + 22696.9)/192.204$$

$$= 2189.7 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= Sallow / (2 \cdot Ap/AT - Rxs/teff)$$

$$= 241.65 / (2 \cdot 2189.704/209.513 - 1302.704/155.5)$$

$$= 192.9 \text{ bars}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= S[t/Rxs]$$

$$= 161.1[155.5/1302.704]$$

$$= 192.3 \text{ bars}$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(Pmax1, Pmax2)$$

$$= \min(192.918, 192.289)$$

$$= 192.289 \text{ bars}$$

Average Primary Membrane Stress [σAvg]:

$$= (fN + fS + fY) / AT$$

$$= (22696.936 + 337804.281 + 68684.312)/209.513$$

$$= 200.892 \text{ N./mm}^2$$

General Primary Membrane Stress [σCirc]:

$$= P \cdot Rxs / teff$$

$$= 192.204 \cdot 1302.704/155.5$$

$$= 161.0 \text{ N./mm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma Avg - \sigma Circ, \sigma Circ)$$

$$= \max(2 \cdot 200.892 - 161.029, 161.029)$$

$$= 240.8 \text{ N./mm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	241.65	N./mm ²
Local Primary Membrane Stress	PL	240.76	N./mm ²
Maximum Allowable Working Pressure	Pmax	192.29	bars

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= (Rnc + tn) / Rnc$$

$$= (26.9 + 78.5)/26.9$$

$$= 3.918 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$= \pi/2 \cdot (Rn + tn)$$

$$= \pi/2 \cdot (26.9 + 78.5)$$

$$= 165.562 \text{ mm.}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm.}$$

Required Weld Throat Dimensions :

$$tc, L41T = 7.071, \text{ must be } \geq \min(0.7 \cdot 78.500, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$= \min(fy \cdot ky, 1.5 \cdot Sn(A2 + A3), \pi/4 \cdot P \cdot Rn^2 \cdot ky^2)$$



$$= \min(68684 \cdot 3.92, 1.5 \cdot 127.7(158.14 + 0.0), \pi/4 \cdot 192.2 \cdot 26.9^2 \cdot 3.92^2)$$

$$= 17101.365 \text{ Kg}$$

Weld Stress Value [τ]:

$$= \text{fwelds} / (L_T(0.49 \cdot L_{41T} + 0.6 \cdot t_{w1} + 0.49 \cdot L_{43T}))$$

$$= 17101.365 / (165.562(0.49 \cdot 7.071 + 0.6 \cdot 155.5 + 0.49 \cdot 0.0))$$

$$= 10.468 < \text{or} = \text{to } 161.100 \text{ Weld Stress Passed}$$

Nozzle Calculations per Section 4.5: External Pressure Case:

Nozzle Radius for Force Calculation [R_{xn}]:

$$= t_n / \ln[1 + t_n/R_n]$$

$$= 78.5 / \ln[1 + 78.5/26.9]$$

$$= 57.482 \text{ mm.}$$

Shell Radius for Force Calculation [R_{xs}]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$

$$= 155.5 / \ln[1 + 155.5/1226.5]$$

$$= 1302.704 \text{ mm.}$$

Allowable Local Primary Membrane Stress [S_{allow}]:

$$= 1.5 \cdot S$$

$$= 1.5 \cdot 161.1$$

$$= 241.7 \text{ N./mm}^2$$

Force acting on the Nozzle [f_N]:

$$= P \cdot R_{xn} \cdot L_H$$

$$= 1.034 \cdot 57.482 \cdot 201.453$$

$$= 122.1 \text{ Kg}$$

Force acting on the Shell [f_S]:

$$= P \cdot R_{xs} (L_R + t_n)$$

$$= 1.034 \cdot 1302.704 (53.8 + 78.5)$$

$$= 1817.6 \text{ Kg}$$

Discontinuity Force from External Pressure [f_Y]:

$$= P \cdot R_{xs} \cdot R_{nc}$$

$$= 1.034 \cdot 1302.704 \cdot 26.9$$

$$= 369.6 \text{ Kg}$$

Area Resisting External Pressure [A_p]:

$$= (f_S + f_Y + f_N) / P$$

$$= (1817.6 + 369.6 + 122.1) / 1.034$$

$$= 2189.7 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [P_{max1}]:

$$= S_{allow} / (2 \cdot A_p / AT - R_{xs} / t_{eff})$$

$$= 241.65 / (2 \cdot 2189.704 / 209.513 - 1302.704 / 155.5)$$

$$= 192.9 \text{ bars}$$

Maximum Allowable Working Pressure Candidate [P_{max2}]:

$$= S / t / R_{xs}$$

$$= 161.1 [155.5 / 1302.704]$$

$$= 192.3 \text{ bars}$$

Maximum Allowable Working Pressure [P_{max}]:

$$= \min(P_{max1}, P_{max2})$$

$$= \min(192.918, 192.289)$$

$$= 192.289 \text{ bars}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$= (f_N + f_S + f_Y) / AT$$

$$= (122.125 + 1817.617 + 369.568) / 209.513$$

$$= 1.081 \text{ N./mm}^2$$

General Primary Membrane Stress [σ_{Circ}]:

$$= P \cdot R_{xs} / t_{eff}$$

$$= 1.034 \cdot 1302.704 / 155.5$$

$$= 0.9 \text{ N./mm}^2$$



Maximum Local Primary Membrane Stress [PL]:

$$\begin{aligned} &= \max(2 \cdot \sigma_{\text{Avg}} - \sigma_{\text{Circ}}, \sigma_{\text{Circ}}) \\ &= \max(2 \cdot 1.081 - 0.866, 0.866) \\ &= 1.3 \text{ N./mm}^2 \end{aligned}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	241.65	N./mm ²
Local Primary Membrane Stress	PL	1.30	N./mm ²
Maximum Allowable External Pressure	Pmax	192.29	bars

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -29 °C

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -29 °C

Flange MDMT including temperature reduction per 3.11:

Flange MDMT with Temp reduction -43 °C

The Drop for this Nozzle is : 4.5427 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 361.5428 mm.

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Input, Nozzle Desc: L32"

From: 60

Pressure for Reinforcement Calculations	P	192.204	bars
Temperature for Internal Pressure	Temp	85	°C
Design External Pressure	Pext	1.03	bars
Temperature for External Pressure	Tempex	85	°C
Parent Material		SA-516 70	+N+IT
Parent Allowable Stress at Temperature	Sv	161.10	N./mm ²
Parent Allowable Stress At Ambient	Sva	175.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	2450.00	mm.
Design Length of Section	L	5498.3330	mm.
Shell Finished (Minimum) Thickness	t	157.0000	mm.
Shell Internal Corrosion Allowance	cas	1.5000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		2200.00	mm.
User Entered Minimum Design Metal Temperature		-40.00	°C

Type of Element Connected to the Parent : Nozzle

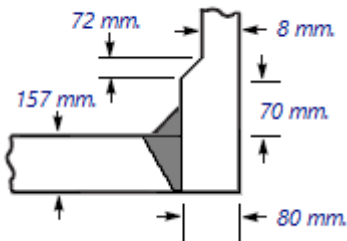
Material		SA-350 LF1	+IT
Material UNS Number		K03009	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	127.70	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	8.0000	mm.
Flange Material		SA-350 LF1	+IT
Hub Height of Integral Nozzle	h	70.0000	mm.
Height of Beveled Transition	L'	72.0000	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		80.0000	mm.
Corrosion Allowance	can	1.5000	mm.
Outside Projection	ho	200.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	157.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.
Flange Class		1500	
Flange Grade		GR 1.4	

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: L32"

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	2.000	in.
Actual Thickness Used in Calculation	0.315	in.



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 53.8 \cdot (\exp(192.204/(127.7 \cdot 1.0)) - 1) + 1.5 + 0.0$$

$$= 4.3696 + 1.5 + 0.0 = 5.8696 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 127.7 \cdot 1.0 \cdot \ln((2 \cdot 6.5 + 53.8) / 53.8) - 0.0$$

$$= 276.365 - 0.0 = 276.3645 \text{ bars}$$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c :	6.020 mm
Min. required shell thickness, internal p + c	157.123 mm.
Min. required shell thickness, external p + c	14.107 mm.

Nozzle neck minimum required thickness	: 6.020 mm.
--	-------------

Minimum nozzle neck thickness 8.001 mm., is greater than required thickness 6.020 mm.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[127.7/161.1, 1]$$

$$= 0.793$$

Thickness of Nozzle at Shell [tn]:

$$= \text{hub thickness} - \text{corrosion allowance}$$

$$= 80.0 - 1.5$$

$$= 78.500 \text{ mm.}$$

Thickness of Nozzle at Top [tn2]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 8.0 - 1.5$$

$$= 6.500 \text{ mm.}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D_i/t$$

$$= 2453.0/155.5$$

$$= 15.775 \text{ must be less than 400.}$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 2450.0/2 + 1.5$$

$$= 1226.500 \text{ mm.}$$

Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$$

$$= \min(\sqrt{1226.5 \cdot 155.5}, 2 \cdot 26.9)$$

$$= 53.800 \text{ mm.}$$

Thickness Limit Candidate [LH1]:

$$= \min[1.5t, t_e] + \sqrt{R_n \cdot t_n}$$



$$= \min[1.5 \cdot 155.5, 0.0] + \sqrt{(26.9 \cdot 78.5)}$$

$$= 45.953 \text{ mm.}$$

Thickness Limit Candidate [LH2]:

$$= L_{pr1}$$

$$= 200.0$$

$$= 200.000 \text{ mm.}$$

Thickness Limit Candidate [LH3]:

$$= 8(t + t_e)$$

$$= 8(155.5 + 0.0)$$

$$= 1244.000 \text{ mm.}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$= \min[LH1, LH2, LH3] + t \text{ for set-in nozzles}$$

$$= \min[45.953, 200.0, 1244.0] + 155.5$$

$$= 201.453 \text{ mm.}$$

Effective Vessel Thickness [teff]:

$$= t + ((A5 \cdot frp) / LR)$$

$$= 155.5 + ((0.0 \cdot 1.0) / 53.8)$$

$$= 155.500 \text{ mm.}$$

Parameter [λ]:

$$= \min(12, (2 \cdot R_n + t_n) / (\sqrt{(D_i + t_{eff}) \cdot t_{eff}}))$$

$$= \min(12, (2 \cdot 26.9 + 78.5) / (\sqrt{(2453.0 + 155.5) \cdot 155.5}))$$

$$= 0.208$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1)$$

$$= 155.5 \cdot 53.8 \cdot \max((0.208/5)^{0.85}, 1)$$

$$= 83.659 \text{ cm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= t_n \cdot LH$$

$$= 78.5 \cdot 201.453$$

$$= 158.140 \text{ cm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot \text{Leg}^2$$

$$= 0.5 \cdot 10.0^2$$

$$= 0.500 \text{ cm}^2$$

The total area contributed by A1 through A43 [AT]:

$$= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43$$

$$= 83.659 + 0.793(158.14 + 0.0) + 0.5 + 0.0 + 0.0$$

$$= 209.513 \text{ cm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n/R_n]$$

$$= 78.5 / \ln[1 + 78.5/26.9]$$

$$= 57.482 \text{ mm.}$$

Shell Radius for Force Calculation [Rxs]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$

$$= 155.5 / \ln[1 + 155.5/1226.5]$$

$$= 1302.704 \text{ mm.}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$

$$= 1.5 \cdot 161.1 \cdot 1.0$$

$$= 241.7 \text{ N./mm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot LH$$

$$= 192.204 \cdot 57.482 \cdot 201.453$$

$$= 22696.9 \text{ Kg}$$

Force acting on the Shell [fS]:

$$= P \cdot Rxs(LR + tn)$$

$$= 192.204 \cdot 1302.704(53.8 + 78.5)$$

$$= 337804.3 \text{ Kgf}$$

Discontinuity Force from Internal Pressure [fY]:

$$= P \cdot Rxs \cdot Rnc$$

$$= 192.204 \cdot 1302.704 \cdot 26.9$$

$$= 68684.3 \text{ Kgf}$$

Area Resisting Internal Pressure [Ap]:

$$= (fS + fY + fN)/P$$

$$= (337804.3 + 68684.3 + 22696.9)/192.204$$

$$= 2189.7 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= Sallow / (2 \cdot Ap/AT - Rxs/teff)$$

$$= 241.65 / (2 \cdot 2189.704/209.513 - 1302.704/155.5)$$

$$= 192.9 \text{ bars}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= S[t/Rxs]$$

$$= 161.1[155.5/1302.704]$$

$$= 192.3 \text{ bars}$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(Pmax1, Pmax2)$$

$$= \min(192.918, 192.289)$$

$$= 192.289 \text{ bars}$$

Average Primary Membrane Stress [σAvg]:

$$= (fN + fS + fY) / AT$$

$$= (22696.936 + 337804.281 + 68684.312)/209.513$$

$$= 200.892 \text{ N./mm}^2$$

General Primary Membrane Stress [σCirc]:

$$= P \cdot Rxs / teff$$

$$= 192.204 \cdot 1302.704/155.5$$

$$= 161.0 \text{ N./mm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma Avg - \sigma Circ, \sigma Circ)$$

$$= \max(2 \cdot 200.892 - 161.029, 161.029)$$

$$= 240.8 \text{ N./mm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	241.65	N./mm ²
Local Primary Membrane Stress	PL	240.76	N./mm ²
Maximum Allowable Working Pressure	Pmax	192.29	bars

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= (Rnc + tn) / Rnc$$

$$= (26.9 + 78.5)/26.9$$

$$= 3.918 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$= \pi/2 \cdot (Rn + tn)$$

$$= \pi/2 \cdot (26.9 + 78.5)$$

$$= 165.562 \text{ mm.}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm.}$$

Required Weld Throat Dimensions :

$$tc, L41T = 7.071, \text{ must be } \geq \min(0.7 \cdot 78.500, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$= \min(fy \cdot ky, 1.5 \cdot Sn(A2 + A3), \pi/4 \cdot P \cdot Rn^2 \cdot ky^2)$$



$$= \min(68684 \cdot 3.92, 1.5 \cdot 127.7(158.14 + 0.0), \pi/4 \cdot 192.2 \cdot 26.9^2 \cdot 3.92^2)$$

$$= 17101.365 \text{ Kg}$$

Weld Stress Value [τ]:

$$= \text{fwelds} / (L_T(0.49 \cdot L_{41T} + 0.6 \cdot t_{w1} + 0.49 \cdot L_{43T}))$$

$$= 17101.365 / (165.562(0.49 \cdot 7.071 + 0.6 \cdot 155.5 + 0.49 \cdot 0.0))$$

$$= 10.468 < \text{or} = \text{to } 161.100 \text{ Weld Stress Passed}$$

Nozzle Calculations per Section 4.5: External Pressure Case:

Nozzle Radius for Force Calculation [R_{xn}]:

$$= t_n / \ln[1 + t_n/R_n]$$

$$= 78.5 / \ln[1 + 78.5/26.9]$$

$$= 57.482 \text{ mm.}$$

Shell Radius for Force Calculation [R_{xs}]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$

$$= 155.5 / \ln[1 + 155.5/1226.5]$$

$$= 1302.704 \text{ mm.}$$

Allowable Local Primary Membrane Stress [S_{allow}]:

$$= 1.5 \cdot S$$

$$= 1.5 \cdot 161.1$$

$$= 241.7 \text{ N./mm}^2$$

Force acting on the Nozzle [f_N]:

$$= P \cdot R_{xn} \cdot L_H$$

$$= 1.034 \cdot 57.482 \cdot 201.453$$

$$= 122.1 \text{ Kg}$$

Force acting on the Shell [f_S]:

$$= P \cdot R_{xs} (L_R + t_n)$$

$$= 1.034 \cdot 1302.704 (53.8 + 78.5)$$

$$= 1817.6 \text{ Kg}$$

Discontinuity Force from External Pressure [f_Y]:

$$= P \cdot R_{xs} \cdot R_{nc}$$

$$= 1.034 \cdot 1302.704 \cdot 26.9$$

$$= 369.6 \text{ Kg}$$

Area Resisting External Pressure [A_p]:

$$= (f_S + f_Y + f_N) / P$$

$$= (1817.6 + 369.6 + 122.1) / 1.034$$

$$= 2189.7 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [P_{max1}]:

$$= S_{allow} / (2 \cdot A_p / AT - R_{xs} / t_{eff})$$

$$= 241.65 / (2 \cdot 2189.704 / 209.513 - 1302.704 / 155.5)$$

$$= 192.9 \text{ bars}$$

Maximum Allowable Working Pressure Candidate [P_{max2}]:

$$= S / t / R_{xs}$$

$$= 161.1 [155.5 / 1302.704]$$

$$= 192.3 \text{ bars}$$

Maximum Allowable Working Pressure [P_{max}]:

$$= \min(P_{max1}, P_{max2})$$

$$= \min(192.918, 192.289)$$

$$= 192.289 \text{ bars}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$= (f_N + f_S + f_Y) / AT$$

$$= (122.125 + 1817.617 + 369.568) / 209.513$$

$$= 1.081 \text{ N./mm}^2$$

General Primary Membrane Stress [σ_{Circ}]:

$$= P \cdot R_{xs} / t_{eff}$$

$$= 1.034 \cdot 1302.704 / 155.5$$

$$= 0.9 \text{ N./mm}^2$$



Maximum Local Primary Membrane Stress [PL]:

$$\begin{aligned} &= \max(2 \cdot \sigma_{\text{Avg}} - \sigma_{\text{Circ}}, \sigma_{\text{Circ}}) \\ &= \max(2 \cdot 1.081 - 0.866, 0.866) \\ &= 1.3 \text{ N./mm}^2 \end{aligned}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	241.65	N./mm ²
Local Primary Membrane Stress	PL	1.30	N./mm ²
Maximum Allowable External Pressure	Pmax	192.29	bars

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -29 °C

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -29 °C

Flange MDMT including temperature reduction per 3.11:

Flange MDMT with Temp reduction -34 °C

The Drop for this Nozzle is : 4.5427 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 361.5428 mm.

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Input, Nozzle Desc: L52"
From: 60

Pressure for Reinforcement Calculations	P	192.204	bars
Temperature for Internal Pressure	Temp	85	°C
Design External Pressure	Pext	1.03	bars
Temperature for External Pressure	Tempex	85	°C
Parent Material		SA-516 70	+N+IT
Parent Allowable Stress at Temperature	Sv	161.10	N./mm ²
Parent Allowable Stress At Ambient	Sva	175.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	2450.00	mm.
Design Length of Section	L	5498.3330	mm.
Shell Finished (Minimum) Thickness	t	157.0000	mm.
Shell Internal Corrosion Allowance	cas	1.5000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		2200.00	mm.
User Entered Minimum Design Metal Temperature		-40.00	°C

Type of Element Connected to the Parent : Nozzle

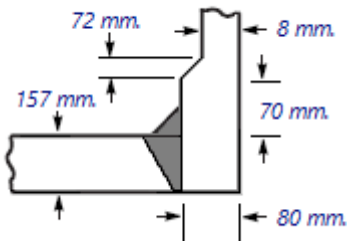
Material		SA-350 LF1	+IT
Material UNS Number		K03009	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	127.70	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	8.0000	mm.
Flange Material		SA-350 LF1	+IT
Hub Height of Integral Nozzle	h	70.0000	mm.
Height of Beveled Transition	L'	72.0000	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		80.0000	mm.
Corrosion Allowance	can	1.5000	mm.
Outside Projection	ho	200.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	157.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.
Flange Class		1500	
Flange Grade		GR 1.4	

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: L52"

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	2.000	in.
Actual Thickness Used in Calculation	0.315	in.



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 53.8 \cdot (\exp(192.204/(127.7 \cdot 1.0)) - 1) + 1.5 + 0.0$$

$$= 4.3696 + 1.5 + 0.0 = 5.8696 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 127.7 \cdot 1.0 \cdot \ln((2 \cdot 6.5 + 53.8) / 53.8) - 0.0$$

$$= 276.365 - 0.0 = 276.3645 \text{ bars}$$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c :	6.020 mm
Min. required shell thickness, internal p + c	157.123 mm.
Min. required shell thickness, external p + c	14.107 mm.

Nozzle neck minimum required thickness	: 6.020 mm.
--	-------------

Minimum nozzle neck thickness 8.001 mm., is greater than required thickness 6.020 mm.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[127.7/161.1, 1]$$

$$= 0.793$$

Thickness of Nozzle at Shell [tn]:

$$= \text{hub thickness} - \text{corrosion allowance}$$

$$= 80.0 - 1.5$$

$$= 78.500 \text{ mm.}$$

Thickness of Nozzle at Top [tn2]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 8.0 - 1.5$$

$$= 6.500 \text{ mm.}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D_i/t$$

$$= 2453.0/155.5$$

$$= 15.775 \text{ must be less than 400.}$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 2450.0/2 + 1.5$$

$$= 1226.500 \text{ mm.}$$

Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$$

$$= \min(\sqrt{1226.5 \cdot 155.5}, 2 \cdot 26.9)$$

$$= 53.800 \text{ mm.}$$

Thickness Limit Candidate [LH1]:

$$= \min[1.5t, t_e] + \sqrt{R_n \cdot t_n}$$



$$= \min[1.5 \cdot 155.5, 0.0] + \sqrt{(26.9 \cdot 78.5)}$$

$$= 45.953 \text{ mm.}$$

Thickness Limit Candidate [LH2]:

$$= L_{pr1}$$

$$= 200.0$$

$$= 200.000 \text{ mm.}$$

Thickness Limit Candidate [LH3]:

$$= 8(t + t_e)$$

$$= 8(155.5 + 0.0)$$

$$= 1244.000 \text{ mm.}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$= \min[LH1, LH2, LH3] + t \text{ for set-in nozzles}$$

$$= \min[45.953, 200.0, 1244.0] + 155.5$$

$$= 201.453 \text{ mm.}$$

Effective Vessel Thickness [teff]:

$$= t + ((A5 \cdot frp) / LR)$$

$$= 155.5 + ((0.0 \cdot 1.0) / 53.8)$$

$$= 155.500 \text{ mm.}$$

Parameter [λ]:

$$= \min(12, (2 \cdot R_n + t_n) / (\sqrt{(D_i + t_{eff}) \cdot t_{eff}}))$$

$$= \min(12, (2 \cdot 26.9 + 78.5) / (\sqrt{(2453.0 + 155.5) \cdot 155.5}))$$

$$= 0.208$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1)$$

$$= 155.5 \cdot 53.8 \cdot \max((0.208/5)^{0.85}, 1)$$

$$= 83.659 \text{ cm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= t_n \cdot LH$$

$$= 78.5 \cdot 201.453$$

$$= 158.140 \text{ cm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot \text{Leg}^2$$

$$= 0.5 \cdot 10.0^2$$

$$= 0.500 \text{ cm}^2$$

The total area contributed by A1 through A43 [AT]:

$$= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43$$

$$= 83.659 + 0.793(158.14 + 0.0) + 0.5 + 0.0 + 0.0$$

$$= 209.513 \text{ cm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n/R_n]$$

$$= 78.5 / \ln[1 + 78.5/26.9]$$

$$= 57.482 \text{ mm.}$$

Shell Radius for Force Calculation [Rxs]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$

$$= 155.5 / \ln[1 + 155.5/1226.5]$$

$$= 1302.704 \text{ mm.}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$

$$= 1.5 \cdot 161.1 \cdot 1.0$$

$$= 241.7 \text{ N./mm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot LH$$

$$= 192.204 \cdot 57.482 \cdot 201.453$$

$$= 22696.9 \text{ Kg}$$

Force acting on the Shell [fS]:

$$= P \cdot Rxs(LR + tn)$$

$$= 192.204 \cdot 1302.704(53.8 + 78.5)$$

$$= 337804.3 \text{ Kgf}$$

Discontinuity Force from Internal Pressure [fY]:

$$= P \cdot Rxs \cdot Rnc$$

$$= 192.204 \cdot 1302.704 \cdot 26.9$$

$$= 68684.3 \text{ Kgf}$$

Area Resisting Internal Pressure [Ap]:

$$= (fS + fY + fN)/P$$

$$= (337804.3 + 68684.3 + 22696.9)/192.204$$

$$= 2189.7 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= Sallow / (2 \cdot Ap/AT - Rxs/teff)$$

$$= 241.65 / (2 \cdot 2189.704/209.513 - 1302.704/155.5)$$

$$= 192.9 \text{ bars}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= S[t/Rxs]$$

$$= 161.1[155.5/1302.704]$$

$$= 192.3 \text{ bars}$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(Pmax1, Pmax2)$$

$$= \min(192.918, 192.289)$$

$$= 192.289 \text{ bars}$$

Average Primary Membrane Stress [σAvg]:

$$= (fN + fS + fY) / AT$$

$$= (22696.936 + 337804.281 + 68684.312)/209.513$$

$$= 200.892 \text{ N./mm}^2$$

General Primary Membrane Stress [σCirc]:

$$= P \cdot Rxs / teff$$

$$= 192.204 \cdot 1302.704/155.5$$

$$= 161.0 \text{ N./mm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma Avg - \sigma Circ, \sigma Circ)$$

$$= \max(2 \cdot 200.892 - 161.029, 161.029)$$

$$= 240.8 \text{ N./mm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	241.65	N./mm ²
Local Primary Membrane Stress	PL	240.76	N./mm ²
Maximum Allowable Working Pressure	Pmax	192.29	bars

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= (Rnc + tn) / Rnc$$

$$= (26.9 + 78.5)/26.9$$

$$= 3.918 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$= \pi/2 \cdot (Rn + tn)$$

$$= \pi/2 \cdot (26.9 + 78.5)$$

$$= 165.562 \text{ mm.}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm.}$$

Required Weld Throat Dimensions :

$$tc, L41T = 7.071, \text{ must be } \geq \min(0.7 \cdot 78.500, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$= \min(fy \cdot ky, 1.5 \cdot Sn(A2 + A3), \pi/4 \cdot P \cdot Rn^2 \cdot ky^2)$$



$$= \min(68684 \cdot 3.92, 1.5 \cdot 127.7(158.14 + 0.0), \pi/4 \cdot 192.2 \cdot 26.9^2 \cdot 3.92^2)$$

$$= 17101.365 \text{ Kg}$$

Weld Stress Value [τ]:

$$= \text{fwelds} / (L_T(0.49 \cdot L_{41T} + 0.6 \cdot t_{w1} + 0.49 \cdot L_{43T}))$$

$$= 17101.365 / (165.562(0.49 \cdot 7.071 + 0.6 \cdot 155.5 + 0.49 \cdot 0.0))$$

$$= 10.468 < \text{or} = \text{to } 161.100 \text{ Weld Stress Passed}$$

Nozzle Calculations per Section 4.5: External Pressure Case:

Nozzle Radius for Force Calculation [R_{xn}]:

$$= t_n / \ln[1 + t_n/R_n]$$

$$= 78.5 / \ln[1 + 78.5/26.9]$$

$$= 57.482 \text{ mm.}$$

Shell Radius for Force Calculation [R_{xs}]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$

$$= 155.5 / \ln[1 + 155.5/1226.5]$$

$$= 1302.704 \text{ mm.}$$

Allowable Local Primary Membrane Stress [S_{allow}]:

$$= 1.5 \cdot S$$

$$= 1.5 \cdot 161.1$$

$$= 241.7 \text{ N./mm}^2$$

Force acting on the Nozzle [f_N]:

$$= P \cdot R_{xn} \cdot L_H$$

$$= 1.034 \cdot 57.482 \cdot 201.453$$

$$= 122.1 \text{ Kg}$$

Force acting on the Shell [f_S]:

$$= P \cdot R_{xs} (L_R + t_n)$$

$$= 1.034 \cdot 1302.704 (53.8 + 78.5)$$

$$= 1817.6 \text{ Kg}$$

Discontinuity Force from External Pressure [f_Y]:

$$= P \cdot R_{xs} \cdot R_{nc}$$

$$= 1.034 \cdot 1302.704 \cdot 26.9$$

$$= 369.6 \text{ Kg}$$

Area Resisting External Pressure [A_p]:

$$= (f_S + f_Y + f_N) / P$$

$$= (1817.6 + 369.6 + 122.1) / 1.034$$

$$= 2189.7 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [P_{max1}]:

$$= S_{allow} / (2 \cdot A_p / AT - R_{xs} / t_{eff})$$

$$= 241.65 / (2 \cdot 2189.704 / 209.513 - 1302.704 / 155.5)$$

$$= 192.9 \text{ bars}$$

Maximum Allowable Working Pressure Candidate [P_{max2}]:

$$= S / t / R_{xs}$$

$$= 161.1 [155.5 / 1302.704]$$

$$= 192.3 \text{ bars}$$

Maximum Allowable Working Pressure [P_{max}]:

$$= \min(P_{max1}, P_{max2})$$

$$= \min(192.918, 192.289)$$

$$= 192.289 \text{ bars}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$= (f_N + f_S + f_Y) / AT$$

$$= (122.125 + 1817.617 + 369.568) / 209.513$$

$$= 1.081 \text{ N./mm}^2$$

General Primary Membrane Stress [σ_{Circ}]:

$$= P \cdot R_{xs} / t_{eff}$$

$$= 1.034 \cdot 1302.704 / 155.5$$

$$= 0.9 \text{ N./mm}^2$$



Maximum Local Primary Membrane Stress [PL]:

$$\begin{aligned} &= \max(2 \cdot \sigma_{\text{Avg}} - \sigma_{\text{Circ}}, \sigma_{\text{Circ}}) \\ &= \max(2 \cdot 1.081 - 0.866, 0.866) \\ &= 1.3 \text{ N./mm}^2 \end{aligned}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	241.65	N./mm ²
Local Primary Membrane Stress	PL	1.30	N./mm ²
Maximum Allowable External Pressure	Pmax	192.29	bars

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -29 °C

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -29 °C

Flange MDMT including temperature reduction per 3.11:

Flange MDMT with Temp reduction -34 °C

The Drop for this Nozzle is : 4.5427 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 361.5428 mm.

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Input, Nozzle Desc: B124"

From: 70

Pressure for Reinforcement Calculations	P	192.204	bars
Temperature for Internal Pressure	Temp	85	°C
Design External Pressure	Pext	1.03	bars
Temperature for External Pressure	Tempex	85	°C
Parent Material		SA-516 70	+N+IT
Parent Allowable Stress at Temperature	Sv	161.14	N./mm ²
Parent Allowable Stress At Ambient	Sva	175.00	N./mm ²
Inside Diameter of Hemispherical Head	D	2545.00	mm.
Head Finished (Minimum) Thickness	t	80.0000	mm.
Head Internal Corrosion Allowance	cas	1.5000	mm.
Head External Corrosion Allowance	co	0.0000	mm.
Distance from Head Centerline	L1	0.0000	mm.
User Entered Minimum Design Metal Temperature		-40.00	°C

Type of Element Connected to the Parent : Nozzle

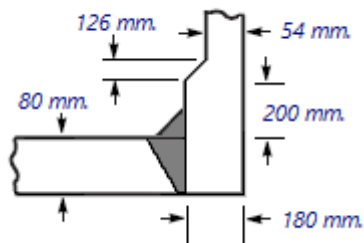
Material		SA-350 LF1	+IT
Material UNS Number		K03009	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	127.70	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		24.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	54.0000	mm.
Hub Height of Integral Nozzle	h	200.0000	mm.
Height of Beveled Transition	L'	126.0000	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		180.0000	mm.
Corrosion Allowance	can	1.5000	mm.
Outside Projection	ho	397.5000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	80.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: B124"

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	24.000	in.
Actual Thickness Used in Calculation	2.126	in.



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 612.6 \cdot (\exp(192.204/(127.7 \cdot 1.0)) - 1) + 1.5 + 0.0$$

$$= 49.7553 + 1.5 + 0.0 = 51.2553 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 127.7 \cdot 1.0 \cdot \ln((2 \cdot 52.5 + 612.6) / 612.6) - 0.0$$

$$= 202.01 - 0.0 = 202.0098 \text{ bars}$$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c : 9.840 mm
 Min. required shell thickness, internal p + c : 79.819 mm.
 Min. required shell thickness, external p + c : 6.561 mm.

Nozzle neck minimum required thickness : 9.840 mm.

Minimum nozzle neck thickness 54.000 mm., is greater than required thickness 9.840 mm.

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 67.0,	Allowable	: 127.7 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 265.1 N./mm ²	Passed
Occasional	: 51.6,	Allowable	: 169.8 N./mm ²	Passed
Shear	: 11.9,	Allowable	: 89.4 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[127.7/161.1, 1]$$

$$= 0.792$$

Thickness of Nozzle at Shell [tn]:

$$= \text{hub thickness} - \text{corrosion allowance}$$

$$= 180.0 - 1.5$$

$$= 178.500 \text{ mm.}$$

Thickness of Nozzle at Top [tn2]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 54.0 - 1.5$$

$$= 52.500 \text{ mm.}$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 2545.0/2 + 1.5$$

$$= 1274.000 \text{ mm.}$$

Effective Length of Vessel Wall [LR]:



$$= \min(\sqrt{(\text{Reff} \cdot t)}, 2 \cdot R_n)$$

$$= \min(\sqrt{(1274.0 \cdot 78.5)}, 2 \cdot 306.3)$$

$$= 316.242 \text{ mm.}$$

Intermediate value [Cn]:

$$= \min((t + t_e)/t_n)^{0.35}, 1)$$

$$= \min((78.5 + 0.0)/178.5)^{0.35}, 1)$$

$$= \min(0.75, 1)$$

$$= 0.750$$

Intermediate value [Cp]:

$$= 9999999.000$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$= \min(t + t_e + F_p \cdot \sqrt{(R_n \cdot t_n)}, L_{pr1} + t)$$

$$= \min(78.5 + 0.0 + 0.75 \cdot \sqrt{(306.3 \cdot 178.5)}, 397.5 + 78.5)$$

$$= 253.898 \text{ mm.}$$

Effective Vessel Thickness [teff]:

$$= t + ((A_5 \cdot f_{rp}) / LR)$$

$$= 78.5 + ((0.0 \cdot 1.0) / 316.242)$$

$$= 78.500 \text{ mm.}$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR$$

$$= 78.5 \cdot 316.242$$

$$= 248.250 \text{ cm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= t_n \cdot LH$$

$$= 178.5 \cdot 253.898$$

$$= 453.207 \text{ cm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot \text{Leg}^2$$

$$= 0.5 \cdot 10.0^2$$

$$= 0.500 \text{ cm}^2$$

The total area contributed by A1 through A43 [AT]:

$$= A_1 + f_{rn}(A_2 + A_3) + A_{41} + A_{42} + A_{43}$$

$$= 248.25 + 0.792(453.207 + 0.0) + 0.5 + 0.0 + 0.0$$

$$= 607.905 \text{ cm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n/R_n]$$

$$= 178.5 / \ln[1 + 178.5/306.3]$$

$$= 388.744 \text{ mm.}$$

Shell Radius for Force Calculation [Rxs]:

$$= t_{eff} / \ln[1 + t_{eff}/\text{Reff}]$$

$$= 78.5 / \ln[1 + 78.5/1274.0]$$

$$= 1312.860 \text{ mm.}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$

$$= 1.5 \cdot 161.141 \cdot 1.0$$

$$= 241.7 \text{ N./mm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot LH$$

$$= 192.204 \cdot 388.744 \cdot 253.898$$

$$= 193455.7 \text{ Kgf}$$

Force acting on the Shell [fS]:

$$= P \cdot R_{xs} \cdot (LR + t_n)/2$$

$$= 192.204 \cdot 1312.86 \cdot (316.242 + 178.5)/2$$

$$= 636541.6 \text{ Kgf}$$

Discontinuity Force from Internal Pressure [FY]:

$$= (P \cdot Rxs \cdot Rnc) / 2$$

$$= (192.204 \cdot 1312.86 \cdot 306.3) / 2$$

$$= 394089.7 \text{ Kgf}$$

Area Resisting Internal Pressure [Ap]:

$$= (fS + fY + fN) / P$$

$$= (636542 + 394089.7 + 193455.7) / 192.204$$

$$= 6245.3 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= Sallow / (2 \cdot Ap / AT - Rxs / (2 \cdot teff))$$

$$= 241.712 / (2 \cdot 6245.291 / 607.905 - 1312.86 / (2 \cdot 78.5))$$

$$= 198.4 \text{ bars}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= 2 \cdot S[t/Rxs]$$

$$= 2 \cdot 161.141[78.5/1312.86]$$

$$= 192.7 \text{ bars}$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(Pmax1, Pmax2)$$

$$= \min(198.36, 192.691)$$

$$= 192.691 \text{ bars}$$

Average Primary Membrane Stress [σAvg]:

$$= (fN + fS + fY) / AT$$

$$= (193455.703 + 636542 + 394089.656) / 607.905$$

$$= 197.472 \text{ N./mm}^2$$

General Primary Membrane Stress [σCirc]:

$$= P \cdot Rxs / (2 \cdot teff)$$

$$= 192.204 \cdot 1312.86 / (2 \cdot 78.5)$$

$$= 160.7 \text{ N./mm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma Avg - \sigma Circ, \sigma Circ)$$

$$= \max(2 \cdot 197.472 - 160.734, 160.734)$$

$$= 234.2 \text{ N./mm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	241.71	N./mm ²
Local Primary Membrane Stress	PL	234.21	N./mm ²
Maximum Allowable Working Pressure	Pmax	192.69	bars

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= (Rnc + tn) / Rnc$$

$$= (306.3 + 178.5) / 306.3$$

$$= 1.583 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$= \pi / 2 \cdot (Rn + tn)$$

$$= \pi / 2 \cdot (306.3 + 178.5)$$

$$= 761.522 \text{ mm.}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm.}$$

Required Weld Throat Dimensions :

$$tc, L41T = 7.071, \text{ must be } \geq \min(0.7 \cdot 178.500, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$= \min(fy \cdot ky, 1.5 \cdot Sn(A2 + A3), \pi / 4 \cdot P \cdot Rn^2 \cdot ky^2)$$

$$= \min(394090 \cdot 1.58, 1.5 \cdot 127.7(557.502 + 0.0), \pi / 4 \cdot 192.2 \cdot 306.3^2 \cdot 1.58^2)$$

$$= 361805.125 \text{ Kgf}$$

Weld Stress Value [τ]:

$$= fwelds / (Lt(0.49 \cdot L41T + 0.6 \cdot tw1 + 0.49 \cdot L43T))$$



$= 361805.125 / (761.522(0.49 \cdot 7.071 + 0.6 \cdot 78.5 + 0.49 \cdot 0.0))$
 $= 92.145 < \text{or} = \text{to } 161.141 \text{ Weld Stress Passed}$

Nozzle Calculations per Section 4.5: External Pressure Case:

Nozzle Radius for Force Calculation [Rxn]:

$= t_n / \ln[1 + t_n/R_n]$
 $= 178.5 / \ln[1 + 178.5/306.3]$
 $= 388.744 \text{ mm.}$

Shell Radius for Force Calculation [Rxs]:

$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$
 $= 78.5 / \ln[1 + 78.5/1274.0]$
 $= 1312.860 \text{ mm.}$

Allowable Local Primary Membrane Stress [Sallow]:

$= 1.5 \cdot S$
 $= 1.5 \cdot 161.141$
 $= 241.7 \text{ N./mm}^2$

Force acting on the Nozzle [fN]:

$= P \cdot R_{xn} \cdot L_H$
 $= 1.034 \cdot 388.744 \cdot 253.898$
 $= 1040.9 \text{ Kg}$

Force acting on the Shell [fS]:

$= P \cdot R_{xs} \cdot (L_R + t_n)/2$
 $= 1.034 \cdot 1312.86 \cdot (316.242 + 178.5)/2$
 $= 3425.0 \text{ Kg}$

Discontinuity Force from External Pressure [fY]:

$= (P \cdot R_{xs} \cdot R_{nc})/2$
 $= (1.034 \cdot 1312.86 \cdot 306.3)/2$
 $= 2120.5 \text{ Kg}$

Area Resisting External Pressure [Ap]:

$= (f_S + f_Y + f_N)/P$
 $= (3425.0 + 2120.5 + 1040.9)/1.034$
 $= 6245.3 \text{ cm}^2$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$= S_{allow} / (2 \cdot A_p / A_T - R_{xs} / (2 \cdot t_{eff}))$
 $= 241.712 / (2 \cdot 6245.291 / 607.905 - 1312.86 / (2 \cdot 78.5))$
 $= 198.4 \text{ bars}$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$= 2 \cdot S[t/R_{xs}]$
 $= 2 \cdot 161.141[78.5/1312.86]$
 $= 192.7 \text{ bars}$

Maximum Allowable Working Pressure [Pmax]:

$= \min(P_{max1}, P_{max2})$
 $= \min(198.36, 192.691)$
 $= 192.691 \text{ bars}$

Average Primary Membrane Stress [σ_{Avg}]:

$= (f_N + f_S + f_Y) / A_T$
 $= (1040.923 + 3425.027 + 2120.471) / 607.905$
 $= 1.063 \text{ N./mm}^2$

General Primary Membrane Stress [σ_{Circ}]:

$= P \cdot R_{xs} / (2 \cdot t_{eff})$
 $= 1.034 \cdot 1312.86 / (2 \cdot 78.5)$
 $= 0.9 \text{ N./mm}^2$

Maximum Local Primary Membrane Stress [PL]:

$= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$
 $= \max(2 \cdot 1.063 - 0.865, 0.865)$
 $= 1.3 \text{ N./mm}^2$



Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	241.71	N./mm ²
Local Primary Membrane Stress	PL	1.26	N./mm ²
Maximum Allowable External Pressure	Pmax	192.69	bars

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -29 °C

The Drop for this Nozzle is : 95.9690 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 573.4690 mm.

Input Echo, WRC107/537 Item 1, Description: B124" :

Diameter Basis for Vessel	Vbasis	ID	
Cylindrical or Spherical Vessel	Cylsph	Spherical	
Internal Corrosion Allowance	Cas	1.5000	mm.
Vessel Diameter	Dv	2545.000	mm.
Vessel Thickness	Tv	80.000	mm.
Design Temperature	T1	85.0	C
Vessel Material		SA-516 70	
Vessel UNS Number		K02700	
Vessel Cold S.I. Allowable	Smc	175.00	N./mm ²
Vessel Hot S.I. Allowable	Smh	161.14	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Attachment Type	Type	Round	
WRC107 Attachment Classification	Holsol	Hollow	
Diameter Basis for Nozzle	Nbasis	ID	
Corrosion Allowance for Nozzle	Can	1.5000	mm.
Nozzle Diameter	Dn	609.600	mm.
Nozzle Thickness	Tn	180.000	mm.
Nozzle Material		SA-350 LF1	
Nozzle UNS Number		K03009	
Nozzle Cold S.I. Allowable	SNmc	138.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh	127.70	N./mm ²
Design Internal Pressure	Dp	192.204	bars
Include Pressure Thrust		No	

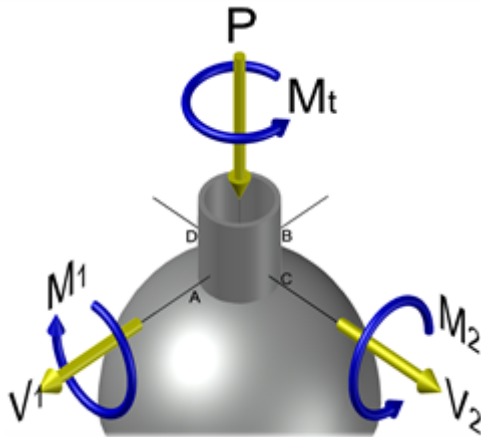
External Forces and Moments in WRC 107/537 Convention:

Radial Load (SUS)	P	-14684.9	Kgf
Longitudinal Shear (SUS)	(V1) V1	12979.7	Kgf
Circumferential Shear (SUS)	(Vc) V2	12979.7	Kgf
Circumferential Moment (SUS)	(Mc) M1	20427.4	Kg-m.
Longitudinal Moment (SUS)	(Ml) M2	20427.4	Kg-m.
Torsional Moment (SUS)	Mt	26422.6	Kg-m.

Use Interactive Control No
 WRC107 Version Version March 1979

Include Pressure Stress Indices per Div. 2 No
 Compute Pressure Stress per WRC-368 No
 Local Loads applied at end of Nozzle/Attachment No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 969.6 + 2 \cdot 1.65 \cdot \sqrt{(1313.25 (80.0 - 1.5))}$$

$$= 2029.152 \text{ mm.}$$

WRC 107 Stress Calculation for SUSTAINED loads:

Radial Load		P	-14684.9	Kgf
Circumferential Shear	(VC)	V2	12979.7	Kgf
Longitudinal Shear	(VL)	V1	12979.7	Kgf
Circumferential Moment	(MC)	M1	20427.4	Kg-m.
Longitudinal Moment	(ML)	M2	20427.4	Kg-m.
Torsional Moment		MT	26422.6	Kg-m.

Dimensionless Param: U = 1.51 TAU = 5.00 (2.22) RHO = 0.44

Dimensionless Loads for Spherical Shells at Attachment Junction:

Curves read for 1979	Figure	Value	Location
$N(x) * T / P$	SP 1	0.02605	(A,B,C,D)
$M(x) / P$	SP 1	0.02431	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / MC$	SM 1	0.02558	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / MC$	SM 1	0.03210	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / ML$	SM 1	0.02558	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / ML$	SM 1	0.03210	(A,B,C,D)
$N(y) * T / P$	SP 1	0.03764	(A,B,C,D)
$M(y) / P$	SP 1	0.00684	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / MC$	SM 1	0.03403	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / MC$	SM 1	0.01022	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / ML$	SM 1	0.03403	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / ML$	SM 1	0.01022	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Memb.	P	0.6	0.6	0.6	0.6	0.6	0.6	0.6	0.6
Rad. Bend.	P	3.4	-3.4	3.4	-3.4	3.4	-3.4	3.4	-3.4
Rad. Memb.	MC	0.0	0.0	0.0	0.0	-2.6	-2.6	2.6	2.6
Rad. Bend.	MC	0.0	0.0	0.0	0.0	-19.5	19.5	19.5	-19.5
Rad. Memb.	ML	-2.6	-2.6	2.6	2.6	0.0	0.0	0.0	0.0
Rad. Bend.	ML	-19.5	19.5	19.5	-19.5	0.0	0.0	0.0	0.0
Tot. Rad. Str.		-18.1	14.1	26.1	-19.7	-18.1	14.1	26.1	-19.7



Tang. Memb. P	0.9	0.9	0.9	0.9	0.9	0.9	0.9	0.9
Tang. Bend. P	1.0	-1.0	1.0	-1.0	1.0	-1.0	1.0	-1.0
Tang. Memb. MC	0.0	0.0	0.0	0.0	-3.4	-3.4	3.4	3.4
Tang. Bend. MC	0.0	0.0	0.0	0.0	-6.2	6.2	6.2	-6.2
Tang. Memb. ML	-3.4	-3.4	3.4	3.4	0.0	0.0	0.0	0.0
Tang. Bend. ML	-6.2	6.2	6.2	-6.2	0.0	0.0	0.0	0.0
Tot. Tang. Str.	-7.8	2.7	11.5	-2.8	-7.8	2.7	11.5	-2.8
Shear VC	1.1	1.1	-1.1	-1.1	0.0	0.0	0.0	0.0
Shear VL	0.0	0.0	0.0	0.0	-1.1	-1.1	1.1	1.1
Shear MT	2.2	2.2	2.2	2.2	2.2	2.2	2.2	2.2
Tot. Shear	3.3	3.3	1.2	1.2	1.2	1.2	3.3	3.3
Str. Int.	19.0	15.0	26.2	19.8	18.2	14.2	26.8	20.3

WRC 107/537 Stress Summations:

Vessel Stress Summation at Attachment Junction (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Pm (SUS)	160.7	160.7	160.7	160.7	160.7	160.7	160.7	160.7
Rad.	Pl (SUS)	-2.0	-2.0	3.2	3.2	-2.0	-2.0	3.2	3.2
Rad.	Q (SUS)	-16.1	16.1	22.9	-22.9	-16.1	16.1	22.9	-22.9
Long.	Pm (SUS)	160.7	160.7	160.7	160.7	160.7	160.7	160.7	160.7
Long.	Pl (SUS)	-2.6	-2.6	4.3	4.3	-2.6	-2.6	4.3	4.3
Long.	Q (SUS)	-5.3	5.3	7.2	-7.2	-5.3	5.3	7.2	-7.2
Shear	Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear	Pl (SUS)	1.1	1.1	-1.1	-1.1	-1.1	-1.1	1.1	1.1
Shear	Q (SUS)	2.2	2.2	2.2	2.2	2.2	2.2	2.2	2.2
Pm (SUS)		160.7	160.7	160.7	160.7	160.7	160.7	160.7	160.7
Pm+Pl (SUS)		159.6	159.6	165.7	165.7	159.6	159.6	165.7	165.7
Pm+Pl+Q (Total)		153.9	175.7	186.9	158.0	153.0	175.0	187.6	158.5

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	160.73	161.14	Passed
Pm+Pl (SUS)	165.70	241.71	Passed
Pm+Pl+Q (TOTAL)	187.55	484.02	Passed

The Pm+Pl+Q allowable above was based on the rules of ASME VIII-2 Part 4, 4.1.6.3 (a)&(b) or per ASME VIII-2, 5.5.6.1(1)&(2) Part 5 as applicable.

The Pm+Pl+Q allowable was based on a temperature range cycling from ambient to design temperature.

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PV Elite[®] Pressure Vessel Design Notes

Project Information	
Project Number	
Client	
Date	
Designer	
Design Code	
Additional Standards	

Design Conditions		
Internal Design Pressure:		
External Design Pressure:		
Design Int. Temperature:		
Design Ext. Temperature:		
Corrosion Allowance:		

Vessel Geometry		
Vessel Type		
Diameter		
Tangent-to-Tangent Length		
Head Type		
Number of Segments		

Material Specifications	
Shell:	
Head:	
Nozzle:	

Nozzle/Opening Details	ID/OD	Size	Rating	Matl.	Reinf.

Miscellaneous		
Support Type		
Number of Supports		
Support Material		
Seismic Code		
Wind Code		
Insulation		
Pressure Test		
Radiography		
Fatigue Analysis		

Analysis Notes: